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\\USER

jon

Saskia

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				dzne_ep3d_Frankfurt_6echoes_minimumMaxBW_optR dzne_ep3d_Frankfurt_6echoes_minimumMaxBW_optR dzne_ep3d_Frankfurt_6echoes_minimumMaxBW_optR
			DZNE_3D_EPI_func	dzne_ep3d_3mm_TE12p2_TR26p7_volTR133p5_ES0p32 dzne_ep3d_1p8mm_TE23p8_TR50p1_volTR450p9_ES0p38 dzne_ep3d_1p2mm_TE24p5_TR51p6_volTR1238p4_ES0p52 dzne_ep3d_1mm_TE22p1_TR44_volTR1496_ES0p58 dzne_ep3d_3mm_TE8p6_TR19p6_volTR98_ES0p22_wRS dzne_ep3d_1p8mm_TE20p2_TR42p9_volTR386p1_ES0p32_wRS dzne_ep3d_1p2mm_TE18p6_TR39p9_volTR1037p4_ES0p39_wRS dzne_ep3d_1mm_TE17p3_TR34p6_volTR1271p6_ES0p45_wRS ep_seg_fid_3p3x3p3x3_cant_match_acceleration ep_seg_fid_1p7x1p7x1p8_cant_match_EPI_factor ep_seg_fid_3p3x3p3x3_cant_match_acceleration dzne_ep3d_3mm_TE8p6_TR19p6_volTR98_ES0p22_wRS dzne_ep3d_3mm_TE10_TR20_volTR100_ES0p25_wRS dzne_ep3d_normal
			3D EPI fQSM	AAhead_scout ep_seg_fid ep_seg_fQSM ep_seg_fQSM ep_seg_fQSM_0p7 ep_seg_fQSM_0p6 ep_seg_fQSM_0p6 dzne_ep3d_0p6mm_TE26p5_TR56p9_volTR_10p9_CAIPi_2x2 gre_3D_0p5mm_bipolar_4echoes_fatsat gre_3D_0p5mm_bipolar_4echoes_waterex dzne_ep3d_0p6mm_TE15p2_TR33p4_volTR_8p55_CAIPi_2x3_2 dzne_ep3d_0p6mm_TE14p3_TR31p6_volTR_8p09_CAIPi_2x3_2 dzne_ep3d_0p6mm_TE15p2_TR33p4_volTR8550p4_C2x3s2 dzne_ep3d_0p6mm_TE14p5_TR32_volTR8192_C2x3s2 gre_3D_0p5mm_monopolar_6echoes_waterex
			QSM_3D_Jingyuan	gre_3D_0p5mm_ps_monopolar_6echoes gre_3D_0p5mm_ps_monopolar_8echoes gre_3D_0p5mm_ps_bipolar_9echoes
			20260107_TOF_unconfuse	

	localizer tof_fl3d_tra_cs8 tof_fl3d_tra_GRAPPA2 tof_fl3d_tra_cs8_seg7 tof_fl3d_tra_GRAPPA2_seg7 tof_fl3d_tra_GRAPPA2 tof_fl3d_tra_cs8_longTR tof_fl3d_tra_GRAPPA1 tof_fl3d_tra_cs8_slabs5 gre
	MP2RAGE
	t1_mp2rage_sag_0p6_Magdeburg t1_mp2rage_sag_0p6_Magdeburg_wip
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	tof_fl3d_tra_0p3iso_TE3p6_TR8p8_GRAPPA_3x1_slabs1 tof_fl3d_tra_0p3iso_TE3p6_TR8p8_GRAPPA_3x1_slabs1
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				dzne_ep3d_3mm_TE8p8_TR19p9_volTR99p5_es_29 dzne_ep3d_1p8mm_1p8mm_TE18p2_TR38P7_volTR_309p6_es2
			260529_THS_ses01	
				localizer AAhead_scout_md gre_b0map_4iso_sag dzne_b1map_5iso_sag head bss_map ptxps_AC_PA7_PW10_4mmISO_89s ptxps_CP_PA4_PW10_4mmISO_20s mp2rage_0p7iso t1_mp2rage_sag_0p7_wip t2_flair_sag_p9_0.7iso_dr ep2d_diff_1p3iso_b2k5_PA ep2d_diff_1p3iso_b1k0_PA ep2d_diff_1p3iso_AP ep2d_bold_1p8mm_TR1000 ep2d_bold_1p0mm_TR1000_TE25_SMS2x6_BW3126_GC ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC dzne_ep3d_vasoNIH tof_fl3d_tra_0p3iso_TE1p52_TR4p81_GRAPPA_3x1_noFC dzne_ep3d_Frankfurt_6echoes_minimumMaxBW
			260601_THS_ses02	
				localizer AAhead_scout_md gre_b0map_4iso_sag dzne_b1map_5iso_sag head bss_map ptxps_AC_PA7_PW10_4mmISO_89s ptxps_CP_PA4_PW10_4mmISO_20s gre_FA10 gre_FA0 mp2rage_0p7iso mp2rage_0p7iso gre_FA10 gre_FA0

			ep2d_diff_1p3iso_b2k5_PA ep2d_diff_1p3iso_b1k0_PA ep2d_diff_1p3iso_AP ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC dzne_ep3d_vasoNIH dzne_ep3d_vasoNIH dzne_ep3d_Frankfurt_6echoes_minimumMaxBW
		260602_THS_ses03	localizer AAhead_scout_md gre_b0map_4iso_sag dzne_b1map_5iso_sag dzne_b1map_5iso_sag localizer AAhead_scout_md gre_b0map_4iso_sag dzne_b1map_5iso_sag head bss_map ptxps_AC_PA7_PW10_4mmISO_89s ptxps_CP_PA4_PW10_4mmISO_20s gre_FA10_acpc gre_FA0_acpc gre_FA10_plus20mm gre_FA0_up mp2rage_0p7iso_patientSpecific mp2rage_0p7iso_TrueForm ep2d_diff_1p3iso_b2k5_PA ep2d_diff_1p3iso_b1k0_PA ep2d_diff_1p3iso_AP ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC dzne_ep3d_vasoNIH dzne_ep3d_vasoNIH dzne_ep3d_Frankfurt_6echoes_minimumMaxBW
		gradient_comparison	ep2d_bold_grad_1p8mm ep2d_bold_grad_1p8mm_SC72 ep2d_bold_1p4mm_TE19_TR1000 ep2d_bold_grad_1p4mm_SC72

		cmrr_mbep2d_bold_1p4mm_TE19_TR1000 ep2d_bold_grad_1p0mm_SC72 ep2d_bold_1p0mm_TR1000_TE25_SMS2x6_BW3126_GC
	260805-S012-2	AAhead_scout ptxps_AC_PA8_PW12_3mmISO_46s ptxps_CP_PA4_PW10_3mmISO_34s ep2d_bold_1p8mm_TE21_TR1000 ep2d_bold_1p8mm_TE21_TR1000_MB1 ep2d_bold_1p0mm_TE19_TR2000 ep2d_bold_1p0mm_TE19_TR2000_MB1 ep2d_bold_1p4mm_TE19_TR1000 ep2d_bold_1p4mm_TE19_TR1000_MB1 tof_fl3d_tra_0p25iso_TE1p67_TR5p09_GRAPPA_3x1_noFC tof_fl3d_tra_0p25iso_TE1p67_TR5p09_GRAPPA_3x1_noFC gre_3D_0p5mm_bipolar_4echoes_waterex ep2d_bold_1p4mm_TE19_TR1000_80slices tof_fl3d_tra_0p25iso_TE1p67_TR5p09_GRAPPA_3x1_noFC gre_3D_0p5mm_bipolar_4echoes_waterex gre_3D_0p5mm_bipolar_4echoes_waterex tof_fl3d_tra_0p25iso_TE1p67_TR5p09_GRAPPA_3x1_noFC tof_fl3d_tra_0p25iso_TE1p67_TR5p09_GRAPPA_3x1_noFC
	epirecon	dzne_ep3d dzne_ep3d dzne_ep3d
	260921-VascProt	AAhead_scout dzne_ep3d_6echoes_R1 t1_mp2rage_sag_0p7_R5 tof_HR_fl3d_tra_0p2_TE1p83_TR10_GRAPPA_3x1_FA15
	THS_S02_ses01	localizer AAhead_scout_md gre_b0map_4iso_sag dzne_b1map_5iso_sag gre_FA10_acpc gre_FA0_acpc

					gre_FA10_plus20mm	
					gre_FA0_plus20mm	
					mp2rage_0p7iso_TrueForm	
					mp2rage_0p7iso_TrueForm	
					ep2d_diff_1p3iso_b2k5_PA	
					ep2d_diff_1p3iso_b1k0_PA	
					ep2d_diff_1p3iso_AP	
					ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC	
					dzne_ep3d_vasoNIH	
					dzne_ep3d_vasoNIH	
					dzne_ep3d_Frankfurt_6echoes_minimumMaxBW_optR	

\\USER\jon\Saskia\Defaults_B1\localizer

TA: 15 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	All Segments
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
TE	3.69 ms
Averages	2
Concatenations	3

Contrast - Common

TR	8.6 ms
TE	3.69 ms

Contrast - Common

TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slice Group	3

Geometry - Common

Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg
Slice Group	2
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
Slice Group	3
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.6 ms
Segments	1
Concatenations	3

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None

Physio - Cardiac

Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	20 deg
Measurements	1
Contrasts	1
TE	3.69 ms
TR	8.6 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	qfl
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Sequence - Part 1

Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\Defaults_B1\bss_map

TA: 2:01 min Coil Selection: Manual Voxel Size: 4.0x4.0x4.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
TR	5.0 ms
TE	2.04 ms
TE2 B0	4.08 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	5.0 ms
TE	2.04 ms
TE2 B0	4.08 ms
Flip Angle	5 deg
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
Base Resolution	64
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
TR	5.0 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	192 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
! Ref. Amplitude 1H	270.000 V

System - Tx/Rx

Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	5 deg

Inline - MapIt

Measurements	1
Contrasts	1
TE	2.04 ms
TE2 B0	4.08 ms
TR	5.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	1000 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off
Phase Stabilisation	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	NC7

Sequence - Special

Map Type	Integrated
Number of shim-sets	8
RO Spoiler	1.00
Measure B0	On
Call B0 Acq	Off
Call B1 Acq	Off
Rel. Mask Threshold	20

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\Defaults_B1\head

TA: 36 sec Coil Selection: Manual Voxel Size: 4.0×4.0×5.0 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	16
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Routine

Slice Group	1
Slices	25
Distance Factor	100 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	3900.0 ms
TE	1.56 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Geometry - Common

Slice Group	1
Slices	25
Distance Factor	100 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	3900.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	3900.0 ms
TE	1.56 ms
Magn. Preparation	None
Flip Angle	5 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Centric

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	256 mm
R >> L	256 mm
F >> H	245 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3900.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
--------------------	----------

Physio - Cardiac

Magn. Preparation	None
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	tfl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Whisper
Flow Compensation	None
Reordering	Centric
Bandwidth	440 Hz/Px
Echo Spacing	3.54 ms
Asymmetric Echo	Allowed
Turbo Factor	64

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Tx Scale Diag. Magn.	0.3
Tx Scale Diag. Phase	0 deg
Tx Scale Off-diag. Magn.	1.0
Tx Scale Off-diag. Phase	0 deg
Rel. B1 mapping	On
Ref. Scan	Off
Use B1 Map Recon.	On
Prep Pulse	Selective
Dummy RF pulses	1000
Dummy RF TRs	0

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\Defaults_B1\ldzne_b1mapping

TA: 7 sec Coil Selection: Auto Voxel Size: 5.0×5.0×5.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 P4.8 F34.1 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Averages	1
Concatenations	1

Contrast - Common

TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Flip Angle 1	60 deg
Flip Angle 2	6 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
Base Resolution	44
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 P4.8 F34.1 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P4.8 F34.1 mm
Orientation	Sagittal

Geometry - AutoAlign

Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	34 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	b1map 9e96902
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	1000 Hz/Px
Asymmetric Echo	Off
Turbo Factor	440
Echo Train Duration	1298 ms

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Mapping Technique	DREAM
Timing Scheme	STE*
Mixing Time	1130 us
Scale risetime	1.2

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\Defaults_B1\gre_field_mapping_tra

TA: 31 sec Coil Selection: Auto Voxel Size: 4.0×4.0×4.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	44
Distance Factor	0 %
Position	L0.0 P8.4 F25.1 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
Averages	1
Concatenations	1

Contrast - Common

TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
MTC	Off
Flip Angle	25 deg
Fat-Water Contrast	Standard
Contrasts	2
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	44
Distance Factor	0 %
Position	L0.0 P8.4 F25.1 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.4 F25.1 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.4 F25.1
L	0.0 mm
P	8.4 mm
F	25.1 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	25 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.4 F25.1 mm
Orientation	Transversal
Rotation	0.00 deg
A >> P	256 mm
R >> L	256 mm
F >> H	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fm_r
Dimension	2D
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	On
Bandwidth	908 Hz/Px
Asymmetric Echo	Off

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\TOF\tof_fl3d_tra_0p3iso_TE3p12_TR7p29_GRAPPA_3x1_slabs1

TA: 5:43 min Coil Selection: Manual Voxel Size: 0.3x0.3x0.3 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
Base Resolution	656
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	L0.3 P17.8 H13.2 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	224
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	7.29 ms
TE	3.12 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Contrast - Common

TR	7.29 ms
TE	3.12 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	12 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	L0.3 P17.8 H13.2 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	224
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	200 mm

Geometry - Common

FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	7.29 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.3 P17.8 H13.2 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.3 P17.8 H13.2
L	0.3 mm
P	17.8 mm
H	13.2 mm
Initial Orientation	Transversal
Initial Rotation	90.35 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	13 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.3 P17.8 H13.2 mm
Orientation	Transversal

System - Adjust Volume

Rotation	90.35 deg
R >> L	175 mm
A >> P	200 mm
F >> H	68 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7.29 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	87.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.12 ms
TR	7.29 ms

Inline - MIP

MIP Sag	On
---------	----

Inline - MIP

MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Low SAR
Gradient Mode	Boost*
Flow Compensation	On
Reordering	Linear
Bandwidth	254 Hz/Px
Echo Spacing	7.29 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\TOF\tof_fl3d_tra_0p3iso_TE1p52_TR4p81_GRAPPA_3x1_noFC

TA: 8:02 min Coil Selection: Manual Voxel Size: 0.3×0.3×0.3 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	L0.3 P17.8 H13.2 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	416
Phase Oversampling	0 %
Slice Oversampling	15.4 %
FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	4.81 ms
TE	1.52 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	4.81 ms
TE	1.52 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	5 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
Base Resolution	656
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	L0.3 P17.8 H13.2 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	416
Phase Oversampling	0 %
Slice Oversampling	15.4 %
FOV Read	200 mm

Geometry - Common

FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	4.81 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.3 P17.8 H13.2 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.3 P17.8 H13.2
L	0.3 mm
P	17.8 mm
H	13.2 mm
Initial Orientation	Transversal
Initial Rotation	90.35 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	13 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.3 P17.8 H13.2 mm
Orientation	Transversal

System - Adjust Volume

Rotation	90.35 deg
R >> L	175 mm
A >> P	200 mm
F >> H	125 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	4.81 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	87.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.52 ms
TR	4.81 ms

Inline - MIP

MIP Sag	On
---------	----

Inline - MIP

MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Fast
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	254 Hz/Px
Echo Spacing	4.81 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\TOF\tof_f13d_tra_0p3iso_TE1p54_TR4p83_GRAPPA_2x2_noFC_sl3

TA: 9:10 min Coil Selection: Manual Voxel Size: 0.3×0.3×0.3 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	3
Distance Factor	-19 %
Position	L0.3 P8.8 H18.6 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	33.3 %
FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	4.83 ms
TE	1.54 ms
Averages	1
Concatenations	3
Coil Elements	NC7

Contrast - Common

TR	4.83 ms
TE	1.54 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
Base Resolution	656
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	48
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-19 %
Position	L0.3 P8.8 H18.6 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	33.3 %

Geometry - Common

FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	4.83 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	L0.3 P8.8 H18.6 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.3 P8.8 H18.6
L	0.3 mm
P	8.8 mm
H	18.6 mm
Initial Orientation	Transversal
Initial Rotation	90.35 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	19 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.3 P8.8 H18.6 mm
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System - Adjust Volume

Orientation	Transversal
Rotation	90.35 deg
R >> L	175 mm
A >> P	200 mm
F >> H	114 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	4.83 ms
Segments	1
Concatenations	3

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	87.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.54 ms
TR	4.83 ms

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Fast
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	254 Hz/Px
Echo Spacing	4.83 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\TOF\tof_fl3d_tra_0p3iso_TE1p54_TR4p83_CS_4_noFC_sl3

TA: 6:39 min Coil Selection: Manual Voxel Size: 0.3×0.3×0.3 mm³ Acc:: 4.0 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	3
Distance Factor	-19 %
Position	L0.3 P17.8 H13.2 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	33.3 %
FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	4.83 ms
TE	1.54 ms
Averages	1
Concatenations	3
Coil Elements	NC7

Contrast - Common

TR	4.83 ms
TE	1.54 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
Base Resolution	656
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	4.0
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-19 %
Position	L0.3 P17.8 H13.2 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	33.3 %
FOV Read	200 mm

Geometry - Common

FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	4.83 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	L0.3 P17.8 H13.2 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.3 P17.8 H13.2
L	0.3 mm
P	17.8 mm
H	13.2 mm
Initial Orientation	Transversal
Initial Rotation	90.35 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	13 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.3 P17.8 H13.2 mm
Orientation	Transversal

System - Adjust Volume

Rotation	90.35 deg
R >> L	175 mm
A >> P	200 mm
F >> H	114 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	4.83 ms
Segments	1
Concatenations	3

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	87.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.54 ms
TR	4.83 ms

Inline - MIP

MIP Sag	On
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Inline - MIP

MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Fast
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	254 Hz/Px
Echo Spacing	4.83 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\TOF\tof_fl3d_tra_0p3iso_TE4p12_TR73p85_cs7p2_slabs1

TA: 4:06 min Coil Selection: Manual Voxel Size: 0.3×0.3×0.3 mm³ Acc:: 7.2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	205 mm
FOV Phase	90.7 %
Slice Thickness	0.30 mm
Base Resolution	688
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	L0.3 P17.8 H13.2 mm
Orientation	T > C-16.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	205 mm
FOV Phase	90.7 %
Slice Thickness	0.30 mm
TR	73.85 ms
TE	4.12 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	7.2
Reference Scans	GRE/Separate
Reference Lines PE	33
Reference Lines 3D	24
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	On

Contrast - Common

TR	73.85 ms
TE	4.12 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	23 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	L0.3 P17.8 H13.2 mm
Orientation	T > C-16.6
Phase Encoding Dir.	R >> L
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	205 mm

Geometry - Common

FOV Phase	90.7 %
Slice Thickness	0.30 mm
TR	73.85 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.3 P17.8 H13.2 mm
Orientation	T > C-16.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.3 P17.8 H13.2
L	0.3 mm
P	17.8 mm
H	13.2 mm
Initial Orientation	T > C
T > C	-16.60
> S	0.00
Initial Rotation	90.35 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	13 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.3 P17.8 H13.2 mm
Orientation	T > C-16.6
Rotation	90.35 deg
R >> L	186 mm
A >> P	205 mm
F >> H	77 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	73.85 ms
Segments	7
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	205 mm
FOV Phase	90.7 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	4.12 ms
TR	73.85 ms

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Low SAR
Gradient Mode	Boost*
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	119 Hz/Px
Echo Spacing	10.55 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	7

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\TOF\tof_fl3d_tra_0p3iso_TE1p53_TR4p82_cs7p2_noFC

TA: 1:15 min Coil Selection: Manual Voxel Size: 0.3×0.3×0.3 mm³ Acc:: 7.2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	L0.3 P17.8 H13.2 mm
Orientation	T > C-16.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	4.82 ms
TE	1.53 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	4.82 ms
TE	1.53 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	23 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
Base Resolution	656
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	7.2
Reference Scans	GRE/Separate
Reference Lines PE	33
Reference Lines 3D	24
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	L0.3 P17.8 H13.2 mm
Orientation	T > C-16.6
Phase Encoding Dir.	R >> L
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	200 mm

Geometry - Common

FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	4.82 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.3 P17.8 H13.2 mm
Orientation	T > C-16.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.3 P17.8 H13.2
L	0.3 mm
P	17.8 mm
H	13.2 mm
Initial Orientation	T > C
T > C	-16.60
> S	0.00
Initial Rotation	90.35 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	13 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.3 P17.8 H13.2 mm
Orientation	T > C-16.6
Rotation	90.35 deg
R >> L	175 mm
A >> P	200 mm
F >> H	53 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	4.82 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	87.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.53 ms
TR	4.82 ms

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Fast
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	254 Hz/Px
Echo Spacing	4.82 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\TOF\tof_fl3d_tra_0p2iso_TE1p76_TR5p34_cs7p2_noFC

TA: 4:03 min Coil Selection: Manual Voxel Size: 0.2x0.2x0.2 mm³ Acc:: 7.2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	-18 %
Position	L0.3 P17.8 H13.2 mm
Orientation	T > C-16.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	88
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	87.1 %
Slice Thickness	0.20 mm
TR	5.34 ms
TE	1.76 ms
Averages	4
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	5.34 ms
TE	1.76 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	23 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	200 mm
FOV Phase	87.1 %
Slice Thickness	0.20 mm
Base Resolution	1024
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	7.2
Reference Scans	GRE/Separate
Reference Lines PE	33
Reference Lines 3D	24
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-18 %
Position	L0.3 P17.8 H13.2 mm
Orientation	T > C-16.6
Phase Encoding Dir.	R >> L
Slices per Slab	88
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	200 mm

Geometry - Common

FOV Phase	87.1 %
Slice Thickness	0.20 mm
TR	5.34 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.3 P17.8 H13.2 mm
Orientation	T > C-16.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.3 P17.8 H13.2
L	0.3 mm
P	17.8 mm
H	13.2 mm
Initial Orientation	T > C
T > C	-16.60
> S	0.00
Initial Rotation	90.35 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	13 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.3 P17.8 H13.2 mm
Orientation	T > C-16.6
Rotation	90.35 deg
R >> L	175 mm
A >> P	200 mm
F >> H	18 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.34 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	87.1 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.76 ms
TR	5.34 ms

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Fast
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	244 Hz/Px
Echo Spacing	5.34 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\TOF\tof_fl3d_tra_0p25iso_TE1p67_TR5p09_CS_4p0_noFC

TA: 4:49 min Coil Selection: Manual Voxel Size: 0.3×0.3×0.3 mm³ Acc:: 4.0 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	200 mm
FOV Phase	88.5 %
Slice Thickness	0.25 mm
Base Resolution	800
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	2
Distance Factor	-20 %
Position	L0.3 P20.2 H36.0 mm
Orientation	T > C-10.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	11.1 %
FOV Read	200 mm
FOV Phase	88.5 %
Slice Thickness	0.25 mm
TR	5.09 ms
TE	1.67 ms
Averages	1
Concatenations	2
Coil Elements	NC7

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	4.0
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Contrast - Common

TR	5.09 ms
TE	1.67 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	5 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	2
Distance Factor	-20 %
Position	L0.3 P20.2 H36.0 mm
Orientation	T > C-10.6
Phase Encoding Dir.	R >> L
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	11.1 %
FOV Read	200 mm

Geometry - Common

FOV Phase	88.5 %
Slice Thickness	0.25 mm
TR	5.09 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	2

Geometry - AutoAlign

Slab Group	1
Position	L0.3 P20.2 H36.0 mm
Orientation	T > C-10.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.3 P20.2 H36.0
L	0.3 mm
P	20.2 mm
H	36.0 mm
Initial Orientation	T > C
T > C	-10.60
> S	0.00
Initial Rotation	90.35 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	36 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.3 P20.2 H36.0 mm
Orientation	T > C-10.6
Rotation	90.35 deg
R >> L	177 mm
A >> P	200 mm
F >> H	65 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.09 ms
Segments	1
Concatenations	2

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	88.5 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	2

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.67 ms
TR	5.09 ms

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Fast
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	250 Hz/Px
Echo Spacing	5.09 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\TOF\tof_fl3d_tra_0p3iso_TE1p44_TR4p45_GRAPPA_3x1_noFC

TA: 3:29 min Coil Selection: Manual Voxel Size: 0.3×0.3×0.3 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	R0.1 P8.5 H8.5 mm
Orientation	T > C-18.3
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	18.2 %
FOV Read	210 mm
FOV Phase	85.6 %
Slice Thickness	0.30 mm
TR	4.45 ms
TE	1.44 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	4.45 ms
TE	1.44 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	210 mm
FOV Phase	85.6 %
Slice Thickness	0.30 mm
Base Resolution	720
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	R0.1 P8.5 H8.5 mm
Orientation	T > C-18.3
Phase Encoding Dir.	R >> L
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	18.2 %
FOV Read	210 mm

Geometry - Common

FOV Phase	85.6 %
Slice Thickness	0.30 mm
TR	4.45 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R0.1 P8.5 H8.5 mm
Orientation	T > C-18.3
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.1 P8.5 H8.5
R	0.1 mm
P	8.5 mm
H	8.5 mm
Initial Orientation	T > C
T > C	-18.30
> S	0.00
Initial Rotation	89.91 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	8 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.1 P8.5 H8.5 mm
Orientation	T > C-18.3
Rotation	89.91 deg
R >> L	180 mm
A >> P	210 mm
F >> H	53 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	4.45 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	210 mm
FOV Phase	85.6 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.44 ms
TR	4.45 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Fast
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	302 Hz/Px
Echo Spacing	4.45 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\TOF\tof_fl3d_tra_0p25iso_TE1p67_TR5p09_GRAPPA_3x1_noFC

TA: 7:46 min Coil Selection: Manual Voxel Size: 0.3x0.3x0.3 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	2
Distance Factor	-20 %
Position	R0.2 P19.2 H40.8 mm
Orientation	T > C-18.3
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	160
Phase Oversampling	0 %
Slice Oversampling	10.0 %
FOV Read	200 mm
FOV Phase	90.0 %
Slice Thickness	0.25 mm
TR	5.09 ms
TE	1.67 ms
Averages	1
Concatenations	2
Coil Elements	NC7

Contrast - Common

TR	5.09 ms
TE	1.67 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	200 mm
FOV Phase	90.0 %
Slice Thickness	0.25 mm
Base Resolution	800
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	2
Distance Factor	-20 %
Position	R0.2 P19.2 H40.8 mm
Orientation	T > C-18.3
Phase Encoding Dir.	R >> L
Slices per Slab	160
Phase Oversampling	0 %
Slice Oversampling	10.0 %
FOV Read	200 mm

Geometry - Common

FOV Phase	90.0 %
Slice Thickness	0.25 mm
TR	5.09 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	2

Geometry - AutoAlign

Slab Group	1
Position	R0.2 P19.2 H40.8 mm
Orientation	T > C-18.3
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.2 P19.2 H40.8
R	0.2 mm
P	19.2 mm
H	40.8 mm
Initial Orientation	T > C
T > C	-18.30
> S	0.00
Initial Rotation	89.91 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	41 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.2 P19.2 H40.8 mm
Orientation	T > C-18.3
Rotation	89.91 deg
R >> L	180 mm
A >> P	200 mm
F >> H	72 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.09 ms
Segments	1
Concatenations	2

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	90.0 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	2

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.67 ms
TR	5.09 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Fast
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	250 Hz/Px
Echo Spacing	5.09 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\TOF\tof_HR_fl3d_tra_0p25iso_TE1p67_TR5p09_GRAPPA_3x1_noFC

TA: 7:46 min Coil Selection: Auto Voxel Size: 0.3x0.3x0.3 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	2
Distance Factor	-20 %
Position	R0.2 P19.2 H40.8 mm
Orientation	T > C-18.3
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	160
Phase Oversampling	0 %
Slice Oversampling	10.0 %
FOV Read	200 mm
FOV Phase	90.0 %
Slice Thickness	0.25 mm
TR	5.09 ms
TE	1.67 ms
Averages	1
Concatenations	2

Contrast - Common

TR	5.09 ms
TE	1.67 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	200 mm
FOV Phase	90.0 %
Slice Thickness	0.25 mm
Base Resolution	800
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	2
Distance Factor	-20 %
Position	R0.2 P19.2 H40.8 mm
Orientation	T > C-18.3
Phase Encoding Dir.	R >> L
Slices per Slab	160
Phase Oversampling	0 %
Slice Oversampling	10.0 %
FOV Read	200 mm
FOV Phase	90.0 %

Geometry - Common

Slice Thickness	0.25 mm
TR	5.09 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	2

Geometry - AutoAlign

Slab Group	1
Position	R0.2 P19.2 H40.8 mm
Orientation	T > C-18.3
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.2 P19.2 H40.8
R	0.2 mm
P	19.2 mm
H	40.8 mm
Initial Orientation	T > C
T > C	-18.30
> S	0.00
Initial Rotation	89.91 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	41 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.2 P19.2 H40.8 mm
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System - Adjust Volume

Orientation	T > C-18.3
Rotation	89.91 deg
R >> L	180 mm
A >> P	200 mm
F >> H	72 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.09 ms
Segments	1
Concatenations	2

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	90.0 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	2

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.67 ms
TR	5.09 ms

Inline - MIP

MIP Sag	Off
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Inline - MIP

MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Fast
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	250 Hz/Px
Echo Spacing	5.09 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\TOF\tof_HR_fl3d_tra_0p17iso_TE1p82_TR5p53_GRAPPA_3x1_noFC

TA: 14:33 min Coil Selection: Auto Voxel Size: 0.2x0.2x0.2 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	200 mm
FOV Phase	90.0 %
Slice Thickness	0.17 mm
Base Resolution	1200
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	2
Distance Factor	-20 %
Position	R0.2 P19.2 H40.8 mm
Orientation	T > C-18.3
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	192
Phase Oversampling	0 %
Slice Oversampling	8.3 %
FOV Read	200 mm
FOV Phase	90.0 %
Slice Thickness	0.17 mm
TR	5.53 ms
TE	1.82 ms
Averages	1
Concatenations	2

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Contrast - Common

TR	5.53 ms
TE	1.82 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	2
Distance Factor	-20 %
Position	R0.2 P19.2 H40.8 mm
Orientation	T > C-18.3
Phase Encoding Dir.	R >> L
Slices per Slab	192
Phase Oversampling	0 %
Slice Oversampling	8.3 %
FOV Read	200 mm
FOV Phase	90.0 %

Geometry - Common

Slice Thickness	0.17 mm
TR	5.53 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	2

Geometry - AutoAlign

Slab Group	1
Position	R0.2 P19.2 H40.8 mm
Orientation	T > C-18.3
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.2 P19.2 H40.8
R	0.2 mm
P	19.2 mm
H	40.8 mm
Initial Orientation	T > C
T > C	-18.30
> S	0.00
Initial Rotation	89.91 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	41 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.2 P19.2 H40.8 mm
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System - Adjust Volume

Orientation	T > C-18.3
Rotation	89.91 deg
R >> L	180 mm
A >> P	200 mm
F >> H	59 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.53 ms
Segments	1
Concatenations	2

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	90.0 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	2

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.82 ms
TR	5.53 ms

Inline - MIP

MIP Sag	Off
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Inline - MIP

MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Fast
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	245 Hz/Px
Echo Spacing	5.53 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\TOF\tof_HR_fl3d_tra_0p2_TE1p71_TR5p23_GRAPPA_3x1_noFC_center

TA: 8:06 min Coil Selection: Auto Voxel Size: 0.2x0.2x0.2 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	L0.0 P5.4 H1.2 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm
TR	5.31 ms
TE	1.75 ms
Averages	1
Concatenations	1

Contrast - Common

TR	5.31 ms
TE	1.75 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
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Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm
Base Resolution	1040
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	L0.0 P5.4 H1.2 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm

Geometry - Common

TR	5.31 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P5.4 H1.2 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.0 P5.4 H1.2
R	0.0 mm
P	5.4 mm
H	1.2 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	1 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P5.4 H1.2 mm
Orientation	Transversal
Rotation	90.00 deg
R >> L	180 mm

System - Adjust Volume

A >> P	208 mm
F >> H	48 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.31 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	208 mm
FOV Phase	86.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.75 ms
TR	5.31 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off

Inline - MIP

Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Fast
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	253 Hz/Px
Echo Spacing	5.31 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\TOF\tof_HR_fl3d_tra_0p2_TE1p71_TR5p23_GRAPPA_3x1_noFC_center

TA: 10:49 min Coil Selection: Auto Voxel Size: 0.2x0.2x0.2 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm
Base Resolution	1040
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	L0.6 P4.4 H2.4 mm
Orientation	T > C5.3 > S-2.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm
TR	8.61 ms
TE	3.68 ms
Averages	1
Concatenations	1

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	4
Reference Lines PE	48
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Strong
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Contrast - Common

TR	8.61 ms
TE	3.68 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	L0.6 P4.4 H2.4 mm
Orientation	T > C5.3 > S-2.6
Phase Encoding Dir.	R >> L
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm

Contrast - Dynamic

Dynamic Mode	Standard
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Geometry - Common

TR	8.61 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.6 P4.4 H2.4 mm
Orientation	T > C5.3 > S-2.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.0 P5.4 H1.2
R	0.0 mm
P	5.4 mm
H	1.2 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.6 P4.4 H2.4 mm
Orientation	T > C5.3 > S-2.6
Rotation	87.59 deg
R >> L	180 mm

System - Adjust Volume

A >> P	208 mm
F >> H	48 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.61 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	208 mm
FOV Phase	86.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.68 ms
TR	8.61 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off

Inline - MIP

Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Fast
Gradient Mode	Boost*
Flow Compensation	On
Reordering	Linear
Bandwidth	150 Hz/Px
Echo Spacing	8.61 ms
Asymmetric Echo	Strong
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\TOF\tof_HR_fl3d_tra_0p2_TE1p71_TR5p23_GRAPPA_3x1_noFC_center

TA: 7:35 min Coil Selection: Auto Voxel Size: 0.2x0.2x0.2 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Centric

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm
Base Resolution	1040
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	L0.6 P4.4 H2.4 mm
Orientation	T > C5.3 > S-2.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	120
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm
TR	9.44 ms
TE	3.12 ms
Averages	1
Concatenations	1

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	48
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Contrast - Common

TR	9.44 ms
TE	3.12 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	15 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	L0.6 P4.4 H2.4 mm
Orientation	T > C5.3 > S-2.6
Phase Encoding Dir.	R >> L
Slices per Slab	120
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm

Contrast - Dynamic

Dynamic Mode	Standard
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Geometry - Common

TR	9.44 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.6 P4.4 H2.4 mm
Orientation	T > C5.3 > S-2.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.0 P5.4 H1.2
R	0.0 mm
P	5.4 mm
H	1.2 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.6 P4.4 H2.4 mm
Orientation	T > C5.3 > S-2.6
Rotation	87.59 deg
R >> L	180 mm

System - Adjust Volume

A >> P	208 mm
F >> H	24 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	9.44 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	208 mm
FOV Phase	86.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.12 ms
TR	9.44 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off

Inline - MIP

Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Reordering	Centric
Bandwidth	150 Hz/Px
Echo Spacing	9.44 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	None

\\USER\jon\Saskia\TOF\tof_HR_f13d_tra_0p2_TE1p81_TR5p81_GRAPPA_3x1_noFC

TA: 9:05 min Coil Selection: Auto Voxel Size: 0.2x0.2x0.2 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Centric

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm
Base Resolution	1040
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	R0.3 P23.3 H37.9 mm
Orientation	T > C-12.1 > S0.1
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm
TR	5.81 ms
TE	1.81 ms
Averages	1
Concatenations	1

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	36
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Contrast - Common

TR	5.81 ms
TE	1.81 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	R0.3 P23.3 H37.9 mm
Orientation	T > C-12.1 > S0.1
Phase Encoding Dir.	R >> L
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm

Contrast - Dynamic

Dynamic Mode	Standard
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Geometry - Common

TR	5.81 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R0.3 P23.3 H37.9 mm
Orientation	T > C-12.1 > S0.1
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.1 P6.2 H44.8
L	0.1 mm
P	6.2 mm
H	44.8 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	38 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.3 P23.3 H37.9 mm
Orientation	T > C-12.1 > S0.1
Rotation	88.22 deg
R >> L	180 mm

System - Adjust Volume

A >> P	208 mm
F >> H	48 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.81 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	208 mm
FOV Phase	86.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.81 ms
TR	5.81 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	On
MIP Time	Off

Inline - MIP

Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Centric
Bandwidth	253 Hz/Px
Echo Spacing	5.81 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\TOF\tof_HR_f13d_tra_0p2_TE1p83_TR10_GRAPPA_3x1_FA15

TA: 6:32 min Coil Selection: Auto Voxel Size: 0.2x0.2x0.2 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	-10 %
Position	R2.6 P25.9 H50.5 mm
Orientation	T > C-12.6 > S1.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	104
Phase Oversampling	0 %
Slice Oversampling	15.4 %
FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm
TR	10.00 ms
TE	1.83 ms
Averages	1
Concatenations	1

Contrast - Common

TR	10.00 ms
TE	1.83 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	15 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
--------------	----------

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Centric

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm
Base Resolution	1040
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	36
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-10 %
Position	R2.6 P25.9 H50.5 mm
Orientation	T > C-12.6 > S1.6
Phase Encoding Dir.	R >> L
Slices per Slab	104
Phase Oversampling	0 %
Slice Oversampling	15.4 %
FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm

Geometry - Common

TR	10.00 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R2.6 P25.9 H50.5 mm
Orientation	T > C-12.6 > S1.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.2 P12.7 H54.6
R	0.2 mm
P	12.7 mm
H	54.6 mm
Initial Orientation	T > C
T > C	-4.50
> S	0.00
Initial Rotation	89.88 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	50 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.6 P25.9 H50.5 mm
Orientation	T > C-12.6 > S1.6

System - Adjust Volume

Rotation	89.51 deg
R >> L	180 mm
A >> P	208 mm
F >> H	21 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	10.00 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	208 mm
FOV Phase	86.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.83 ms
TR	10.00 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Centric
Bandwidth	253 Hz/Px
Echo Spacing	5.83 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\DZNE_3D_EPI\dzne_ep3d_HERTZ_0p4mm_QSM

TA: 4:57 min Coil Selection: Auto Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	416
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	204 mm
FOV Phase	100.0 %
Slice Thickness	0.40 mm
TR	34.4 ms
Vol. TR	143.104 s
TE 1	16.70 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	34.4 ms
Vol. TR	143.104 s
TE 1	16.70 ms
Multi-echo spacing	30.40 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Fast Water Excitation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	2
Reordering	Linear

Resolution - Common

FOV Read	204 mm
FOV Phase	100.0 %
Slice Thickness	0.40 mm
Base Resolution	510
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	48
Acceleration Factor 3D	3
Reference Lines 3D	48
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	416
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	204 mm
FOV Phase	100.0 %
Slice Thickness	0.40 mm
TR	34.4 ms
Vol. TR	143.104 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	34 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	204 mm
F >> H	204 mm
R >> L	167 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	654 Hz/Px
Echo Spacing	1.73 ms
Asymmetric Echo	Off
Turbo Factor	4160
Segmentation	30
EPI Factor	17

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	Single-RECT
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI\ep_seg_fid_HERTZ_0p4mm_QSM

TA: 2:37 min Coil Selection: Auto Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	416
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	204 mm
FOV Phase	100.0 %
Slice Thickness	0.40 mm
TR	41.0 ms
TE	19.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	41.0 ms
TE	19.00 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Water Excitation
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	204 mm
FOV Phase	100.0 %
Slice Thickness	0.40 mm
Base Resolution	512
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	48
Acceleration Factor 3D	3
Reference Lines 3D	36
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	3D
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	416
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	204 mm
FOV Phase	100.0 %
Slice Thickness	0.40 mm
TR	41.0 ms
Multi-Slice Mode	Interleaved
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	204 mm
F >> H	204 mm
R >> L	167 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Image Scaling	1.000
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Physio - Signal

1st Signal/Mode	None
TR	41.0 ms
Segments	9
Concatenations	1

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	652 Hz/Px
Echo Spacing	1.68 ms
Free Echo Spacing	Off
Segments	9
EPI Factor	19

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI\dzne_ep3d_Frankfurt_ref_minimumSameBW

TA: 5:09 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	39.4 ms
Vol. TR	302.592 s
TE 1	6.70 ms
TE 2	17.40 ms
TE 3	28.10 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	39.4 ms
Vol. TR	302.592 s
TE 1	6.70 ms
TE 2	17.40 ms
TE 3	28.10 ms
Multi-echo spacing	10.70 ms
MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fast Fat Saturation
Contrasts	3
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
Base Resolution	500
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	36
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	39.4 ms
Vol. TR	302.592 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
------------	---

Geometry - AutoAlign

Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
-----------------	-------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	22 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Rotation	90.00 deg
R >> L	175 mm
A >> P	210 mm
F >> H	135 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	626 Hz/Px
Echo Spacing	1.90 ms
Asymmetric Echo	Off
Turbo Factor	7680
Segmentation	40
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI\ldzne_ep3d_Frankfurt_6echoes_minimumMaxBW

TA: 6:29 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	49.9 ms
Vol. TR	383.232 s
TE 1	4.70 ms
TE 2	11.90 ms
TE 3	19.10 ms
TE 4	26.30 ms
TE 5	33.50 ms
TE 6	40.70 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	49.9 ms
Vol. TR	383.232 s
TE 1	4.70 ms
TE 2	11.90 ms
TE 3	19.10 ms
TE 4	26.30 ms
TE 5	33.50 ms
TE 6	40.70 ms
Multi-echo spacing	7.20 ms

Contrast - Common

MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fast Fat Saturation
Contrasts	6
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
Base Resolution	500
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	36
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %

Geometry - Common

Slice Thickness	0.35 mm
TR	49.9 ms
Vol. TR	383.232 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
-----------------	-------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	22 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Rotation	90.00 deg
R >> L	175 mm
A >> P	210 mm
F >> H	135 mm

System - Adjust Volume

Reset	Off
-------	-----

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.20 ms
Asymmetric Echo	Off
Turbo Factor	7680
Segmentation	40
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI\dzne_ep3d_HERTZ_0p4mm_QSM_tra_R-L

TA: 4:35 min Coil Selection: Auto Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P7.8 F22.1 mm
Orientation	T > C-12.5
Phase Encoding Dir.	R >> L
AutoAlign	---
Slices per Slab	384
Phase Oversampling	1.4 %
Slice Oversampling	0.0 %
FOV Read	206 mm
FOV Phase	85.2 %
Slice Thickness	0.40 mm
TR	38.4 ms
Vol. TR	127.79520 s
TE 1	17.00 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	38.4 ms
Vol. TR	127.79520 s
TE 1	17.00 ms
Multi-echo spacing	31.40 ms
MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fast Fat Saturation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	2
Reordering	Linear

Resolution - Common

FOV Read	206 mm
FOV Phase	85.2 %
Slice Thickness	0.40 mm
Base Resolution	512
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	48
Acceleration Factor 3D	3
Reference Lines 3D	48
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P7.8 F22.1 mm
Orientation	T > C-12.5
Phase Encoding Dir.	R >> L
Slices per Slab	384
Phase Oversampling	1.4 %
Slice Oversampling	0.0 %
FOV Read	206 mm
FOV Phase	85.2 %
Slice Thickness	0.40 mm
TR	38.4 ms
Vol. TR	127.79520 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P7.8 F22.1 mm
Orientation	T > C-12.5
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
-----------------	-------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	22 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P7.8 F22.1 mm
Orientation	T > C-12.5
Rotation	90.00 deg
R >> L	176 mm
A >> P	206 mm
F >> H	154 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	610 Hz/Px
Echo Spacing	1.79 ms
Asymmetric Echo	Off
Turbo Factor	3328
Segmentation	26
EPI Factor	17

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI\dzne_ep3d_HERTZ_0p4mm_QSM_tra_AP

TA: 5:07 min Coil Selection: Auto Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P18.0 H14.0 mm
Orientation	T > C-12.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	320
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	160 mm
FOV Phase	127.5 %
Slice Thickness	0.40 mm
TR	44.7 ms
Vol. TR	143.04 s
TE 1	20.20 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	44.7 ms
Vol. TR	143.04 s
TE 1	20.20 ms
Multi-echo spacing	37.30 ms
MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fast Fat Saturation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	2
Reordering	Linear

Resolution - Common

FOV Read	160 mm
FOV Phase	127.5 %
Slice Thickness	0.40 mm
Base Resolution	400
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	48
Acceleration Factor 3D	3
Reference Lines 3D	48
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P18.0 H14.0 mm
Orientation	T > C-12.5
Phase Encoding Dir.	A >> P
Slices per Slab	320
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	160 mm
FOV Phase	127.5 %
Slice Thickness	0.40 mm
TR	44.7 ms
Vol. TR	143.04 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P18.0 H14.0 mm
Orientation	T > C-12.5
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
-----------------	-------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	14 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P18.0 H14.0 mm
Orientation	T > C-12.5
Rotation	90.00 deg
R >> L	160 mm
A >> P	204 mm
F >> H	128 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	520 Hz/Px
Echo Spacing	2.14 ms
Asymmetric Echo	Off
Turbo Factor	3200
Segmentation	30
EPI Factor	17

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI\dzne_ep3d_Frankfurt_ref_maxBW_3TE

TA: 2:57 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	384
Phase Oversampling	3.5 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	38.8 ms
Vol. TR	171.34080 s
TE 1	6.70 ms
TE 2	17.40 ms
TE 3	28.10 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	38.8 ms
Vol. TR	171.34080 s
TE 1	6.70 ms
TE 2	17.40 ms
TE 3	28.10 ms
Multi-echo spacing	10.70 ms
MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fast Fat Saturation
Contrasts	3
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
Base Resolution	500
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	36
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	384
Phase Oversampling	3.5 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	38.8 ms
Vol. TR	171.34080 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
------------	---

Geometry - AutoAlign

Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
-----------------	-------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	22 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Rotation	90.00 deg
R >> L	175 mm
A >> P	210 mm
F >> H	135 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1250 Hz/Px
Echo Spacing	1.01 ms
Asymmetric Echo	Off
Turbo Factor	4416
Segmentation	23
EPI Factor	9

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI\ldzne_ep3d_Frankfurt_6echoes_minimumMaxBW_optR

TA: 6:52 min Coil Selection: Auto Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	4.2 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	50.6 ms
Vol. TR	404.8 s
TE 1	5.00 ms
TE 2	12.20 ms
TE 3	19.40 ms
TE 4	26.60 ms
TE 5	33.80 ms
TE 6	41.00 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	50.6 ms
Vol. TR	404.8 s
TE 1	5.00 ms
TE 2	12.20 ms
TE 3	19.40 ms
TE 4	26.60 ms
TE 5	33.80 ms
TE 6	41.00 ms
Multi-echo spacing	7.20 ms

Contrast - Common

MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fat Saturation
Contrasts	6
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
Base Resolution	500
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	36
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	4.2 %
FOV Read	175 mm
FOV Phase	120.0 %

Geometry - Common

Slice Thickness	0.35 mm
TR	50.6 ms
Vol. TR	404.8 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	90.00 deg
R >> L	175 mm
A >> P	210 mm
F >> H	135 mm

System - Adjust Volume

Reset	Off
-------	-----

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Optimized
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.20 ms
Asymmetric Echo	Off
Turbo Factor	8000
Segmentation	40
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI\ldzne_ep3d_Frankfurt_6echoes_minimumMaxBW_optR

TA: 6:52 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P13.8 H22.1 mm
Orientation	T > C-12.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	4.2 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	50.6 ms
Vol. TR	404.8 s
TE 1	5.00 ms
TE 2	12.20 ms
TE 3	19.40 ms
TE 4	26.60 ms
TE 5	33.80 ms
TE 6	41.00 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	50.6 ms
Vol. TR	404.8 s
TE 1	5.00 ms
TE 2	12.20 ms
TE 3	19.40 ms
TE 4	26.60 ms
TE 5	33.80 ms
TE 6	41.00 ms
Multi-echo spacing	7.20 ms

Contrast - Common

MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fat Saturation
Contrasts	6
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
Base Resolution	500
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	36
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P13.8 H22.1 mm
Orientation	T > C-12.9
Phase Encoding Dir.	A >> P
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	4.2 %
FOV Read	175 mm
FOV Phase	120.0 %

Geometry - Common

Slice Thickness	0.35 mm
TR	50.6 ms
Vol. TR	404.8 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P13.8 H22.1 mm
Orientation	T > C-12.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P13.8 H22.1
L	0.0 mm
P	13.8 mm
H	22.1 mm
Initial Orientation	T > C
T > C	-12.90
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	22 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P15.0 H29.3 mm
! Orientation	T > C-12.9
! Rotation	90.00 deg
! R >> L	175 mm

System - Adjust Volume

! A >> P	210 mm
! F >> H	96 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Optimized
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.20 ms
Asymmetric Echo	Off
Turbo Factor	8000
Segmentation	40
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI\ldzne_ep3d_Frankfurt_6echoes_minimumMaxBW_optR

TA: 6:52 min Coil Selection: Auto Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P13.8 H22.1 mm
Orientation	T > C-12.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	4.2 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	50.6 ms
Vol. TR	404.8 s
TE 1	5.00 ms
TE 2	12.20 ms
TE 3	19.40 ms
TE 4	26.60 ms
TE 5	33.80 ms
TE 6	41.00 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	50.6 ms
Vol. TR	404.8 s
TE 1	5.00 ms
TE 2	12.20 ms
TE 3	19.40 ms
TE 4	26.60 ms
TE 5	33.80 ms
TE 6	41.00 ms
Multi-echo spacing	7.20 ms

Contrast - Common

MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fat Saturation
Contrasts	6
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
Base Resolution	500
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	36
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P13.8 H22.1 mm
Orientation	T > C-12.9
Phase Encoding Dir.	A >> P
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	4.2 %
FOV Read	175 mm
FOV Phase	120.0 %

Geometry - Common

Slice Thickness	0.35 mm
TR	50.6 ms
Vol. TR	404.8 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P13.8 H22.1 mm
Orientation	T > C-12.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P13.8 H22.1
L	0.0 mm
P	13.8 mm
H	22.1 mm
Initial Orientation	T > C
T > C	-12.90
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	22 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P13.8 H22.1 mm
Orientation	T > C-12.9
Rotation	90.00 deg
R >> L	175 mm

System - Adjust Volume

A >> P	210 mm
F >> H	135 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Optimized
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.20 ms
Asymmetric Echo	Off
Turbo Factor	8000
Segmentation	40
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI\ldzne_ep3d_Frankfurt_6echoes_minimumMaxBW_optR

TA: 6:52 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P13.8 H22.1 mm
Orientation	T > C-12.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	4.2 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	50.6 ms
Vol. TR	404.8 s
TE 1	5.00 ms
TE 2	12.20 ms
TE 3	19.40 ms
TE 4	26.60 ms
TE 5	33.80 ms
TE 6	41.00 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	50.6 ms
Vol. TR	404.8 s
TE 1	5.00 ms
TE 2	12.20 ms
TE 3	19.40 ms
TE 4	26.60 ms
TE 5	33.80 ms
TE 6	41.00 ms
Multi-echo spacing	7.20 ms

Contrast - Common

MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fat Saturation
Contrasts	6
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
Base Resolution	500
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	36
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P13.8 H22.1 mm
Orientation	T > C-12.9
Phase Encoding Dir.	A >> P
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	4.2 %
FOV Read	175 mm
FOV Phase	120.0 %

Geometry - Common

Slice Thickness	0.35 mm
TR	50.6 ms
Vol. TR	404.8 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P13.8 H22.1 mm
Orientation	T > C-12.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P13.8 H22.1
L	0.0 mm
P	13.8 mm
H	22.1 mm
Initial Orientation	T > C
T > C	-12.90
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	22 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P13.8 H22.1 mm
Orientation	T > C-12.9
Rotation	90.00 deg
R >> L	175 mm

System - Adjust Volume

A >> P	210 mm
F >> H	135 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Optimized
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.20 ms
Asymmetric Echo	Off
Turbo Factor	8000
Segmentation	40
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI_func\dzne_ep3d_3mm_TE12p2_TR26p7_volTR133p5_ES0p32

TA: 3 sec Coil Selection: Auto Voxel Size: 3.0×3.0×3.0 mm³ Acc.: 9 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	15.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	26.7 ms
Vol. TR	133.5 ms
TE 1	12.20 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	26.7 ms
Vol. TR	133.5 ms
TE 1	12.20 ms
Multi-echo spacing	22.70 ms
MTC	Off
Flip Angle	9 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
Base Resolution	70
Phase Resolution	100 %
Slice Resolution	96 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	24
Acceleration Factor 3D	9
Reference Lines 3D	24
Reordering Shift 3D	3
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	15.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	26.7 ms
Vol. TR	133.5 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	7142 Hz/Px
Echo Spacing	0.32 ms
Asymmetric Echo	Off
Turbo Factor	5
Segmentation	1
EPI Factor	70

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI_func\dzne_ep3d_1p8mm_TE23p8_TR50p1_volTR450p9_ES0p38

TA: 4 sec Coil Selection: Auto Voxel Size: 1.8×1.8×1.8 mm³ Acc.: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	64
Phase Oversampling	0.0 %
Slice Oversampling	12.5 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.80 mm
TR	50.1 ms
Vol. TR	450.9 ms
TE 1	23.80 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	50.1 ms
Vol. TR	450.9 ms
TE 1	23.80 ms
Multi-echo spacing	39.90 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.80 mm
Base Resolution	120
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	24
Acceleration Factor 3D	8
Reference Lines 3D	48
Reordering Shift 3D	2
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	64
Phase Oversampling	0.0 %
Slice Oversampling	12.5 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.80 mm
TR	50.1 ms
Vol. TR	450.9 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	216 mm
R >> L	216 mm
F >> H	116 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	4166 Hz/Px
Echo Spacing	0.33 ms
Asymmetric Echo	Off
Turbo Factor	9
Segmentation	1
EPI Factor	120

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI_func\dzne_ep3d_1p2mm_TE24p5_TR51p6_volTR1238p4_ES0p52

TA: 7 sec Coil Selection: Auto Voxel Size: 1.2×1.2×1.2 mm³ Acc.: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	8.3 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.20 mm
TR	51.6 ms
Vol. TR	1341.6 ms
TE 1	24.50 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	51.6 ms
Vol. TR	1341.6 ms
TE 1	24.50 ms
Multi-echo spacing	39.10 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.20 mm
Base Resolution	180
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	4
Reference Lines 3D	48
Reordering Shift 3D	2
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	8.3 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.20 mm
TR	51.6 ms
Vol. TR	1341.6 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	116 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	2778 Hz/Px
Echo Spacing	0.43 ms
Asymmetric Echo	Off
Turbo Factor	26
Segmentation	1
EPI Factor	90

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI_func\dzne_ep3d_1mm_TE22p1_TR44_vol\TR1496_ES0p58

TA: 8 sec Coil Selection: Auto Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 9 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	6.3 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	44.0 ms
Vol. TR	1496 ms
TE 1	22.10 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	44.0 ms
Vol. TR	1496 ms
TE 1	22.10 ms
Multi-echo spacing	37.20 ms
MTC	Off
Flip Angle	15 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	216
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	48
Acceleration Factor 3D	3
Reference Lines 3D	48
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	6.3 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	44.0 ms
Vol. TR	1496 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	216 mm
R >> L	216 mm
F >> H	96 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	2314 Hz/Px
Echo Spacing	0.51 ms
Asymmetric Echo	Off
Turbo Factor	34
Segmentation	1
EPI Factor	72

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI_func\dzne_ep3d_3mm_TE8p6_TR19p6_volTR98_ES0p22_wRS

TA: 3 sec Coil Selection: Auto Voxel Size: 3.0×3.0×3.0 mm³ Acc:: 9 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	10.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	19.6 ms
Vol. TR	98 ms
TE 1	9.50 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	19.6 ms
Vol. TR	98 ms
TE 1	9.50 ms
Multi-echo spacing	17.80 ms
MTC	Off
Flip Angle	9 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
Base Resolution	70
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	24
Acceleration Factor 3D	9
Reference Lines 3D	24
Reordering Shift 3D	3
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	10.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	19.6 ms
Vol. TR	98 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	7142 Hz/Px
Echo Spacing	0.25 ms
Asymmetric Echo	Off
Turbo Factor	5
Segmentation	1
EPI Factor	70

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI_func\dzne_ep3d_1p8mm_TE20p2_TR42p9_volTR386p1_ES0p32_wRS

TA: 4 sec Coil Selection: Auto Voxel Size: 1.8×1.8×1.8 mm³ Acc.: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	64
Phase Oversampling	0.0 %
Slice Oversampling	12.5 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.80 mm
TR	42.9 ms
Vol. TR	386.1 ms
TE 1	21.20 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	42.9 ms
Vol. TR	386.1 ms
TE 1	21.20 ms
Multi-echo spacing	41.10 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.80 mm
Base Resolution	120
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	24
Acceleration Factor 3D	8
Reference Lines 3D	48
Reordering Shift 3D	2
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	64
Phase Oversampling	0.0 %
Slice Oversampling	12.5 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.80 mm
TR	42.9 ms
Vol. TR	386.1 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	216 mm
R >> L	216 mm
F >> H	116 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	4166 Hz/Px
Echo Spacing	0.34 ms
Asymmetric Echo	Off
Turbo Factor	9
Segmentation	1
EPI Factor	120

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI_func\ldzne_ep3d_1p2mm_TE18p6_TR39p9_volTR1037p4_ES0p39_wRS

TA: 7 sec Coil Selection: Auto Voxel Size: 1.2×1.2×1.2 mm³ Acc.: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	8.3 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.20 mm
TR	39.9 ms
Vol. TR	1037.4 ms
TE 1	19.40 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	39.9 ms
Vol. TR	1037.4 ms
TE 1	19.40 ms
Multi-echo spacing	37.30 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.20 mm
Base Resolution	180
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	4
Reference Lines 3D	48
Reordering Shift 3D	2
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	8.3 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.20 mm
TR	39.9 ms
Vol. TR	1037.4 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	116 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	3086 Hz/Px
Echo Spacing	0.41 ms
Asymmetric Echo	Off
Turbo Factor	26
Segmentation	1
EPI Factor	90

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI_funcldzne_ep3d_1mm_TE17p3_TR34p6_volTR1271p6_ES0p45_wRS

TA: 7 sec Coil Selection: Auto Voxel Size: 1.0×1.0×1.0 mm³ Acc.: 9 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	6.3 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	37.4 ms
Vol. TR	1271.6 ms
TE 1	17.30 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	37.4 ms
Vol. TR	1271.6 ms
TE 1	17.30 ms
Multi-echo spacing	30.70 ms
MTC	Off
Flip Angle	15 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	216
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	48
Acceleration Factor 3D	3
Reference Lines 3D	48
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	6.3 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	37.4 ms
Vol. TR	1271.6 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	216 mm
R >> L	216 mm
F >> H	96 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	2894 Hz/Px
Echo Spacing	0.42 ms
Asymmetric Echo	Off
Turbo Factor	34
Segmentation	1
EPI Factor	72

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI_funclep_seg_fid_3p3x3p3x3_cant_match_acceleration

TA: 5 sec Coil Selection: Auto Voxel Size: 3.3×3.3×3.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	10.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	120.0 ms
TE	60.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	120.0 ms
TE	60.00 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Water Excitation
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
Base Resolution	64
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	48
Acceleration Factor 3D	4
Reference Lines 3D	36
Reordering Shift 3D	0
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	3D
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	10.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	120.0 ms
Multi-Slice Mode	Interleaved
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Image Scaling	1.000
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Physio - Signal

1st Signal/Mode	None
TR	120.0 ms
Segments	1
Concatenations	1

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	3906 Hz/Px
Echo Spacing	0.31 ms
Free Echo Spacing	Off
Segments	1
EPI Factor	71

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI_funclep_seg_fid_1p7x1p7x1p8_cant_match_EPI_factor

TA: 8 sec Coil Selection: Auto Voxel Size: 1.7x1.7x1.8 mm³ Acc.: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.80 mm
TR	120.0 ms
TE	60.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	120.0 ms
TE	60.00 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Standard
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.80 mm
Base Resolution	128
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	48
Acceleration Factor 3D	8
Reference Lines 3D	40
Reordering Shift 3D	4
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	3D
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.80 mm
TR	120.0 ms
Multi-Slice Mode	Interleaved
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	216 mm
R >> L	216 mm
F >> H	116 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Image Scaling	1.000
---------------	-------

Physio - Signal

1st Signal/Mode	None
TR	120.0 ms
Segments	1
Concatenations	1

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	3906 Hz/Px
Echo Spacing	0.31 ms
Free Echo Spacing	Off
Segments	1
EPI Factor	75

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI_funclep_seg_fid_3p3x3p3x3_cant_match_acceleration

TA: 10 sec Coil Selection: Auto Voxel Size: 1.1×1.1×1.2 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	96
Phase Oversampling	0 %
Slice Oversampling	8.3 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.20 mm
TR	120.0 ms
TE	60.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	120.0 ms
TE	60.00 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Water Excitation
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.20 mm
Base Resolution	192
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	4
Reference Lines 3D	36
Reordering Shift 3D	2
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	3D
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	96
Phase Oversampling	0 %
Slice Oversampling	8.3 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.20 mm
TR	120.0 ms
Multi-Slice Mode	Interleaved
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	116 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Image Scaling	1.000
---------------	-------

Physio - Signal

1st Signal/Mode	None
TR	120.0 ms
Segments	1
Concatenations	1

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	3720 Hz/Px
Echo Spacing	0.39 ms
Free Echo Spacing	Off
Segments	1
EPI Factor	71

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI_func\dzne_ep3d_3mm_TE8p6_TR19p6_volTR98_ES0p22_wRS

TA: 4 sec Coil Selection: Auto Voxel Size: 3.0×3.0×3.0 mm³ Acc:: 9 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	10.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	18.8 ms
Vol. TR	94 ms
TE 1	9.50 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	18.8 ms
Vol. TR	94 ms
TE 1	9.50 ms
Multi-echo spacing	17.80 ms
MTC	Off
Flip Angle	9 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	2
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
Base Resolution	70
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	24
Acceleration Factor 3D	9
Reference Lines 3D	24
Reordering Shift 3D	3
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	10.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	18.8 ms
Vol. TR	94 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	22 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	7142 Hz/Px
Echo Spacing	0.25 ms
Asymmetric Echo	Off
Turbo Factor	5
Segmentation	1
EPI Factor	70

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI_func\dzne_ep3d_3mm_TE10_TR20_volTR100_ESOp25_wRS

TA: 4 sec Coil Selection: Auto Voxel Size: 3.0×3.0×3.0 mm³ Acc.: 9 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	10.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	20.9 ms
Vol. TR	104.5 ms
TE 1	9.50 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	20.9 ms
Vol. TR	104.5 ms
TE 1	9.50 ms
Multi-echo spacing	17.80 ms
MTC	Off
Flip Angle	9 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	2
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
Base Resolution	70
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	24
Acceleration Factor 3D	9
Reference Lines 3D	24
Reordering Shift 3D	3
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	10.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	20.9 ms
Vol. TR	104.5 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	22 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	7142 Hz/Px
Echo Spacing	0.25 ms
Asymmetric Echo	Off
Turbo Factor	5
Segmentation	1
EPI Factor	70

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\DZNE_3D_EPI_func\ldzne_ep3d_normal

TA: 14 sec Coil Selection: Auto Voxel Size: 3.0x3.0x3.0 mm³ Acc:: 9 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L2.5 P11.3 H1.4 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	10.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	18.3 ms
Vol. TR	91.5 ms
TE 1	9.20 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	18.3 ms
Vol. TR	91.5 ms
TE 1	9.20 ms
Multi-echo spacing	16.40 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	100
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
Base Resolution	70
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	24
Acceleration Factor 3D	9
Reference Lines 3D	24
Reordering Shift 3D	3
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L2.5 P11.3 H1.4 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	10.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	18.3 ms
Vol. TR	91.5 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L2.5 P11.3 H1.4 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	1 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L2.5 P11.3 H1.4 mm
Orientation	T > C-4.2 > S2.0
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Optimized
Gradient Mode	Normal
Reordering	Linear
Bandwidth	7142 Hz/Px
Echo Spacing	0.23 ms
Asymmetric Echo	Off
Turbo Factor	5
Segmentation	1
EPI Factor	70

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\3D EPI fQSM\AAhead_scout

TA: 14 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.25 ms
TE	1.53 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.25 ms
TE	1.53 ms
Flip Angle	16 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.3 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.25 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Initial Position	Isocenter
L	0.0 mm

Geometry - AutoAlign

P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
AutoAlign	Head

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off
AutoAlign MPR	On

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	16 deg
Measurements	1
Contrasts	1
TE	1.53 ms
TR	3.25 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\USER\jon\Saskia\3D EPI fQSM\lep_seg_fid

TA: 3 sec Coil Selection: Auto Voxel Size: 3.1×3.1×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	400 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	19.0 ms
TE	4.50 ms
Averages	1
Concatenations	1

Contrast - Common

TR	19.0 ms
TE	4.50 ms
MTC	Off
Flip Angle	90 deg
Fat-Water Contrast	Fat Saturation
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	400 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Normalize	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	400 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	19.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	19.0 ms
Segments	1
Concatenations	1

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	752 Hz/Px
Echo Spacing	1.46 ms
Free Echo Spacing	Off
Segments	1
EPI Factor	3

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\3D EPI fQSM\lep_seg_fQSM

TA: 13 sec Coil Selection: Auto Voxel Size: 0.8x0.8x0.8 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L3.2 P3.7 H40.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	120
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	204 mm
FOV Phase	100.0 %
Slice Thickness	0.80 mm
TR	53.0 ms
TE	20.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	53.0 ms
TE	20.00 ms
MTC	Off
Flip Angle	14 deg
Fat-Water Contrast	Fat Saturation
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	204 mm
FOV Phase	100.0 %
Slice Thickness	0.80 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	48
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L3.2 P3.7 H40.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	204 mm
FOV Phase	100.0 %
Slice Thickness	0.80 mm
TR	53.0 ms
Multi-Slice Mode	Interleaved
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L3.2 P3.7 H40.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	40 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Image Scaling	1.000
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Physio - Signal

1st Signal/Mode	None
TR	53.0 ms
Segments	1
Concatenations	1

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	2790 Hz/Px
Echo Spacing	0.50 ms
Free Echo Spacing	Off
Segments	1
EPI Factor	75

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\3D EPI fQSM\lep_seg_fQSM

TA: 13 sec Coil Selection: Auto Voxel Size: 0.8x0.8x0.8 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L3.2 P3.7 H40.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	120
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	204 mm
FOV Phase	100.0 %
Slice Thickness	0.80 mm
TR	53.0 ms
TE	20.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	53.0 ms
TE	20.00 ms
MTC	Off
Flip Angle	14 deg
Fat-Water Contrast	Fat Saturation
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	204 mm
FOV Phase	100.0 %
Slice Thickness	0.80 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	48
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L3.2 P3.7 H40.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	120
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	204 mm
FOV Phase	100.0 %
Slice Thickness	0.80 mm
TR	53.0 ms
Multi-Slice Mode	Interleaved
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L3.2 P3.7 H40.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	40 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Image Scaling	1.000
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Physio - Signal

1st Signal/Mode	None
TR	53.0 ms
Segments	1
Concatenations	1

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	2442 Hz/Px
Echo Spacing	0.50 ms
Free Echo Spacing	Off
Segments	1
EPI Factor	75

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\3D EPI fQSM\lep_seg_fQSM_0p7

TA: 21 sec Coil Selection: Auto Voxel Size: 0.7×0.7×0.7 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L3.2 P3.7 H40.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	202 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	55.0 ms
TE	21.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	55.0 ms
TE	21.00 ms
MTC	Off
Flip Angle	14 deg
Fat-Water Contrast	Fat Saturation
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	2
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	202 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
Base Resolution	288
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	48
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L3.2 P3.7 H40.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	202 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	55.0 ms
Multi-Slice Mode	Interleaved
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L3.2 P3.7 H40.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	40 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Transversal
! Rotation	0.00 deg
! A >> P	263 mm
! R >> L	350 mm
! F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Image Scaling	1.000
---------------	-------

Physio - Signal

1st Signal/Mode	None
TR	55.0 ms
Segments	1
Concatenations	1

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	2170 Hz/Px
Echo Spacing	0.53 ms
Free Echo Spacing	Off
Segments	1
EPI Factor	73

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\3D EPI fQSM\lep_seg_fQSM_0p6

TA: 27 sec Coil Selection: Auto Voxel Size: 0.6x0.6x0.6 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L3.2 P3.7 H40.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	202 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	68.0 ms
TE	27.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	68.0 ms
TE	27.00 ms
MTC	Off
Flip Angle	14 deg
Fat-Water Contrast	Fat Saturation
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	2
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	202 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
Base Resolution	336
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	48
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L3.2 P3.7 H40.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	202 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	68.0 ms
Multi-Slice Mode	Interleaved
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L3.2 P3.7 H40.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	40 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Transversal
! Rotation	0.00 deg
! A >> P	263 mm
! R >> L	350 mm
! F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Image Scaling	1.000
---------------	-------

Physio - Signal

1st Signal/Mode	None
TR	68.0 ms
Segments	1
Concatenations	1

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	1860 Hz/Px
Echo Spacing	0.60 ms
Free Echo Spacing	Off
Segments	1
EPI Factor	85

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\3D EPI fQSM\lep_seg_fQSM_0p6

TA: 25 sec Coil Selection: Auto Voxel Size: 0.6x0.6x0.6 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L3.2 P3.7 H40.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	202 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	51.0 ms
TE	19.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	51.0 ms
TE	19.00 ms
MTC	Off
Flip Angle	14 deg
Fat-Water Contrast	Fat Saturation
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	2
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	202 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
Base Resolution	336
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	3
Reference Lines 3D	48
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L3.2 P3.7 H40.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	202 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	51.0 ms
Multi-Slice Mode	Interleaved
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L3.2 P3.7 H40.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	40 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Transversal
! Rotation	0.00 deg
! A >> P	263 mm
! R >> L	350 mm
! F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Image Scaling	1.000
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Physio - Signal

1st Signal/Mode	None
TR	51.0 ms
Segments	1
Concatenations	1

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	1860 Hz/Px
Echo Spacing	0.60 ms
Free Echo Spacing	Off
Segments	1
EPI Factor	57

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\3D EPI fQSM\ldzne_ep3d_0p6mm_TE26p5_TR56p9_volTR_10p9_CAIPi_2x2

TA: 15 sec Coil Selection: Auto Voxel Size: 0.6x0.6x0.6 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P4.2 F6.6 mm
Orientation	T > C-10.9
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	192
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	202 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	56.9 ms
Vol. TR	10.92480 s
TE 1	27.60 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	56.9 ms
Vol. TR	10.92480 s
TE 1	27.60 ms
Multi-echo spacing	53.60 ms
MTC	Off
Flip Angle	15 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	202 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
Base Resolution	336
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPiRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	24
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P4.2 F6.6 mm
Orientation	T > C-10.9
Phase Encoding Dir.	A >> P
Slices per Slab	192
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	202 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	56.9 ms
Vol. TR	10.92480 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P4.2 F6.6 mm
Orientation	T > C-10.9
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	7 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P4.2 F6.6 mm
Orientation	T > C-10.9
Rotation	0.00 deg
A >> P	202 mm
R >> L	202 mm
F >> H	116 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1860 Hz/Px
Echo Spacing	0.63 ms
Asymmetric Echo	Off
Turbo Factor	192
Segmentation	2
EPI Factor	84

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\3D EPI fQSM\gre_3D_0p5mm_bipolar_4echoes_fatsat

TA: 9:34 min Coil Selection: Manual Voxel Size: 0.5×0.5×0.5 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Common

Contrasts	4
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
Base Resolution	440
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	R >> L
AutoAlign	---
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
TR	26.0 ms
TE 1	2.00 ms
TE 2	4.24 ms
TE 3	6.48 ms
TE 4	8.72 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	48
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	R >> L
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm

Contrast - Common

TR	26.0 ms
TE 1	2.00 ms
TE 2	4.24 ms
TE 3	6.48 ms
TE 4	8.72 ms
MTC	Off
Magn. Preparation	None
Flip Angle	9 deg
Fat-Water Contrast	Fat Saturation
Dark Blood	Off

Geometry - Common

TR	26.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	22 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Rotation	90.00 deg
R >> L	169 mm
A >> P	220 mm

System - Adjust Volume

F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	26.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fat Saturation
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	76.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Sequence - Nuclei

Coil Elements	NC7
---------------	-----

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	9 deg
Measurements	1
Contrasts	4
TE 1	2.00 ms
TE 2	4.24 ms
TE 3	6.48 ms
TE 4	8.72 ms
TR	26.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Fast
Readout Mode	Bipolar
Gradient Mode	Boost
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Bandwidth 1	470 Hz/Px
Bandwidth 2	470 Hz/Px
Bandwidth 3	470 Hz/Px
Bandwidth 4	470 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Phase Stabilisation	On
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

\\USER\jon\Saskia\3D EPI fQSM\gre_3D_0p5mm_bipolar_4echoes_waterex

TA: 5:54 min Coil Selection: Manual Voxel Size: 0.5×0.5×0.5 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	R >> L
AutoAlign	---
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
TR	16.0 ms
TE 1	2.00 ms
TE 2	4.08 ms
TE 3	6.16 ms
TE 4	8.24 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	16.0 ms
TE 1	2.00 ms
TE 2	4.08 ms
TE 3	6.16 ms
TE 4	8.24 ms
MTC	Off
Magn. Preparation	None
Flip Angle	7 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off

Contrast - Common

Contrasts	4
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
Base Resolution	440
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	48
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	R >> L
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm

Geometry - Common

TR	16.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	22 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Rotation	90.00 deg
R >> L	169 mm
A >> P	220 mm

System - Adjust Volume

F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	16.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	76.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Sequence - Nuclei

Coil Elements	NC7
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Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	7 deg
Measurements	1
Contrasts	4
TE 1	2.00 ms
TE 2	4.08 ms
TE 3	6.16 ms
TE 4	8.24 ms
TR	16.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Fast
Readout Mode	Bipolar
Gradient Mode	Fast
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Bandwidth 1	520 Hz/Px
Bandwidth 2	520 Hz/Px
Bandwidth 3	520 Hz/Px
Bandwidth 4	520 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Phase Stabilisation	On
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

\\USER\jon\Saskia\3D EPI fQSM\ldzne_ep3d_0p6mm_TE15p2_TR33p4_volTR_8p55_CAIPi_2x3_2

TA: 1:33 min Coil Selection: Auto Voxel Size: 0.6x0.6x0.6 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	192
Phase Oversampling	0.8 %
Slice Oversampling	0.0 %
FOV Read	168 mm
FOV Phase	130.4 %
Slice Thickness	0.60 mm
TR	33.4 ms
Vol. TR	8550.4 ms
TE 1	15.20 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	33.4 ms
Vol. TR	8550.4 ms
TE 1	15.20 ms
Multi-echo spacing	28.70 ms
MTC	Off
Flip Angle	10 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Reordering	Linear

Resolution - Common

FOV Read	168 mm
FOV Phase	130.4 %
Slice Thickness	0.60 mm
Base Resolution	280
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPiRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	3
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
Slices per Slab	192
Phase Oversampling	0.8 %
Slice Oversampling	0.0 %
FOV Read	168 mm
FOV Phase	130.4 %
Slice Thickness	0.60 mm
TR	33.4 ms
Vol. TR	8550.4 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	22 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Rotation	90.00 deg
R >> L	168 mm
A >> P	219 mm
F >> H	116 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	2232 Hz/Px
Echo Spacing	0.61 ms
Asymmetric Echo	Off
Turbo Factor	256
Segmentation	4
EPI Factor	46

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\3D EPI fQSM\ldzne_ep3d_0p6mm_TE14p3_TR31p6_volTR_8p09_CAIPi_2x3_2

TA: 1:28 min Coil Selection: Auto Voxel Size: 0.6x0.6x0.6 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	192
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	31.6 ms
Vol. TR	8089.6 ms
TE 1	14.30 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	31.6 ms
Vol. TR	8089.6 ms
TE 1	14.30 ms
Multi-echo spacing	27.00 ms
MTC	Off
Flip Angle	10 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
Base Resolution	352
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPiRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	3
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
Slices per Slab	192
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	31.6 ms
Vol. TR	8089.6 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	22 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L2.5 P8.3 H21.7 mm
Orientation	T > C-4.2 > S2.0
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	116 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	2030 Hz/Px
Echo Spacing	0.60 ms
Asymmetric Echo	Off
Turbo Factor	256
Segmentation	4
EPI Factor	44

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\3D EPI fQSM\ldzne_ep3d_0p6mm_TE15p2_TR33p4_volTR8550p4_C2x3s2

TA: 12 sec Coil Selection: Auto Voxel Size: 0.6x0.6x0.6 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L2.5 P11.3 H1.4 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	192
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	33.4 ms
Vol. TR	8550.4 ms
TE 1	15.20 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	33.4 ms
Vol. TR	8550.4 ms
TE 1	15.20 ms
Multi-echo spacing	26.60 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
Base Resolution	352
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	3
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L2.5 P11.3 H1.4 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
Slices per Slab	192
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	33.4 ms
Vol. TR	8550.4 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L2.5 P11.3 H1.4 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	1 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L2.5 P11.3 H1.4 mm
Orientation	T > C-4.2 > S2.0
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	116 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Optimized
Gradient Mode	Fast
Reordering	Linear
Bandwidth	2030 Hz/Px
Echo Spacing	0.59 ms
Asymmetric Echo	Off
Turbo Factor	256
Segmentation	4
EPI Factor	44

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\3D EPI fQSM\ldzne_ep3d_0p6mm_TE14p5_TR32_volTR8192_C2x3s2

TA: 12 sec Coil Selection: Auto Voxel Size: 0.6x0.6x0.6 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L2.5 P11.3 H1.4 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	192
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	32.0 ms
Vol. TR	8192 ms
TE 1	14.50 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	32.0 ms
Vol. TR	8192 ms
TE 1	14.50 ms
Multi-echo spacing	26.60 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
Base Resolution	352
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	3
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L2.5 P11.3 H1.4 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
Slices per Slab	192
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	32.0 ms
Vol. TR	8192 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L2.5 P11.3 H1.4 mm
Orientation	T > C-4.2 > S2.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	1 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L2.5 P11.3 H1.4 mm
Orientation	T > C-4.2 > S2.0
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	116 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Optimized
Gradient Mode	Fast
Reordering	Linear
Bandwidth	2030 Hz/Px
Echo Spacing	0.59 ms
Asymmetric Echo	Off
Turbo Factor	256
Segmentation	4
EPI Factor	44

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\3D EPI fQSM\gre_3D_0p5mm_monopolar_6echoes_waterex

TA: 8:50 min Coil Selection: Manual Voxel Size: 0.5×0.5×0.5 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L2.2 P13.0 H5.9 mm
Orientation	T > C-17.1 > S2.4
Phase Encoding Dir.	R >> L
AutoAlign	---
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
TR	24.0 ms
TE 1	2.00 ms
TE 2	4.84 ms
TE 3	7.68 ms
TE 4	10.52 ms
TE 5	13.36 ms
TE 6	16.20 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	24.0 ms
TE 1	2.00 ms
TE 2	4.84 ms
TE 3	7.68 ms
TE 4	10.52 ms
TE 5	13.36 ms
TE 6	16.20 ms
MTC	Off

Contrast - Common

Magn. Preparation	None
Flip Angle	9 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	6
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
Base Resolution	440
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	48
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L2.2 P13.0 H5.9 mm
Orientation	T > C-17.1 > S2.4
Phase Encoding Dir.	R >> L
Slices per Slab	240
Phase Oversampling	0 %

Geometry - Common

Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
TR	24.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L2.2 P13.0 H5.9 mm
Orientation	T > C-17.1 > S2.4
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	6 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L2.2 P13.0 H5.9 mm
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System - Adjust Volume

Orientation	T > C-17.1 > S2.4
Rotation	90.00 deg
R >> L	169 mm
A >> P	220 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	24.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	76.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off

Inline - MIP

MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	9 deg
Measurements	1
Contrasts	6
TE 1	2.00 ms
TE 2	4.84 ms
TE 3	7.68 ms
TE 4	10.52 ms
TE 5	13.36 ms
TE 6	16.20 ms
TR	24.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Fast
Readout Mode	Monopolar
Gradient Mode	Fast
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Flow Compensation 5	None
Flow Compensation 6	None
Bandwidth 1	520 Hz/Px
Bandwidth 2	520 Hz/Px
Bandwidth 3	520 Hz/Px
Bandwidth 4	520 Hz/Px
Bandwidth 5	520 Hz/Px
Bandwidth 6	520 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
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Sequence - Part 2

RF Spoiling	On
Incr. Gradient Spoiling	Off
Phase Stabilisation	On
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	NC7

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\QSM_3D_Jingyuan\gre_3D_0p5mm_ps_monopolar_6echoes

TA: 13:27 min Coil Selection: Manual Voxel Size: 0.5×0.5×0.5 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L2.5 P15.5 F20.1 mm
Orientation	T > C-32.7 > S2.0
Phase Encoding Dir.	R >> L
AutoAlign	---
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	27.3 %
FOV Read	240 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
TR	39.0 ms
TE 1	2.53 ms
TE 2	6.36 ms
TE 3	10.19 ms
TE 4	14.02 ms
TE 5	17.85 ms
TE 6	21.68 ms
Averages	1
Concatenations	1
Coil Elements	R32

Contrast - Common

TR	39.0 ms
TE 1	2.53 ms
TE 2	6.36 ms
TE 3	10.19 ms
TE 4	14.02 ms
TE 5	17.85 ms
TE 6	21.68 ms
MTC	Off

Contrast - Common

Magn. Preparation	None
Flip Angle	12 deg
Fat-Water Contrast	Fat Saturation
Dark Blood	Off
Contrasts	6
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	240 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
Base Resolution	440
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	48
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L2.5 P15.5 F20.1 mm
Orientation	T > C-32.7 > S2.0
Phase Encoding Dir.	R >> L
Slices per Slab	176
Phase Oversampling	0 %

Geometry - Common

Slice Oversampling	27.3 %
FOV Read	240 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
TR	39.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L2.5 P15.5 F20.1 mm
Orientation	T > C-32.7 > S2.0
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	20 mm
Table Position	F

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L2.5 P15.5 F20.1 mm
----------	---------------------

System - Adjust Volume

Orientation	T > C-32.7 > S2.0
Rotation	90.00 deg
R >> L	185 mm
A >> P	240 mm
F >> H	88 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	39.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fat Saturation
Magn. Preparation	None
Dark Blood	Off
FOV Read	240 mm
FOV Phase	76.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off

Inline - MIP

MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	12 deg
Measurements	1
Contrasts	6
TE 1	2.53 ms
TE 2	6.36 ms
TE 3	10.19 ms
TE 4	14.02 ms
TE 5	17.85 ms
TE 6	21.68 ms
TR	39.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Fast
Readout Mode	Monopolar
Gradient Mode	Boost
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Flow Compensation 5	None
Flow Compensation 6	None
Bandwidth 1	310 Hz/Px
Bandwidth 2	310 Hz/Px
Bandwidth 3	310 Hz/Px
Bandwidth 4	310 Hz/Px
Bandwidth 5	310 Hz/Px
Bandwidth 6	310 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
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Sequence - Part 2

RF Spoiling	On
Incr. Gradient Spoiling	Off
Phase Stabilisation	On
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	R32

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off

\\USER\jon\Saskia\QSM_3D_Jingyuan\gre_3D_0p5mm_ps_monopolar_8echoes

TA: 16:12 min Coil Selection: Manual Voxel Size: 0.5×0.5×0.5 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Common

TE 6	21.68 ms
TE 7	25.51 ms
TE 8	29.34 ms
MTC	Off
Magn. Preparation	None
Flip Angle	12 deg
Fat-Water Contrast	Fat Saturation
Dark Blood	Off
Contrasts	8
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L2.5 P15.5 F20.1 mm
Orientation	T > C-32.7 > S2.0
Phase Encoding Dir.	R >> L
AutoAlign	---
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	27.3 %
FOV Read	240 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
TR	47.0 ms
TE 1	2.53 ms
TE 2	6.36 ms
TE 3	10.19 ms
TE 4	14.02 ms
TE 5	17.85 ms
TE 6	21.68 ms
TE 7	25.51 ms
TE 8	29.34 ms
Averages	1
Concatenations	1
Coil Elements	R32

Resolution - Common

FOV Read	240 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
Base Resolution	440
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	48
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Contrast - Common

TR	47.0 ms
TE 1	2.53 ms
TE 2	6.36 ms
TE 3	10.19 ms
TE 4	14.02 ms
TE 5	17.85 ms

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L2.5 P15.5 F20.1 mm

Geometry - Common

Orientation	T > C-32.7 > S2.0
Phase Encoding Dir.	R >> L
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	27.3 %
FOV Read	240 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
TR	47.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L2.5 P15.5 F20.1 mm
Orientation	T > C-32.7 > S2.0
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	20 mm
Table Position	F

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never

System - Adjustments

Assume Silicone	Off
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System - Adjust Volume

Position	L2.5 P15.5 F20.1 mm
Orientation	T > C-32.7 > S2.0
Rotation	90.00 deg
R >> L	185 mm
A >> P	240 mm
F >> H	88 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	47.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fat Saturation
Magn. Preparation	None
Dark Blood	Off
FOV Read	240 mm
FOV Phase	76.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off

Inline - MIP

MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	12 deg
Measurements	1
Contrasts	8
TE 1	2.53 ms
TE 2	6.36 ms
TE 3	10.19 ms
TE 4	14.02 ms
TE 5	17.85 ms
TE 6	21.68 ms
TE 7	25.51 ms
TE 8	29.34 ms
TR	47.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Fast
Readout Mode	Monopolar
Gradient Mode	Boost
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Flow Compensation 5	None
Flow Compensation 6	None
Flow Compensation 7	None
Flow Compensation 8	None
Bandwidth 1	310 Hz/Px
Bandwidth 2	310 Hz/Px
Bandwidth 3	310 Hz/Px

Sequence - Part 1

Bandwidth 4	310 Hz/Px
Bandwidth 5	310 Hz/Px
Bandwidth 6	310 Hz/Px
Bandwidth 7	310 Hz/Px
Bandwidth 8	310 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Phase Stabilisation	On
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	R32

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off

\\USER\jon\Saskia\QSM_3D_Jingyuan\gre_3D_0p5mm_ps_bipolar_9echoes

TA: 15:52 min Coil Selection: Manual Voxel Size: 0.5×0.5×0.5 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L2.5 P15.5 F20.1 mm
Orientation	T > C-32.7 > S2.0
Phase Encoding Dir.	R >> L
AutoAlign	---
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	27.3 %
FOV Read	240 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
TR	46.0 ms
TE 1	2.53 ms
TE 2	5.85 ms
TE 3	9.17 ms
TE 4	12.49 ms
TE 5	15.81 ms
TE 6	19.13 ms
TE 7	22.45 ms
TE 8	25.77 ms
TE 9	29.09 ms
Averages	1
Concatenations	1
Coil Elements	R32

Contrast - Common

TR	46.0 ms
TE 1	2.53 ms
TE 2	5.85 ms
TE 3	9.17 ms
TE 4	12.49 ms

Contrast - Common

TE 5	15.81 ms
TE 6	19.13 ms
TE 7	22.45 ms
TE 8	25.77 ms
TE 9	29.09 ms
MTC	Off
Magn. Preparation	None
Flip Angle	12 deg
Fat-Water Contrast	Fat Saturation
Dark Blood	Off
Contrasts	9
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	240 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
Base Resolution	440
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	48
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1

Geometry - Common

Distance Factor	20 %
Position	L2.5 P15.5 F20.1 mm
Orientation	T > C-32.7 > S2.0
Phase Encoding Dir.	R >> L
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	27.3 %
FOV Read	240 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
TR	46.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L2.5 P15.5 F20.1 mm
Orientation	T > C-32.7 > S2.0
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	20 mm
Table Position	F

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm

System - Adjustments

Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L2.5 P15.5 F20.1 mm
Orientation	T > C-32.7 > S2.0
Rotation	90.00 deg
R >> L	185 mm
A >> P	240 mm
F >> H	88 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	46.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fat Saturation
Magn. Preparation	None
Dark Blood	Off
FOV Read	240 mm
FOV Phase	76.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
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Inline - MIP

MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	12 deg
Measurements	1
Contrasts	9
TE 1	2.53 ms
TE 2	5.85 ms
TE 3	9.17 ms
TE 4	12.49 ms
TE 5	15.81 ms
TE 6	19.13 ms
TE 7	22.45 ms
TE 8	25.77 ms
TE 9	29.09 ms
TR	46.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Fast
Readout Mode	Bipolar
Gradient Mode	Boost
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Flow Compensation 5	None
Flow Compensation 6	None
Flow Compensation 7	None
Flow Compensation 8	None

Sequence - Part 1

Flow Compensation 9	None
Bandwidth 1	310 Hz/Px
Bandwidth 2	310 Hz/Px
Bandwidth 3	310 Hz/Px
Bandwidth 4	310 Hz/Px
Bandwidth 5	310 Hz/Px
Bandwidth 6	310 Hz/Px
Bandwidth 7	310 Hz/Px
Bandwidth 8	310 Hz/Px
Bandwidth 9	310 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Phase Stabilisation	On
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	R32

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off

\\USER\jon\Saskia\20260107_TOF_unconfuse\localizer

TA: 9 sec Coil Selection: Manual Voxel Size: 0.5×0.5×5.0 mm³ Acc.: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	All Segments
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Common

TE	3.69 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
TE	3.69 ms
Averages	2
Concatenations	3
Coil Elements	R32

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	24
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %

Contrast - Common

TR	8.6 ms
----	--------

Geometry - Common

Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg
Slice Group	2
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
Slice Group	3
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.6 ms
Segments	1
Concatenations	3

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	20 deg
Measurements	1
Contrasts	1
TE	3.69 ms
TR	8.6 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	qfl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	R32

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\20260107_TOF_unconfuseltof_fl3d_tra_cs8

TA: 58 sec Coil Selection: Manual Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 7.2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	512
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	48
Phase Oversampling	10 %
Slice Oversampling	16.7 %
FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	12.49 ms
TE	5.97 ms
Averages	1
Concatenations	1
Coil Elements	R32

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	7.2
Reference Scans	GRE/Separate
Reference Lines PE	32
Reference Lines 3D	32
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	On

Contrast - Common

TR	12.49 ms
TE	5.97 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	23 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	48
Phase Oversampling	10 %
Slice Oversampling	16.7 %
FOV Read	206 mm

Geometry - Common

FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	12.49 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal

System - Adjust Volume

Rotation	90.00 deg
R >> L	187 mm
A >> P	206 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	12.49 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	206 mm
FOV Phase	90.6 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	5.97 ms
TR	12.49 ms

Inline - MIP

MIP Sag	On
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Inline - MIP

MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	610 Hz/Px
Echo Spacing	8.94 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	None

\\USER\jon\Saskia\20260107_TOF_unconfuse\tof_fl3d_tra_GRAPPA2

TA: 3:14 min Coil Selection: Manual Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	512
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	48
Phase Oversampling	10 %
Slice Oversampling	16.7 %
FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	12.49 ms
TE	5.97 ms
Averages	1
Concatenations	1
Coil Elements	R32

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	On

Contrast - Common

TR	12.49 ms
TE	5.97 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	22 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	48
Phase Oversampling	10 %
Slice Oversampling	16.7 %
FOV Read	206 mm

Geometry - Common

FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	12.49 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal

System - Adjust Volume

Rotation	90.00 deg
R >> L	187 mm
A >> P	206 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	12.49 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	206 mm
FOV Phase	90.6 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	5.97 ms
TR	12.49 ms

Inline - MIP

MIP Sag	On
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Inline - MIP

MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	610 Hz/Px
Echo Spacing	8.94 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	None

\\USER\jon\Saskia\20260107_TOF_unconfuseltf_fl3d_tra_cs8_seg7

TA: 44 sec Coil Selection: Manual Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 7.2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	512
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	48
Phase Oversampling	10 %
Slice Oversampling	16.7 %
FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	62.58 ms
TE	5.97 ms
Averages	1
Concatenations	1
Coil Elements	R32

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	7.2
Reference Scans	GRE/Separate
Reference Lines PE	32
Reference Lines 3D	32
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	On

Contrast - Common

TR	62.58 ms
TE	5.97 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	48
Phase Oversampling	10 %
Slice Oversampling	16.7 %
FOV Read	206 mm

Geometry - Common

FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	62.58 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal

System - Adjust Volume

Rotation	90.00 deg
R >> L	187 mm
A >> P	206 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	62.58 ms
Segments	7
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	206 mm
FOV Phase	90.6 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	5.97 ms
TR	62.58 ms

Inline - MIP

MIP Sag	On
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Inline - MIP

MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	610 Hz/Px
Echo Spacing	8.94 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	7

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	None

\\USER\jon\Saskia\20260107_TOF_unconfuse\tof_fl3d_tra_GRAPPA2_seg7

TA: 2:21 min Coil Selection: Manual Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	512
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	48
Phase Oversampling	10 %
Slice Oversampling	16.7 %
FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	62.58 ms
TE	5.97 ms
Averages	1
Concatenations	1
Coil Elements	R32

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	On

Contrast - Common

TR	62.58 ms
TE	5.97 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	19 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	48
Phase Oversampling	10 %
Slice Oversampling	16.7 %
FOV Read	206 mm

Geometry - Common

FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	62.58 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal

System - Adjust Volume

Rotation	90.00 deg
R >> L	187 mm
A >> P	206 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	62.58 ms
Segments	7
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	206 mm
FOV Phase	90.6 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	5.97 ms
TR	62.58 ms

Inline - MIP

MIP Sag	On
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Inline - MIP

MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	610 Hz/Px
Echo Spacing	8.94 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	7

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	None

\\USER\jon\Saskia\20260107_TOF_unconfuse\tof_fl3d_tra_GRAPPA2

TA: 3:23 min Coil Selection: Manual Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	512
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	33.3 %
FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	12.49 ms
TE	5.97 ms
Averages	1
Concatenations	1
Coil Elements	R32

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	On

Contrast - Common

TR	12.49 ms
TE	5.97 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	22 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	33.3 %
FOV Read	206 mm

Geometry - Common

FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	12.49 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal

System - Adjust Volume

Rotation	90.00 deg
R >> L	187 mm
A >> P	206 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	12.49 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	206 mm
FOV Phase	90.6 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	5.97 ms
TR	12.49 ms

Inline - MIP

MIP Sag	On
---------	----

Inline - MIP

MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	610 Hz/Px
Echo Spacing	8.94 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	None

\\USER\jon\Saskia\20260107_TOF_unconfuse\tof_fl3d_tra_cs8_longTR

TA: 4:19 min Coil Selection: Manual Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 7.2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	512
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	48
Phase Oversampling	10 %
Slice Oversampling	16.7 %
FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	62.58 ms
TE	5.97 ms
Averages	1
Concatenations	1
Coil Elements	R32

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	7.2
Reference Scans	GRE/Separate
Reference Lines PE	32
Reference Lines 3D	32
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	On

Contrast - Common

TR	62.58 ms
TE	5.97 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	23 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	48
Phase Oversampling	10 %
Slice Oversampling	16.7 %
FOV Read	206 mm

Geometry - Common

FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	62.58 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal

System - Adjust Volume

Rotation	90.00 deg
R >> L	187 mm
A >> P	206 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	62.58 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	206 mm
FOV Phase	90.6 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	5.97 ms
TR	62.58 ms

Inline - MIP

MIP Sag	On
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Inline - MIP

MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	610 Hz/Px
Echo Spacing	8.94 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	None

\\USER\jon\Saskia\20260107_TOF_unconfuse\tof_fl3d_tra_GRAPPA1

TA: 6:15 min Coil Selection: Manual Voxel Size: 0.4×0.4×0.4 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	512
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	33.3 %
FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	12.49 ms
TE	5.97 ms
Averages	1
Concatenations	1
Coil Elements	R32

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	33.3 %
FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	12.49 ms
Multi-Slice Mode	Sequential

Contrast - Common

TR	12.49 ms
TE	5.97 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	22 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	90.00 deg
R >> L	187 mm
A >> P	206 mm
F >> H	20 mm

System - Adjust Volume

Reset	Off
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System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	12.49 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	206 mm
FOV Phase	90.6 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	5.97 ms
TR	12.49 ms

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off

Inline - MIP

Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	610 Hz/Px
Echo Spacing	8.94 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	None

\\USER\jon\Saskia\20260107_TOF_unconfuse\tof_fl3d_tra_cs8_slabs5

TA: 4:45 min Coil Selection: Manual Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 7.2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	5
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	48
Phase Oversampling	10 %
Slice Oversampling	16.7 %
FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	12.49 ms
TE	5.97 ms
Averages	1
Concatenations	5
Coil Elements	R32

Contrast - Common

TR	12.49 ms
TE	5.97 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	23 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	206 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	512
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	7.2
Reference Scans	GRE/Separate
Reference Lines PE	32
Reference Lines 3D	32
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	5
Distance Factor	-19 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	48
Phase Oversampling	10 %
Slice Oversampling	16.7 %
FOV Read	206 mm

Geometry - Common

FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	12.49 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	5

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal

System - Adjust Volume

Rotation	90.00 deg
R >> L	187 mm
A >> P	206 mm
F >> H	82 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	12.49 ms
Segments	1
Concatenations	5

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	206 mm
FOV Phase	90.6 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	5

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	5.97 ms
TR	12.49 ms

Inline - MIP

MIP Sag	On
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Inline - MIP

MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	610 Hz/Px
Echo Spacing	8.94 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	None

\\USER\jon\Saskia\20260107_TOF_unconfuse\gre

TA: 3:07 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	16.7 %
FOV Read	206 mm
FOV Phase	93.8 %
Slice Thickness	0.40 mm
TR	13.0 ms
TE	5.97 ms
Averages	1
Concatenations	1

Contrast - Common

TR	13.0 ms
TE	5.97 ms
MTC	Off
Magn. Preparation	None
Flip Angle	22 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Each Measurement
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Resolution - Common

FOV Read	206 mm
FOV Phase	93.8 %
Slice Thickness	0.40 mm
Base Resolution	512
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	16.7 %
FOV Read	206 mm
FOV Phase	93.8 %
Slice Thickness	0.40 mm
TR	13.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
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Geometry - AutoAlign

Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	13.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	206 mm
FOV Phase	93.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline - MapIt

MapIt	None
Flip Angle	22 deg
Measurements	1
Contrasts	1
TE	5.97 ms
TR	13.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Bandwidth	610 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\MP2RAGE\t1_mp2rage_sag_0p6_Magdeburg

TA: 10:57 min Coil Selection: Manual Voxel Size: 0.6×0.6×0.6 mm³ Acc:: 4.0 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.6 P3.9 F19.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Slices per Slab	320
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	222 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	6000.0 ms
TE	1.79 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	6000.0 ms
TE	1.79 ms
Magn. Preparation	Non-sel. IR
TI 1	813 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	222 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
Base Resolution	384
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	4.0
Reference Scans	GRE/Separate
Reference Lines PE	32
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.6 P3.9 F19.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	320
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	222 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	6000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.6 P3.9 F19.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	L0.6 P3.9 F19.8
L	0.6 mm
P	3.9 mm
F	19.8 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	20 mm
Table Position	F

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.6 P3.9 F19.8 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	222 mm
F >> H	222 mm
R >> L	192 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	6000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI 1	813 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	222 mm
FOV Phase	100.0 %
Phase Resolution	100 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	1.79 ms
TR	6000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	320 Hz/Px
Echo Spacing	5.26 ms
Asymmetric Echo	Allowed
Turbo Factor	300

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\MP2RAGE\t1_mp2rage_sag_0p6_Magdeburg_wip

TA: 10:57 min Coil Selection: Manual Voxel Size: 0.6×0.6×0.6 mm³ Acc:: 4.0 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.6 P3.9 F19.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Slices per Slab	320
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	222 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	6000.0 ms
TE	1.79 ms
Averages	1
Concatenations	1

Contrast - Common

TR	6000.0 ms
TE	1.79 ms
Magn. Preparation	Non-sel. IR
TI 1	813 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	222 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
Base Resolution	384
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	4.0
Reference Scans	GRE/Separate
Reference Lines PE	32
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.6 P3.9 F19.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	320
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	222 mm
FOV Phase	100.0 %
Slice Thickness	0.60 mm
TR	6000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Contrast - Dynamic

Dynamic Mode	Standard
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Geometry - AutoAlign

Slab Group	1
Position	L0.6 P3.9 F19.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	L0.6 P3.9 F19.8
L	0.6 mm
P	3.9 mm
F	19.8 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	20 mm
Table Position	F

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.6 P3.9 F19.8 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	222 mm
F >> H	222 mm
R >> L	192 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	6000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI 1	813 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	222 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	1.79 ms
TR	6000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	320 Hz/Px
Echo Spacing	5.26 ms
Asymmetric Echo	Allowed
Turbo Factor	300

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Custom pTx Pulse Duration	Off
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Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\FIRE_VB_test\tof_fl3d_tra_0p3iso_TE3p6_TR8p8_GRAPPA_3x1_slabs1

TA: 5:54 min Coil Selection: Manual Voxel Size: 0.3×0.3×0.3 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	L0.3 P8.7 H4.3 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	192
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	8.80 ms
TE	3.60 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	8.80 ms
TE	3.60 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	12 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
Base Resolution	656
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	L0.3 P8.7 H4.3 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	192
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	200 mm

Geometry - Common

FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	8.80 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.3 P8.7 H4.3 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.3 P8.7 H4.3
L	0.3 mm
P	8.7 mm
H	4.3 mm
Initial Orientation	Transversal
Initial Rotation	90.35 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	4 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.3 P8.7 H4.3 mm
Orientation	Transversal

System - Adjust Volume

Rotation	90.35 deg
R >> L	175 mm
A >> P	200 mm
F >> H	58 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.80 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	87.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.60 ms
TR	8.80 ms

Inline - MIP

MIP Sag	On
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Inline - MIP

MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Low SAR
Gradient Mode	Whisper
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	305 Hz/Px
Echo Spacing	8.80 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	None

\\USER\jon\Saskia\FIRE_VB_test\tof_fl3d_tra_0p3iso_TE3p6_TR8p8_GRAPPA_3x1_slabs1

TA: 5:54 min Coil Selection: Manual Voxel Size: 0.3×0.3×0.3 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
Base Resolution	656
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	L0.3 P8.7 H4.3 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	192
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	8.80 ms
TE	3.60 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Contrast - Common

TR	8.80 ms
TE	3.60 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	12 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	L0.3 P8.7 H4.3 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	192
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	200 mm

Geometry - Common

FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	8.80 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.3 P8.7 H4.3 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.3 P8.7 H4.3
L	0.3 mm
P	8.7 mm
H	4.3 mm
Initial Orientation	Transversal
Initial Rotation	90.35 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	4 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.3 P8.7 H4.3 mm
Orientation	Transversal

System - Adjust Volume

Rotation	90.35 deg
R >> L	175 mm
A >> P	200 mm
F >> H	58 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.80 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	87.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.60 ms
TR	8.80 ms

Inline - MIP

MIP Sag	On
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Inline - MIP

MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Low SAR
Gradient Mode	Whisper
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	305 Hz/Px
Echo Spacing	8.80 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	None

\\USER\jon\Saskia\THS\localizer

TA: 15 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	All Segments
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
TE	3.69 ms
Averages	2
Concatenations	3

Contrast - Common

TR	8.6 ms
TE	3.69 ms

Contrast - Common

TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slice Group	3

Geometry - Common

Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg
Slice Group	2
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
Slice Group	3
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.6 ms
Segments	1
Concatenations	3

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None

Physio - Cardiac

Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	20 deg
Measurements	1
Contrasts	1
TE	3.69 ms
TR	8.6 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	qfl
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Sequence - Part 1

Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\THS\AAhead_scout_md

TA: 15 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.53 ms
TE	1.53 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.53 ms
TE	1.53 ms
Flip Angle	16 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.7 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.53 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Initial Position	Isocenter
L	0.0 mm

Geometry - AutoAlign

P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
AutoAlign	Head

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off
AutoAlign MPR	On

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	16 deg
Measurements	1
Contrasts	1
TE	1.53 ms
TR	3.53 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS\gre_b0map_4iso_sag

TA: 31 sec Coil Selection: Auto Voxel Size: 4.0×4.0×4.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	44
Distance Factor	0 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
Averages	1
Concatenations	1

Contrast - Common

TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
MTC	Off
Flip Angle	25 deg
Fat-Water Contrast	Standard
Contrasts	2
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	44
Distance Factor	0 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fm_r
Dimension	2D
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	On
Bandwidth	908 Hz/Px
Asymmetric Echo	Off

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS\ldzne_b1map_5iso_sag

TA: 7 sec Coil Selection: Auto Voxel Size: 5.0×5.0×5.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Averages	1
Concatenations	1

Contrast - Common

TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Flip Angle 1	60 deg
Flip Angle 2	6 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
Base Resolution	44
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal

Geometry - AutoAlign

Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	b1map 9e96902
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	1000 Hz/Px
Asymmetric Echo	Off
Turbo Factor	440
Echo Train Duration	1298 ms

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Mapping Technique	DREAM
Timing Scheme	STE*
Mixing Time	1130 us
Scale risetime	1.2

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS\head

TA: 5 sec Coil Selection: Auto Voxel Size: 4.0×4.0×5.0 mm³ Acc.: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	16
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Routine

Slice Group	1
Slices	25
Distance Factor	100 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	3900.0 ms
TE	1.56 ms
Averages	1
Concatenations	1

Geometry - Common

Slice Group	1
Slices	25
Distance Factor	100 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	3900.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	3900.0 ms
TE	1.56 ms
Magn. Preparation	None
Flip Angle	5 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Centric

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3900.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None

Physio - Cardiac

Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	tfl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Whisper
Flow Compensation	None
Reordering	Centric
Bandwidth	440 Hz/Px
Echo Spacing	3.54 ms
Asymmetric Echo	Allowed
Turbo Factor	64

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Tx Scale Diag. Magn.	0.3
Tx Scale Diag. Phase	0 deg
Tx Scale Off-diag. Magn.	1.0
Tx Scale Off-diag. Phase	0 deg
Rel. B1 mapping	On
Ref. Scan	Off
Use B1 Map Recon.	On
Prep Pulse	Selective
Dummy RF pulses	1000
Dummy RF TRs	0

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS\bss_map

TA: 2:01 min Coil Selection: Manual Voxel Size: 4.0x4.0x4.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
TR	5.0 ms
TE	2.04 ms
TE2 B0	4.08 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	5.0 ms
TE	2.04 ms
TE2 B0	4.08 ms
Flip Angle	5 deg
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
Base Resolution	64
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
TR	5.0 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
! Ref. Amplitude 1H	270.000 V

System - Tx/Rx

Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	5 deg

Inline - MapIt

Measurements	1
Contrasts	1
TE	2.04 ms
TE2 B0	4.08 ms
TR	5.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	1000 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off
Phase Stabilisation	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	NC7

Sequence - Special

Map Type	Integrated
Number of shim-sets	8
RO Spoiler	1.00
Measure B0	On
Call B0 Acq	Off
Call B1 Acq	Off
Rel. Mask Threshold	20

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\THS\ptxps_AC_PA7_PW10_4mmISO_89s

TA: 23 sec Coil Selection: Auto Voxel Size: 4.0×4.0×3.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

Slice Thickness	3.0 mm
Base Resolution	48
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	30
Distance Factor	33 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
TE	23.00 ms
Averages	1
Concatenations	1

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	33 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	3300.0 ms
TE	23.00 ms
MTC	Off
Flip Angle	30 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	6
Delay in TR	0.00 ms

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
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Geometry - Tim Planning Suite

Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	Patient-specific
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	192 mm
R >> L	192 mm
F >> H	119 mm
Reset	Off

System - pTx

B1 Shim	Patient-specific
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3300.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On

BOLD

Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	6
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	4166 Hz/Px
Echo Spacing	0.29 ms
Free Echo Spacing	On
EPI Factor	48

Sequence - Part 2

Introduction	Off
Ghost Reduction	Off

Sequence - Special

ABS Flip Angle	7.00 degree
ABS Pulse Width	10 ms
CP Mode	Off
Delta TE	1.00 ms
ABS Spoilers Width	2520 us
ABS Spoilers Mask Bit	7
ABS Spoilers M1 Null Mode	On
No of Pre Dummy TRs	4

Sequence - Special

No of Post Dummy TRs	0
Ramp Sampling Factor	12.50 %
Grad Stim Ratio	58.16 %
Ref Amplitude	0.00 V
ABS Tx Ref Amp	0.00 V

Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\THS\ptxps_CP_PA4_PW10_4mmISO_20s

TA: 40 sec Coil Selection: Auto Voxel Size: 4.0×4.0×3.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

Slice Thickness	3.0 mm
Base Resolution	48
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	30
Distance Factor	33 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
TE	23.00 ms
Averages	1
Concatenations	1

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	33 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	3300.0 ms
TE	23.00 ms
MTC	Off
Flip Angle	30 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	11
Delay in TR	0.00 ms

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
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Geometry - Tim Planning Suite

Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	Patient-specific
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	192 mm
R >> L	192 mm
F >> H	119 mm
Reset	Off

System - pTx

B1 Shim	Patient-specific
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3300.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On

BOLD

Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	11
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	4166 Hz/Px
Echo Spacing	0.29 ms
Free Echo Spacing	On
EPI Factor	48

Sequence - Part 2

Introduction	Off
Ghost Reduction	Off

Sequence - Special

ABS Flip Angle	4.00 degree
ABS Pulse Width	10 ms
CP Mode	On
Delta TE	1.00 ms
ABS Spoilers Width	2520 us
ABS Spoilers Mask Bit	7
ABS Spoilers M1 Null Mode	On
No of Pre Dummy TRs	4

Sequence - Special

No of Post Dummy TRs	0
Ramp Sampling Factor	12.50 %
Grad Stim Ratio	58.16 %
Ref Amplitude	0.00 V
ABS Tx Ref Amp	0.00 V

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS\gre_FA10_acpc

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\THS\gre_FA0_acpc

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\THS\gre_FA10_up

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\THS\gre_FA0_up

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\THS\mp2rage_0p7iso_patientSpecific

TA: 8:27 min Coil Selection: Auto Voxel Size: 0.7×0.7×0.7 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
TE	3.37 ms
Averages	1
Concatenations	1

Contrast - Common

TR	5000.0 ms
TE	3.37 ms
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
--------------	----------

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	6/8
Slice Partial Fourier	7/8
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	Patient-specific
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	224 mm
F >> H	224 mm
R >> L	180 mm
Reset	Off

System - pTx

B1 Shim	Patient-specific
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	224 mm
FOV Phase	100.0 %
Phase Resolution	100 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	3.37 ms
TR	5000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	200 Hz/Px
Echo Spacing	7.68 ms
Asymmetric Echo	Off
Turbo Factor	224

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS\mp2rage_0p7iso_TrueForm

TA: 8:27 min Coil Selection: Auto Voxel Size: 0.7×0.7×0.7 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
TE	3.37 ms
Averages	1
Concatenations	1

Contrast - Common

TR	5000.0 ms
TE	3.37 ms
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
--------------	----------

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	6/8
Slice Partial Fourier	7/8
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	224 mm
F >> H	224 mm
R >> L	180 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	224 mm
FOV Phase	100.0 %
Phase Resolution	100 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	3.37 ms
TR	5000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	200 Hz/Px
Echo Spacing	7.68 ms
Asymmetric Echo	Off
Turbo Factor	224

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS\ep2d_diff_1p3iso_b2k5_PA

TA: 9:18 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	-180.00 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
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Physio - PACE

Concatenations	1
----------------	---

Diff

Diffusion Mode	MDDW
Diff. Directions	64
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	2500 s/mm ²
Averages 1	8
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\jon\Saskia\THS\step2d_diff_1p3iso_b1k0_PA

TA: 5:20 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	-180.00 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
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Physio - PACE

Concatenations	1
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Diff

Diffusion Mode	MDDW
Diff. Directions	30
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	1000 s/mm ²
Averages 1	8
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\jon\Saskia\THS\lep2d_diff_1p3iso_AP

TA: 1:50 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	4 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Rotation	175.41 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
---------------	-----

Physio - PACE

Concatenations	1
----------------	---

Diff

Diffusion Mode	MDDW
Diff. Directions	30
Diffusion Scheme	Monopolar
Diff. Weightings	1
b-value	0 s/mm ²
Averages	8
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\jon\Saskia\THS\lep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC

TA: 6:37 min Coil Selection: Auto Voxel Size: 1.4×1.4×1.4 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
TE	24.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	24.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	380
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
----------	--------

Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	66
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H25.0
L	0.0 mm
P	0.0 mm
H	25.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	25 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	200 mm
R >> L	200 mm
F >> H	101 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	380
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	2480 Hz/Px
Echo Spacing	0.50 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS\ldzne_ep3d_vasoNIH

TA: 3:14 min Coil Selection: Auto Voxel Size: 0.8x0.8x0.9 mm³ Acc.: 16 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Multi-echo spacing	32.50 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	87
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
Base Resolution	236
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	80
Acceleration Factor 3D	4
Reference Lines 3D	36
Reordering Shift 3D	2
Phase Partial Fourier	7/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	13 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Rotation	0.00 deg
A >> P	194 mm
R >> L	200 mm
F >> H	177 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1766 Hz/Px
Echo Spacing	0.64 ms
Asymmetric Echo	Off
Turbo Factor	52
Segmentation	1
EPI Factor	50

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS\ldzne_ep3d_Frankfurt_6echoes_minimumMaxBW

TA: 6:29 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	49.9 ms
Vol. TR	383.232 s
TE 1	4.70 ms
TE 2	11.90 ms
TE 3	19.10 ms
TE 4	26.30 ms
TE 5	33.50 ms
TE 6	40.70 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	49.9 ms
Vol. TR	383.232 s
TE 1	4.70 ms
TE 2	11.90 ms
TE 3	19.10 ms
TE 4	26.30 ms
TE 5	33.50 ms
TE 6	40.70 ms
Multi-echo spacing	7.20 ms

Contrast - Common

MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fast Fat Saturation
Contrasts	6
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
Base Resolution	500
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	36
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %

Geometry - Common

Slice Thickness	0.35 mm
TR	49.9 ms
Vol. TR	383.232 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
-----------------	-------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	22 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P9.0 F22.1 mm
Orientation	Transversal
Rotation	90.00 deg
R >> L	175 mm
A >> P	210 mm
F >> H	135 mm

System - Adjust Volume

Reset	Off
-------	-----

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.20 ms
Asymmetric Echo	Off
Turbo Factor	7680
Segmentation	40
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS\ep2d_bold_1p8mm_TR1000

TA: 27 sec Coil Selection: Auto Voxel Size: 1.8×1.8×1.8 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	80
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.4 %
Slice Thickness	1.8 mm
TR	1000.0 ms
TE	21.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	21.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
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Resolution - Common

FOV Phase	96.4 %
Slice Thickness	1.8 mm
Base Resolution	112
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	1
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	80
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.4 %
Slice Thickness	1.8 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H25.0
L	0.0 mm
P	0.0 mm
H	25.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	25 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	193 mm
R >> L	200 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	10
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	4464 Hz/Px
Echo Spacing	0.29 ms
Free Echo Spacing	Off
EPI Factor	108

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS\lep2d_bold_1p0mm_TR1000_TE25_SMS2x6_BW3126_GC

TA: 33 sec Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc.: 12 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
Base Resolution	200
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	48
SMS Factor	6
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	102
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
TR	1000.0 ms
TE	25.00 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Geometry - Common

Slice Group	1
Slices	102
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	25.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Delay in TR	0.00 ms

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H25.0
L	0.0 mm
P	0.0 mm
H	25.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	25 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	192 mm
R >> L	200 mm
F >> H	102 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	10
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	Off
EPI Factor	192

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS\ldzne_ep3d_3mm_TE8p8_TR19p9_volTR99p5_es_29

TA: 3 sec Coil Selection: Auto Voxel Size: 3.0×3.0×3.0 mm³ Acc:: 9 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	10.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	19.9 ms
Vol. TR	99.5 ms
TE 1	8.80 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	19.9 ms
Vol. TR	99.5 ms
TE 1	8.80 ms
Multi-echo spacing	16.40 ms
MTC	Off
Flip Angle	9 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
Base Resolution	70
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	24
Acceleration Factor 3D	9
Reference Lines 3D	24
Reordering Shift 3D	3
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	10.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	19.9 ms
Vol. TR	99.5 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	7142 Hz/Px
Echo Spacing	0.23 ms
Asymmetric Echo	Off
Turbo Factor	5
Segmentation	1
EPI Factor	70

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS\ldzne_ep3d_1p8mm_1p8mm_TE18p2_TR38P7_volTR_309p6_es29

TA: 38 sec Coil Selection: Auto Voxel Size: 1.8×1.8×1.8 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P11.8 F8.6 mm
Orientation	T > C-14.1
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	64
Phase Oversampling	0.0 %
Slice Oversampling	12.5 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.80 mm
TR	38.7 ms
Vol. TR	348.3 ms
TE 1	18.20 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	38.7 ms
Vol. TR	348.3 ms
TE 1	18.20 ms
Multi-echo spacing	35.10 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	100
Reordering	Linear

Resolution - Common

FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.80 mm
Base Resolution	120
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	24
Acceleration Factor 3D	8
Reference Lines 3D	24
Reordering Shift 3D	2
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P11.8 F8.6 mm
Orientation	T > C-14.1
Phase Encoding Dir.	A >> P
Slices per Slab	64
Phase Oversampling	0.0 %
Slice Oversampling	12.5 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	1.80 mm
TR	38.7 ms
Vol. TR	348.3 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P11.8 F8.6 mm
Orientation	T > C-14.1
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	9 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P11.8 F8.6 mm
Orientation	T > C-14.1
Rotation	0.00 deg
A >> P	216 mm
R >> L	216 mm
F >> H	116 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	5208 Hz/Px
Echo Spacing	0.29 ms
Asymmetric Echo	Off
Turbo Factor	9
Segmentation	1
EPI Factor	120

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260529_THS_ses01\localizer

TA: 15 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	All Segments
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
TE	3.69 ms
Averages	2
Concatenations	3

Contrast - Common

TR	8.6 ms
TE	3.69 ms

Contrast - Common

TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slice Group	3

Geometry - Common

Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg
Slice Group	2
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
Slice Group	3
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.6 ms
Segments	1
Concatenations	3

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None

Physio - Cardiac

Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	20 deg
Measurements	1
Contrasts	1
TE	3.69 ms
TR	8.6 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	qfl
---------------	-----

Sequence - Part 1

Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\260529_THS_ses01\AAhead_scout_md

TA: 15 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.53 ms
TE	1.53 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.53 ms
TE	1.53 ms
Flip Angle	16 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.7 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.53 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Initial Position	Isocenter
L	0.0 mm

Geometry - AutoAlign

P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
AutoAlign	Head

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off
AutoAlign MPR	On

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	16 deg
Measurements	1
Contrasts	1
TE	1.53 ms
TR	3.53 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260529_THS_ses01\gre_b0map_4iso_sag

TA: 31 sec Coil Selection: Auto Voxel Size: 4.0×4.0×4.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	44
Distance Factor	0 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
Averages	1
Concatenations	1

Contrast - Common

TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
MTC	Off
Flip Angle	25 deg
Fat-Water Contrast	Standard
Contrasts	2
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	44
Distance Factor	0 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fm_r
Dimension	2D
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	On
Bandwidth	908 Hz/Px
Asymmetric Echo	Off

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260529_THS_ses01\dzne_b1map_5iso_sag

TA: 7 sec Coil Selection: Auto Voxel Size: 5.0×5.0×5.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Averages	1
Concatenations	1

Contrast - Common

TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Flip Angle 1	60 deg
Flip Angle 2	6 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
Base Resolution	44
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal

Geometry - AutoAlign

Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	b1map 9e96902
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	1000 Hz/Px
Asymmetric Echo	Off
Turbo Factor	440
Echo Train Duration	1298 ms

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Mapping Technique	DREAM
Timing Scheme	STE*
Mixing Time	1130 us
Scale risetime	1.2

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260529_THS_ses01\head

TA: 5 sec Coil Selection: Auto Voxel Size: 4.0×4.0×5.0 mm³ Acc.: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	16
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Routine

Slice Group	1
Slices	25
Distance Factor	100 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	3900.0 ms
TE	1.56 ms
Averages	1
Concatenations	1

Geometry - Common

Slice Group	1
Slices	25
Distance Factor	100 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	3900.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	3900.0 ms
TE	1.56 ms
Magn. Preparation	None
Flip Angle	5 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Centric

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3900.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None

Physio - Cardiac

Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	tfl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Whisper
Flow Compensation	None
Reordering	Centric
Bandwidth	440 Hz/Px
Echo Spacing	3.54 ms
Asymmetric Echo	Allowed
Turbo Factor	64

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Tx Scale Diag. Magn.	0.3
Tx Scale Diag. Phase	0 deg
Tx Scale Off-diag. Magn.	1.0
Tx Scale Off-diag. Phase	0 deg
Rel. B1 mapping	On
Ref. Scan	Off
Use B1 Map Recon.	On
Prep Pulse	Selective
Dummy RF pulses	1000
Dummy RF TRs	0

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260529_THS_ses01\bss_map

TA: 2:01 min Coil Selection: Manual Voxel Size: 4.0x4.0x4.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
TR	5.0 ms
TE	2.04 ms
TE2 B0	4.08 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	5.0 ms
TE	2.04 ms
TE2 B0	4.08 ms
Flip Angle	5 deg
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
Base Resolution	64
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
TR	5.0 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
! Ref. Amplitude 1H	270.000 V

System - Tx/Rx

Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	5 deg

Inline - MapIt

Measurements	1
Contrasts	1
TE	2.04 ms
TE2 B0	4.08 ms
TR	5.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	1000 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off
Phase Stabilisation	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	NC7

Sequence - Special

Map Type	Integrated
Number of shim-sets	8
RO Spoiler	1.00
Measure B0	On
Call B0 Acq	Off
Call B1 Acq	Off
Rel. Mask Threshold	20

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\260529_THS_ses01\ptxps_AC_PA7_PW10_4mmISO_89s

TA: 23 sec Coil Selection: Auto Voxel Size: 4.0×4.0×3.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

Slice Thickness	3.0 mm
Base Resolution	48
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	30
Distance Factor	33 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
TE	23.00 ms
Averages	1
Concatenations	1

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	33 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	3300.0 ms
TE	23.00 ms
MTC	Off
Flip Angle	30 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	6
Delay in TR	0.00 ms

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %

Geometry - AutoAlign

Slice Group	1
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
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Geometry - Tim Planning Suite

Label	
Table Position	4 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P0.0 H0.0 mm
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3300.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On

BOLD

Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	6
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	4166 Hz/Px
Echo Spacing	0.29 ms
Free Echo Spacing	On
EPI Factor	48

Sequence - Part 2

Introduction	Off
Ghost Reduction	Off

Sequence - Special

ABS Flip Angle	7.00 degree
ABS Pulse Width	10 ms
CP Mode	Off
Delta TE	1.00 ms
ABS Spoilers Width	2520 us
ABS Spoilers Mask Bit	7
ABS Spoilers M1 Null Mode	On
No of Pre Dummy TRs	4

Sequence - Special

No of Post Dummy TRs	0
Ramp Sampling Factor	12.50 %
Grad Stim Ratio	58.16 %
Ref Amplitude	0.00 V
ABS Tx Ref Amp	0.00 V

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260529_THS_ses01\ptxps_CP_PA4_PW10_4mmISO_20s

TA: 40 sec Coil Selection: Auto Voxel Size: 4.0×4.0×3.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

Slice Thickness	3.0 mm
Base Resolution	48
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	30
Distance Factor	33 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
TE	23.00 ms
Averages	1
Concatenations	1

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	33 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	3300.0 ms
TE	23.00 ms
MTC	Off
Flip Angle	30 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	11
Delay in TR	0.00 ms

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %

Geometry - AutoAlign

Slice Group	1
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
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Geometry - Tim Planning Suite

Label	
Table Position	4 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P0.0 H0.0 mm
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3300.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On

BOLD

Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	11
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	4166 Hz/Px
Echo Spacing	0.29 ms
Free Echo Spacing	On
EPI Factor	48

Sequence - Part 2

Introduction	Off
Ghost Reduction	Off

Sequence - Special

ABS Flip Angle	4.00 degree
ABS Pulse Width	10 ms
CP Mode	On
Delta TE	1.00 ms
ABS Spoilers Width	2520 us
ABS Spoilers Mask Bit	7
ABS Spoilers M1 Null Mode	On
No of Pre Dummy TRs	4

Sequence - Special

No of Post Dummy TRs	0
Ramp Sampling Factor	12.50 %
Grad Stim Ratio	58.16 %
Ref Amplitude	0.00 V
ABS Tx Ref Amp	0.00 V

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260529_THS_ses01\mp2rage_0p7iso

TA: 8:27 min Coil Selection: Auto Voxel Size: 0.7×0.7×0.7 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
TE	3.37 ms
Averages	1
Concatenations	1

Contrast - Common

TR	5000.0 ms
TE	3.37 ms
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
--------------	----------

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	6/8
Slice Partial Fourier	7/8
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	Patient-specific
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	224 mm
F >> H	224 mm
R >> L	180 mm
Reset	Off

System - pTx

B1 Shim	Patient-specific
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	224 mm
FOV Phase	100.0 %
Phase Resolution	100 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	3.37 ms
TR	5000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	200 Hz/Px
Echo Spacing	7.68 ms
Asymmetric Echo	Off
Turbo Factor	224

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260529_THS_ses01\t1_mp2rage_sag_0p7_wip

TA: 7:33 min Coil Selection: Manual Voxel Size: 0.7×0.7×0.7 mm³ Acc:: 5.0 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Slices per Slab	224
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	6000.0 ms
TE	1.99 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	6000.0 ms
TE	1.99 ms
Magn. Preparation	Non-sel. IR
TI 1	870 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	5.0
Reference Scans	GRE/Separate
Reference Lines PE	32
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	224
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	6000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	R4.8 P4.0 H26.4
R	4.8 mm
P	4.0 mm
H	26.4 mm
Initial Orientation	S > C
S > C	4.50
> T	-1.10
Initial Rotation	-0.10 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	Patient-specific
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	2.87 deg
A >> P	224 mm
F >> H	224 mm
R >> L	157 mm
Reset	Off

System - pTx

B1 Shim	Patient-specific
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	6000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI 1	870 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	224 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	1.99 ms
TR	6000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	5.86 ms
Asymmetric Echo	Allowed
Turbo Factor	208

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Custom pTx Pulse Duration	Off
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Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\260529_THS_ses01\t2_flair_sag_p9_0.7iso_dr

TA: 6:29 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.7 mm³ Acc:: 9 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	6.7 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	9000.0 ms
TE	300.00 ms
Averages	1.0
Concatenations	1

Contrast - Common

TR	9000.0 ms
TE	300.00 ms
MTC Mode	Off
Magn. Preparation	Non-sel. T2 prep. IR
TI	2500 ms
Flip Angle Mode	T2 Var
Flip Angle 1	120 deg
Flip Angle 2	90 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Total Factor	9
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	3
Reference Lines 3D	24
Reordering Shift 3D	2
Deep Resolve / ID Gain	On
Phase Partial Fourier	Allowed
Slice Partial Fourier	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	6.7 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	9000.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.01 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	Patient-specific
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.01 deg
A >> P	224 mm
F >> H	224 mm
R >> L	168 mm
Reset	Off

System - pTx

B1 Shim	Patient-specific
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System - pTx

Excitation	Non-sel.
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System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	9000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Non-sel. T2 prep. IR
TI	2500 ms
T2 Prep. Duration	125 ms
Dark Blood	Off
FOV Read	224 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	spcir
Dimension	3D

Sequence - Part 1

Excitation	Non-sel.
RF Pulse Type	Low SAR
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	710 Hz/Px
Echo Spacing	3.12 ms
Turbo Factor	240
Echo Train Duration	671 ms

Sequence - Part 2

Introduction	On
Incr. Gradient Spoiling	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	30 s

\\USER\jon\Saskia\260529_THS_ses01\ep2d_diff_1p3iso_b2k5_PA

TA: 9:18 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	4 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Rotation	175.41 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
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Physio - PACE

Concatenations	1
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Diff

Diffusion Mode	MDDW
Diff. Directions	64
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	2500 s/mm ²
Averages 1	8
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\jon\Saskia\260529_THS_ses01\ep2d_diff_1p3iso_b1k0_PA

TA: 5:20 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	4 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Rotation	175.41 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
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Physio - PACE

Concatenations	1
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Diff

Diffusion Mode	MDDW
Diff. Directions	30
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	1000 s/mm ²
Averages 1	8
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\jon\Saskia\260529_THS_ses01\ep2d_diff_1p3iso_AP

TA: 1:50 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	4 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Rotation	175.41 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
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Physio - PACE

Concatenations	1
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Diff

Diffusion Mode	MDDW
Diff. Directions	30
Diffusion Scheme	Monopolar
Diff. Weightings	1
b-value	0 s/mm ²
Averages	8
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\jon\Saskia\260529_THS_ses01\ep2d_bold_1p8mm_TR1000

TA: 27 sec Coil Selection: Auto Voxel Size: 1.8×1.8×1.8 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	80
Distance Factor	0 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.4 %
Slice Thickness	1.8 mm
TR	1000.0 ms
TE	21.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	21.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
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Resolution - Common

FOV Phase	96.4 %
Slice Thickness	1.8 mm
Base Resolution	112
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	1
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	80
Distance Factor	0 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.4 %
Slice Thickness	1.8 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	4 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Rotation	-4.59 deg
A >> P	193 mm
R >> L	200 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	10
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	4464 Hz/Px
Echo Spacing	0.29 ms
Free Echo Spacing	Off
EPI Factor	108

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260529_THS_ses01\ep2d_bold_1p0mm_TR1000_TE25_SMS2x6_BW3126_GC

TA: 33 sec Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc.: 12 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
Base Resolution	200
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	48
SMS Factor	6
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	102
Distance Factor	0 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
TR	1000.0 ms
TE	25.00 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Geometry - Common

Slice Group	1
Slices	102
Distance Factor	0 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	25.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Delay in TR	0.00 ms

Geometry - AutoAlign

Slice Group	1
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	4 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

I Position	L4.2 P8.3 H4.3 mm
I Orientation	T > C-12.8 > S-0.3
I Rotation	-4.59 deg
I A >> P	193 mm
I R >> L	200 mm
I F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	10
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	Off
EPI Factor	192

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260529_THS_ses01\ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC

TA: 37 sec Coil Selection: Auto Voxel Size: 1.4×1.4×1.4 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
TE	24.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	24.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
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Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	66
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L4.2 P8.3 H4.3 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	4 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

I Position	L4.2 P8.3 H4.3 mm
I Orientation	T > C-12.8 > S-0.3
I Rotation	-4.59 deg
I A >> P	193 mm
I R >> L	200 mm
I F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	2480 Hz/Px
Echo Spacing	0.50 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\260529_THS_ses01\ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC

TA: 4:17 min Coil Selection: Auto Voxel Size: 1.4×1.4×1.4 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	L4.2 P12.8 H19.6 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
TE	24.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	24.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	240
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
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Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	66
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	L4.2 P12.8 H19.6 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L4.2 P12.8 H19.6 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.0 P1.0 H16.0
R	0.0 mm
P	1.0 mm
H	16.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	20 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L4.2 P12.8 H19.6 mm
Orientation	T > C-12.8 > S-0.3
Rotation	-4.59 deg
A >> P	200 mm
R >> L	200 mm
F >> H	101 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	240
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	2480 Hz/Px
Echo Spacing	0.50 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\260529_THS_ses01\ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC

TA: 4:17 min Coil Selection: Auto Voxel Size: 1.4×1.4×1.4 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	L4.2 P12.8 H19.6 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
TE	24.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	24.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	240
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
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Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	66
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	L4.2 P12.8 H19.6 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L4.2 P12.8 H19.6 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.0 P1.0 H16.0
R	0.0 mm
P	1.0 mm
H	16.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	20 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L4.2 P12.8 H19.6 mm
Orientation	T > C-12.8 > S-0.3
Rotation	-4.59 deg
A >> P	200 mm
R >> L	200 mm
F >> H	101 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	240
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	2480 Hz/Px
Echo Spacing	0.50 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260529_THS_ses01\ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC

TA: 4:17 min Coil Selection: Auto Voxel Size: 1.4×1.4×1.4 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	L4.2 P12.8 H19.6 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
TE	24.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	24.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	240
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
----------	--------

Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	66
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	L4.2 P12.8 H19.6 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L4.2 P12.8 H19.6 mm
Orientation	T > C-12.8 > S-0.3
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.0 P1.0 H16.0
R	0.0 mm
P	1.0 mm
H	16.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	20 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L4.2 P12.8 H19.6 mm
Orientation	T > C-12.8 > S-0.3
Rotation	-4.59 deg
A >> P	200 mm
R >> L	200 mm
F >> H	101 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	240
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	2480 Hz/Px
Echo Spacing	0.50 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260529_THS_ses01\dzne_ep3d_vasoNIH

TA: 3:14 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.9 mm³ Acc.: 16 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Multi-echo spacing	32.50 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	87
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
Base Resolution	236
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	80
Acceleration Factor 3D	4
Reference Lines 3D	36
Reordering Shift 3D	2
Phase Partial Fourier	7/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	13 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Rotation	0.00 deg
A >> P	194 mm
R >> L	200 mm
F >> H	177 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1766 Hz/Px
Echo Spacing	0.64 ms
Asymmetric Echo	Off
Turbo Factor	52
Segmentation	1
EPI Factor	50

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260529_THS_ses01\tof_fl3d_tra_0p3iso_TE1p52_TR4p81_GRAPPA_3x1_noFC

TA: 3:48 min Coil Selection: Manual Voxel Size: 0.3x0.3x0.3 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	A >> P
TONE Ramp	100 %

Resolution - Common

FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
Base Resolution	656
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	L3.2 P68.2 H0.5 mm
Orientation	C > T-14.5 > S-0.3
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	192
Phase Oversampling	0 %
Slice Oversampling	16.7 %
FOV Read	200 mm
FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	4.82 ms
TE	1.53 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Contrast - Common

TR	4.82 ms
TE	1.53 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	5 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-19 %
Position	L3.2 P68.2 H0.5 mm
Orientation	C > T-14.5 > S-0.3
Phase Encoding Dir.	R >> L
Slices per Slab	192
Phase Oversampling	0 %
Slice Oversampling	16.7 %
FOV Read	200 mm

Geometry - Common

FOV Phase	87.2 %
Slice Thickness	0.30 mm
TR	4.82 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L3.2 P68.2 H0.5 mm
Orientation	C > T-14.5 > S-0.3
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L3.6 P59.1 H9.6
L	3.6 mm
P	59.1 mm
H	9.6 mm
Initial Orientation	C > T
C > T	-27.40
> S	-4.30
Initial Rotation	-3.95 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	1 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L3.2 P68.2 H0.5 mm
Orientation	C > T-14.5 > S-0.3
Rotation	-4.24 deg
R >> L	175 mm
F >> H	200 mm
A >> P	58 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	4.82 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	87.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.53 ms
TR	4.82 ms

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Fast
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	254 Hz/Px
Echo Spacing	4.82 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\260529_THS_ses01\ldzne_ep3d_Frankfurt_6echoes_minimumMaxBW

TA: 6:29 min Coil Selection: Auto Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P11.5 H6.2 mm
Orientation	T > C-16.3
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	49.9 ms
Vol. TR	383.232 s
TE 1	4.70 ms
TE 2	11.90 ms
TE 3	19.10 ms
TE 4	26.30 ms
TE 5	33.50 ms
TE 6	40.70 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	49.9 ms
Vol. TR	383.232 s
TE 1	4.70 ms
TE 2	11.90 ms
TE 3	19.10 ms
TE 4	26.30 ms
TE 5	33.50 ms
TE 6	40.70 ms
Multi-echo spacing	7.20 ms

Contrast - Common

MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fast Fat Saturation
Contrasts	6
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
Base Resolution	500
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	36
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P11.5 H6.2 mm
Orientation	T > C-16.3
Phase Encoding Dir.	A >> P
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %

Geometry - Common

Slice Thickness	0.35 mm
TR	49.9 ms
Vol. TR	383.232 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P11.5 H6.2 mm
Orientation	T > C-16.3
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
-----------------	-------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	6 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P15.3 H18.9 mm
! Orientation	T > C-16.3
! Rotation	90.00 deg
! R >> L	175 mm
! A >> P	210 mm
! F >> H	105 mm

System - Adjust Volume

Reset	Off
-------	-----

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.20 ms
Asymmetric Echo	Off
Turbo Factor	7680
Segmentation	40
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260601_THS_ses02\localizer

TA: 15 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	All Segments
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
TE	3.69 ms
Averages	2
Concatenations	3

Contrast - Common

TR	8.6 ms
TE	3.69 ms

Contrast - Common

TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slice Group	3

Geometry - Common

Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg
Slice Group	2
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
Slice Group	3
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.6 ms
Segments	1
Concatenations	3

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None

Physio - Cardiac

Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	20 deg
Measurements	1
Contrasts	1
TE	3.69 ms
TR	8.6 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	qfl
---------------	-----

Sequence - Part 1

Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\260601_THS_ses02\AAhead_scout_md

TA: 15 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.53 ms
TE	1.53 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.53 ms
TE	1.53 ms
Flip Angle	16 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.7 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.53 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Initial Position	Isocenter
L	0.0 mm

Geometry - AutoAlign

P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
AutoAlign	Head

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off
AutoAlign MPR	On

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	16 deg
Measurements	1
Contrasts	1
TE	1.53 ms
TR	3.53 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260601_THS_ses02\gre_b0map_4iso_sag

TA: 31 sec Coil Selection: Auto Voxel Size: 4.0×4.0×4.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	44
Distance Factor	0 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
Averages	1
Concatenations	1

Contrast - Common

TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
MTC	Off
Flip Angle	25 deg
Fat-Water Contrast	Standard
Contrasts	2
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	44
Distance Factor	0 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fm_r
Dimension	2D
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	On
Bandwidth	908 Hz/Px
Asymmetric Echo	Off

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260601_THS_ses02\ldzne_b1map_5iso_sag

TA: 7 sec Coil Selection: Auto Voxel Size: 5.0×5.0×5.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Averages	1
Concatenations	1

Contrast - Common

TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Flip Angle 1	60 deg
Flip Angle 2	6 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
Base Resolution	44
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal

Geometry - AutoAlign

Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	b1map 9e96902
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	1000 Hz/Px
Asymmetric Echo	Off
Turbo Factor	440
Echo Train Duration	1298 ms

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Mapping Technique	DREAM
Timing Scheme	STE*
Mixing Time	1130 us
Scale risetime	1.2

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260601_THS_ses02\head

TA: 5 sec Coil Selection: Auto Voxel Size: 4.0×4.0×5.0 mm³ Acc.: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	16
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Routine

Slice Group	1
Slices	25
Distance Factor	100 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	3900.0 ms
TE	1.56 ms
Averages	1
Concatenations	1

Geometry - Common

Slice Group	1
Slices	25
Distance Factor	100 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	3900.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	3900.0 ms
TE	1.56 ms
Magn. Preparation	None
Flip Angle	5 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Centric

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3900.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None

Physio - Cardiac

Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	tfl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Whisper
Flow Compensation	None
Reordering	Centric
Bandwidth	440 Hz/Px
Echo Spacing	3.54 ms
Asymmetric Echo	Allowed
Turbo Factor	64

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Tx Scale Diag. Magn.	0.3
Tx Scale Diag. Phase	0 deg
Tx Scale Off-diag. Magn.	1.0
Tx Scale Off-diag. Phase	0 deg
Rel. B1 mapping	On
Ref. Scan	Off
Use B1 Map Recon.	On
Prep Pulse	Selective
Dummy RF pulses	1000
Dummy RF TRs	0

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260601_THS_ses02\bss_map

TA: 2:01 min Coil Selection: Manual Voxel Size: 4.0×4.0×4.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
TR	5.0 ms
TE	2.04 ms
TE2 B0	4.08 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	5.0 ms
TE	2.04 ms
TE2 B0	4.08 ms
Flip Angle	5 deg
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
Base Resolution	64
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
TR	5.0 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	192 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
! Ref. Amplitude 1H	270.000 V

System - Tx/Rx

Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	5 deg

Inline - MapIt

Measurements	1
Contrasts	1
TE	2.04 ms
TE2 B0	4.08 ms
TR	5.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	1000 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off
Phase Stabilisation	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	NC7

Sequence - Special

Map Type	Integrated
Number of shim-sets	8
RO Spoiler	1.00
Measure B0	On
Call B0 Acq	Off
Call B1 Acq	Off
Rel. Mask Threshold	20

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\260601_THS_ses02\ptxps_AC_PA7_PW10_4mmISO_89s

TA: 23 sec Coil Selection: Auto Voxel Size: 4.0×4.0×3.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

Slice Thickness	3.0 mm
Base Resolution	48
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	30
Distance Factor	33 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
TE	23.00 ms
Averages	1
Concatenations	1

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	33 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	3300.0 ms
TE	23.00 ms
MTC	Off
Flip Angle	30 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	6
Delay in TR	0.00 ms

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %

Geometry - AutoAlign

Slice Group	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
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Geometry - Tim Planning Suite

Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Rotation	-1.37 deg
A >> P	192 mm
R >> L	192 mm
F >> H	119 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3300.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On

BOLD

Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	6
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	4166 Hz/Px
Echo Spacing	0.29 ms
Free Echo Spacing	On
EPI Factor	48

Sequence - Part 2

Introduction	Off
Ghost Reduction	Off

Sequence - Special

ABS Flip Angle	7.00 degree
ABS Pulse Width	10 ms
CP Mode	Off
Delta TE	1.00 ms
ABS Spoilers Width	2520 us
ABS Spoilers Mask Bit	7
ABS Spoilers M1 Null Mode	On
No of Pre Dummy TRs	4

Sequence - Special

No of Post Dummy TRs	0
Ramp Sampling Factor	12.50 %
Grad Stim Ratio	58.16 %
Ref Amplitude	0.00 V
ABS Tx Ref Amp	0.00 V

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260601_THS_ses02\ptxps_CP_PA4_PW10_4mmISO_20s

TA: 40 sec Coil Selection: Auto Voxel Size: 4.0×4.0×3.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

Slice Thickness	3.0 mm
Base Resolution	48
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	30
Distance Factor	33 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
TE	23.00 ms
Averages	1
Concatenations	1

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	33 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	3300.0 ms
TE	23.00 ms
MTC	Off
Flip Angle	30 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	11
Delay in TR	0.00 ms

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %

Geometry - AutoAlign

Slice Group	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
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Geometry - Tim Planning Suite

Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Rotation	-1.37 deg
A >> P	192 mm
R >> L	192 mm
F >> H	119 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3300.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On

BOLD

Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	11
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	4166 Hz/Px
Echo Spacing	0.29 ms
Free Echo Spacing	On
EPI Factor	48

Sequence - Part 2

Introduction	Off
Ghost Reduction	Off

Sequence - Special

ABS Flip Angle	4.00 degree
ABS Pulse Width	10 ms
CP Mode	On
Delta TE	1.00 ms
ABS Spoilers Width	2520 us
ABS Spoilers Mask Bit	7
ABS Spoilers M1 Null Mode	On
No of Pre Dummy TRs	4

Sequence - Special

No of Post Dummy TRs	0
Ramp Sampling Factor	12.50 %
Grad Stim Ratio	58.16 %
Ref Amplitude	0.00 V
ABS Tx Ref Amp	0.00 V

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260601_THS_ses02\gre_FA10

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Rotation	-1.37 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\260601_THS_ses02\gre_FA0

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Rotation	-1.37 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\260601_THS_ses02\mp2rage_0p7iso

TA: 8:27 min Coil Selection: Auto Voxel Size: 0.7×0.7×0.7 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
TE	3.37 ms
Averages	1
Concatenations	1

Contrast - Common

TR	5000.0 ms
TE	3.37 ms
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
--------------	----------

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	6/8
Slice Partial Fourier	7/8
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	Patient-specific
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	224 mm
F >> H	224 mm
R >> L	180 mm
Reset	Off

System - pTx

B1 Shim	Patient-specific
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	224 mm
FOV Phase	100.0 %
Phase Resolution	100 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	3.37 ms
TR	5000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	200 Hz/Px
Echo Spacing	7.68 ms
Asymmetric Echo	Off
Turbo Factor	224

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260601_THS_ses02\mp2rage_0p7iso

TA: 8:27 min Coil Selection: Auto Voxel Size: 0.7×0.7×0.7 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
TE	3.37 ms
Averages	1
Concatenations	1

Contrast - Common

TR	5000.0 ms
TE	3.37 ms
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
--------------	----------

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	6/8
Slice Partial Fourier	7/8
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	224 mm
F >> H	224 mm
R >> L	180 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	224 mm
FOV Phase	100.0 %
Phase Resolution	100 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	3.37 ms
TR	5000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	200 Hz/Px
Echo Spacing	7.68 ms
Asymmetric Echo	Off
Turbo Factor	224

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260601_THS_ses02\gre_FA10

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Rotation	-1.37 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\260601_THS_ses02\gre_FA0

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Rotation	-1.37 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
! Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\260601_THS_ses02\ep2d_diff_1p3iso_b2k5_PA

TA: 9:18 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Rotation	178.63 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
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Physio - PACE

Concatenations	1
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Diff

Diffusion Mode	MDDW
Diff. Directions	64
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	2500 s/mm ²
Averages 1	8
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\jon\Saskia\260601_THS_ses02\ep2d_diff_1p3iso_b1k0_PA

TA: 5:20 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Rotation	178.63 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
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Physio - PACE

Concatenations	1
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Diff

Diffusion Mode	MDDW
Diff. Directions	30
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	1000 s/mm ²
Averages 1	8
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\jon\Saskia\260601_THS_ses02\ep2d_diff_1p3iso_AP

TA: 1:50 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Rotation	178.63 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
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Physio - PACE

Concatenations	1
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Diff

Diffusion Mode	MDDW
Diff. Directions	30
Diffusion Scheme	Monopolar
Diff. Weightings	1
b-value	0 s/mm ²
Averages	8
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\jon\Saskia\260601_THS_ses02\lp2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC

TA: 6:37 min Coil Selection: Auto Voxel Size: 1.4x1.4x1.4 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R2.2 P18.4 H26.1 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
TE	24.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	24.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	380
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
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Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	66
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R2.2 P18.4 H26.1 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R2.2 P18.4 H26.1 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H25.0
L	0.0 mm
P	0.0 mm
H	25.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	26 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.2 P18.4 H26.1 mm
Orientation	T > C-13.2 > S1.9
Rotation	-1.37 deg
A >> P	200 mm
R >> L	200 mm
F >> H	101 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	380
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	2480 Hz/Px
Echo Spacing	0.50 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\260601_THS_ses02\ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC

TA: 6:37 min Coil Selection: Auto Voxel Size: 1.4x1.4x1.4 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R2.2 P18.4 H26.1 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
TE	24.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	24.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	380
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
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Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	66
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R2.2 P18.4 H26.1 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R2.2 P18.4 H26.1 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H25.0
L	0.0 mm
P	0.0 mm
H	25.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	26 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.2 P18.4 H26.1 mm
Orientation	T > C-13.2 > S1.9
Rotation	-1.37 deg
A >> P	200 mm
R >> L	200 mm
F >> H	101 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	380
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	2480 Hz/Px
Echo Spacing	0.50 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260601_THS_ses02\lp2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC

TA: 6:37 min Coil Selection: Auto Voxel Size: 1.4x1.4x1.4 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R2.2 P18.4 H26.1 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
TE	24.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	24.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	380
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
----------	--------

Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	66
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R2.2 P18.4 H26.1 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R2.2 P18.4 H26.1 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H25.0
L	0.0 mm
P	0.0 mm
H	25.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	26 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.2 P18.4 H26.1 mm
Orientation	T > C-13.2 > S1.9
Rotation	-1.37 deg
A >> P	200 mm
R >> L	200 mm
F >> H	101 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	380
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	2480 Hz/Px
Echo Spacing	0.50 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260601_THS_ses02\ldzne_ep3d_vasoNIH

TA: 3:14 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.9 mm³ Acc:: 16 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Multi-echo spacing	32.50 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	87
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
Base Resolution	236
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	80
Acceleration Factor 3D	4
Reference Lines 3D	36
Reordering Shift 3D	2
Phase Partial Fourier	7/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Rotation	-1.37 deg
A >> P	194 mm
R >> L	200 mm
F >> H	177 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1766 Hz/Px
Echo Spacing	0.64 ms
Asymmetric Echo	Off
Turbo Factor	52
Segmentation	1
EPI Factor	50

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260601_THS_ses02\ldzne_ep3d_vasoNIH

TA: 3:14 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.9 mm³ Acc.: 16 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Multi-echo spacing	32.50 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	87
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
Base Resolution	236
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	80
Acceleration Factor 3D	4
Reference Lines 3D	36
Reordering Shift 3D	2
Phase Partial Fourier	7/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Rotation	-1.37 deg
A >> P	194 mm
R >> L	200 mm
F >> H	177 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1766 Hz/Px
Echo Spacing	0.64 ms
Asymmetric Echo	Off
Turbo Factor	52
Segmentation	1
EPI Factor	50

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260601_THS_ses02\ldzne_ep3d_Frankfurt_6echoes_minimumMaxBW

TA: 6:29 min Coil Selection: Auto Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	49.9 ms
Vol. TR	383.232 s
TE 1	4.70 ms
TE 2	11.90 ms
TE 3	19.10 ms
TE 4	26.30 ms
TE 5	33.50 ms
TE 6	40.70 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	49.9 ms
Vol. TR	383.232 s
TE 1	4.70 ms
TE 2	11.90 ms
TE 3	19.10 ms
TE 4	26.30 ms
TE 5	33.50 ms
TE 6	40.70 ms
Multi-echo spacing	7.20 ms

Contrast - Common

MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fast Fat Saturation
Contrasts	6
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
Base Resolution	500
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	36
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %

Geometry - Common

Slice Thickness	0.35 mm
TR	49.9 ms
Vol. TR	383.232 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	R1.4 P12.7 H1.7 mm
Orientation	T > C-13.2 > S1.9
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R1.4 P17.5 H14.3 mm
! Orientation	T > C-13.2 > S1.9
! Rotation	88.63 deg
! R >> L	175 mm
! A >> P	210 mm
! F >> H	87 mm

System - Adjust Volume

Reset	Off
-------	-----

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.20 ms
Asymmetric Echo	Off
Turbo Factor	7680
Segmentation	40
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260602_THS_ses03\localizer

TA: 15 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	All Segments
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
TE	3.69 ms
Averages	2
Concatenations	3

Contrast - Common

TR	8.6 ms
TE	3.69 ms

Contrast - Common

TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slice Group	3

Geometry - Common

Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg
Slice Group	2
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
Slice Group	3
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.6 ms
Segments	1
Concatenations	3

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None

Physio - Cardiac

Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	20 deg
Measurements	1
Contrasts	1
TE	3.69 ms
TR	8.6 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	qfl
---------------	-----

Sequence - Part 1

Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\260602_THS_ses03\AAhead_scout_md

TA: 15 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.53 ms
TE	1.53 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.53 ms
TE	1.53 ms
Flip Angle	16 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.7 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.53 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Initial Position	Isocenter
L	0.0 mm

Geometry - AutoAlign

P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
AutoAlign	Head

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off
AutoAlign MPR	On

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	16 deg
Measurements	1
Contrasts	1
TE	1.53 ms
TR	3.53 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260602_THS_ses03\gre_b0map_4iso_sag

TA: 31 sec Coil Selection: Auto Voxel Size: 4.0×4.0×4.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	44
Distance Factor	0 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
Averages	1
Concatenations	1

Contrast - Common

TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
MTC	Off
Flip Angle	25 deg
Fat-Water Contrast	Standard
Contrasts	2
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	44
Distance Factor	0 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fm_r
Dimension	2D
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	On
Bandwidth	908 Hz/Px
Asymmetric Echo	Off

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260602_THS_ses03\dzne_b1map_5iso_sag

TA: 7 sec Coil Selection: Auto Voxel Size: 5.0×5.0×5.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Averages	1
Concatenations	1

Contrast - Common

TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Flip Angle 1	60 deg
Flip Angle 2	6 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
Base Resolution	44
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal

Geometry - AutoAlign

Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	b1map 9e96902
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	1000 Hz/Px
Asymmetric Echo	Off
Turbo Factor	440
Echo Train Duration	1298 ms

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Mapping Technique	DREAM
Timing Scheme	STE*
Mixing Time	1130 us
Scale risetime	1.2

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260602_THS_ses03\dzne_b1map_5iso_sag

TA: 7 sec Coil Selection: Auto Voxel Size: 5.0×5.0×5.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Averages	1
Concatenations	1

Contrast - Common

TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Flip Angle 1	60 deg
Flip Angle 2	6 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
Base Resolution	44
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal

Geometry - AutoAlign

Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	b1map 9e96902
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	1000 Hz/Px
Asymmetric Echo	Off
Turbo Factor	440
Echo Train Duration	1298 ms

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Mapping Technique	DREAM
Timing Scheme	STE*
Mixing Time	1130 us
Scale risetime	1.2

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260602_THS_ses03\localizer

TA: 15 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	All Segments
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
TE	3.69 ms
Averages	2
Concatenations	3

Contrast - Common

TR	8.6 ms
TE	3.69 ms

Contrast - Common

TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slice Group	3

Geometry - Common

Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg
Slice Group	2
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
Slice Group	3
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.6 ms
Segments	1
Concatenations	3

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None

Physio - Cardiac

Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	20 deg
Measurements	1
Contrasts	1
TE	3.69 ms
TR	8.6 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	qfl
---------------	-----

Sequence - Part 1

Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\260602_THS_ses03\AAhead_scout_md

TA: 15 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.53 ms
TE	1.53 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.53 ms
TE	1.53 ms
Flip Angle	16 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.7 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.53 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Initial Position	Isocenter
L	0.0 mm

Geometry - AutoAlign

P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
AutoAlign	Head

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off
AutoAlign MPR	On

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	16 deg
Measurements	1
Contrasts	1
TE	1.53 ms
TR	3.53 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260602_THS_ses03\gre_b0map_4iso_sag

TA: 31 sec Coil Selection: Auto Voxel Size: 4.0×4.0×4.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	44
Distance Factor	0 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
Averages	1
Concatenations	1

Contrast - Common

TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
MTC	Off
Flip Angle	25 deg
Fat-Water Contrast	Standard
Contrasts	2
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	44
Distance Factor	0 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fm_r
Dimension	2D
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	On
Bandwidth	908 Hz/Px
Asymmetric Echo	Off

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260602_THS_ses03\dzne_b1map_5iso_sag

TA: 7 sec Coil Selection: Auto Voxel Size: 5.0×5.0×5.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Averages	1
Concatenations	1

Contrast - Common

TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Flip Angle 1	60 deg
Flip Angle 2	6 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
Base Resolution	44
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal

Geometry - AutoAlign

Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	b1map 9e96902
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	1000 Hz/Px
Asymmetric Echo	Off
Turbo Factor	440
Echo Train Duration	1298 ms

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Mapping Technique	DREAM
Timing Scheme	STE*
Mixing Time	1130 us
Scale risetime	1.2

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260602_THS_ses03\head

TA: 5 sec Coil Selection: Auto Voxel Size: 4.0×4.0×5.0 mm³ Acc.: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	16
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Routine

Slice Group	1
Slices	25
Distance Factor	100 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	3900.0 ms
TE	1.56 ms
Averages	1
Concatenations	1

Geometry - Common

Slice Group	1
Slices	25
Distance Factor	100 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	3900.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	3900.0 ms
TE	1.56 ms
Magn. Preparation	None
Flip Angle	5 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Centric

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3900.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None

Physio - Cardiac

Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	tfl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Whisper
Flow Compensation	None
Reordering	Centric
Bandwidth	440 Hz/Px
Echo Spacing	3.54 ms
Asymmetric Echo	Allowed
Turbo Factor	64

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Tx Scale Diag. Magn.	0.3
Tx Scale Diag. Phase	0 deg
Tx Scale Off-diag. Magn.	1.0
Tx Scale Off-diag. Phase	0 deg
Rel. B1 mapping	On
Ref. Scan	Off
Use B1 Map Recon.	On
Prep Pulse	Selective
Dummy RF pulses	1000
Dummy RF TRs	0

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260602_THS_ses03\bss_map

TA: 2:01 min Coil Selection: Manual Voxel Size: 4.0×4.0×4.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
TR	5.0 ms
TE	2.04 ms
TE2 B0	4.08 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	5.0 ms
TE	2.04 ms
TE2 B0	4.08 ms
Flip Angle	5 deg
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
Base Resolution	64
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.00 mm
TR	5.0 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
! Ref. Amplitude 1H	270.000 V

System - Tx/Rx

Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	5 deg

Inline - MapIt

Measurements	1
Contrasts	1
TE	2.04 ms
TE2 B0	4.08 ms
TR	5.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	1000 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off
Phase Stabilisation	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	NC7

Sequence - Special

Map Type	Integrated
Number of shim-sets	8
RO Spoiler	1.00
Measure B0	On
Call B0 Acq	Off
Call B1 Acq	Off
Rel. Mask Threshold	20

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\260602_THS_ses03\ptxps_AC_PA7_PW10_4mmISO_89s

TA: 23 sec Coil Selection: Auto Voxel Size: 4.0×4.0×3.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

Slice Thickness	3.0 mm
Base Resolution	48
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	30
Distance Factor	33 %
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
TE	23.00 ms
Averages	1
Concatenations	1

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	33 %
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	3300.0 ms
TE	23.00 ms
MTC	Off
Flip Angle	30 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	6
Delay in TR	0.00 ms

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %

Geometry - AutoAlign

Slice Group	1
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
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Geometry - Tim Planning Suite

Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	Patient-specific
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Rotation	0.54 deg
A >> P	192 mm
R >> L	192 mm
F >> H	119 mm
Reset	Off

System - pTx

B1 Shim	Patient-specific
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3300.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On

BOLD

Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	6
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	4166 Hz/Px
Echo Spacing	0.29 ms
Free Echo Spacing	On
EPI Factor	48

Sequence - Part 2

Introduction	Off
Ghost Reduction	Off

Sequence - Special

ABS Flip Angle	7.00 degree
ABS Pulse Width	10 ms
CP Mode	Off
Delta TE	1.00 ms
ABS Spoilers Width	2520 us
ABS Spoilers Mask Bit	7
ABS Spoilers M1 Null Mode	On
No of Pre Dummy TRs	4

Sequence - Special

No of Post Dummy TRs	0
Ramp Sampling Factor	12.50 %
Grad Stim Ratio	58.16 %
Ref Amplitude	0.00 V
ABS Tx Ref Amp	0.00 V

Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\260602_THS_ses03\ptxps_CP_PA4_PW10_4mmISO_20s

TA: 40 sec Coil Selection: Auto Voxel Size: 4.0×4.0×3.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

Slice Thickness	3.0 mm
Base Resolution	48
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	30
Distance Factor	33 %
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
TE	23.00 ms
Averages	1
Concatenations	1

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	33 %
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	3300.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	3300.0 ms
TE	23.00 ms
MTC	Off
Flip Angle	30 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	11
Delay in TR	0.00 ms

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %

Geometry - AutoAlign

Slice Group	1
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
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Geometry - Tim Planning Suite

Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	Patient-specific
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Rotation	0.54 deg
A >> P	192 mm
R >> L	192 mm
F >> H	119 mm
Reset	Off

System - pTx

B1 Shim	Patient-specific
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3300.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On

BOLD

Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	11
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	4166 Hz/Px
Echo Spacing	0.29 ms
Free Echo Spacing	On
EPI Factor	48

Sequence - Part 2

Introduction	Off
Ghost Reduction	Off

Sequence - Special

ABS Flip Angle	4.00 degree
ABS Pulse Width	10 ms
CP Mode	On
Delta TE	1.00 ms
ABS Spoilers Width	2520 us
ABS Spoilers Mask Bit	7
ABS Spoilers M1 Null Mode	On
No of Pre Dummy TRs	4

Sequence - Special

No of Post Dummy TRs	0
Ramp Sampling Factor	12.50 %
Grad Stim Ratio	58.16 %
Ref Amplitude	0.00 V
ABS Tx Ref Amp	0.00 V

Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\260602_THS_ses03\gre_FA10_acpc

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Rotation	0.54 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\260602_THS_ses03\gre_FA0_acpc

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5x0.5x5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Rotation	0.54 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
! Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\260602_THS_ses03\gre_FA10_plus20mm

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.3 P13.5 H21.2 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.3 P13.5 H21.2 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R2.3 P13.5 H21.2 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.0 A0.0 H20.0
R	0.0 mm
A	0.0 mm
H	20.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	21 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.3 P13.5 H21.2 mm
Orientation	T > C-12.1 > S1.7
Rotation	0.54 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\260602_THS_ses03\gre_FA0_up

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.3 P13.5 H21.2 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.3 P13.5 H21.2 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R2.3 P13.5 H21.2 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.0 A0.0 H20.0
R	0.0 mm
A	0.0 mm
H	20.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	21 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.3 P13.5 H21.2 mm
Orientation	T > C-12.1 > S1.7
Rotation	0.54 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
! Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\260602_THS_ses03\mp2rage_0p7iso_patientSpecific

TA: 8:27 min Coil Selection: Auto Voxel Size: 0.7×0.7×0.7 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
TE	3.37 ms
Averages	1
Concatenations	1

Contrast - Common

TR	5000.0 ms
TE	3.37 ms
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
--------------	----------

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	6/8
Slice Partial Fourier	7/8
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	Patient-specific
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	224 mm
F >> H	224 mm
R >> L	180 mm
Reset	Off

System - pTx

B1 Shim	Patient-specific
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	224 mm
FOV Phase	100.0 %
Phase Resolution	100 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	3.37 ms
TR	5000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	200 Hz/Px
Echo Spacing	7.68 ms
Asymmetric Echo	Off
Turbo Factor	224

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260602_THS_ses03\mp2rage_0p7iso_TrueForm

TA: 8:27 min Coil Selection: Auto Voxel Size: 0.7×0.7×0.7 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
TE	3.37 ms
Averages	1
Concatenations	1

Contrast - Common

TR	5000.0 ms
TE	3.37 ms
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
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Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	6/8
Slice Partial Fourier	7/8
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	224 mm
F >> H	224 mm
R >> L	180 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	224 mm
FOV Phase	100.0 %
Phase Resolution	100 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	3.37 ms
TR	5000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	200 Hz/Px
Echo Spacing	7.68 ms
Asymmetric Echo	Off
Turbo Factor	224

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260602_THS_ses03\lep2d_diff_1p3iso_b2k5_PA

TA: 9:18 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Rotation	-179.46 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
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Physio - PACE

Concatenations	1
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Diff

Diffusion Mode	MDDW
Diff. Directions	64
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	2500 s/mm ²
Averages 1	8
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\jon\Saskia\260602_THS_ses03\ep2d_diff_1p3iso_b1k0_PA

TA: 5:20 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Rotation	-179.46 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
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Physio - PACE

Concatenations	1
----------------	---

Diff

Diffusion Mode	MDDW
Diff. Directions	30
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	1000 s/mm ²
Averages 1	8
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\jon\Saskia\260602_THS_ses03\ep2d_diff_1p3iso_AP

TA: 1:50 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Rotation	-179.46 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
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Physio - PACE

Concatenations	1
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Diff

Diffusion Mode	MDDW
Diff. Directions	30
Diffusion Scheme	Monopolar
Diff. Weightings	1
b-value	0 s/mm ²
Averages	8
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\jon\Saskia\260602_THS_ses03\lep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC

TA: 6:37 min Coil Selection: Auto Voxel Size: 1.4x1.4x1.4 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R2.4 P14.5 H26.0 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
TE	24.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	24.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	380
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
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Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	66
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R2.4 P14.5 H26.0 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R2.4 P14.5 H26.0 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H25.0
L	0.0 mm
P	0.0 mm
H	25.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	26 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.4 P14.5 H26.0 mm
Orientation	T > C-12.1 > S1.7
Rotation	0.54 deg
A >> P	200 mm
R >> L	200 mm
F >> H	101 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	380
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	2480 Hz/Px
Echo Spacing	0.50 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260602_THS_ses03\ldzne_ep3d_vasoNIH

TA: 3:14 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.9 mm³ Acc.: 16 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Multi-echo spacing	32.50 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	87
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
Base Resolution	236
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	80
Acceleration Factor 3D	4
Reference Lines 3D	36
Reordering Shift 3D	2
Phase Partial Fourier	7/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	13 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Rotation	0.00 deg
A >> P	194 mm
R >> L	200 mm
F >> H	177 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1766 Hz/Px
Echo Spacing	0.64 ms
Asymmetric Echo	Off
Turbo Factor	52
Segmentation	1
EPI Factor	50

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260602_THS_ses03\ldzne_ep3d_vasoNIH

TA: 3:14 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.9 mm³ Acc.: 16 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Multi-echo spacing	32.50 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	87
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
Base Resolution	236
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	80
Acceleration Factor 3D	4
Reference Lines 3D	36
Reordering Shift 3D	2
Phase Partial Fourier	7/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	13 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Rotation	0.00 deg
A >> P	194 mm
R >> L	200 mm
F >> H	177 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1766 Hz/Px
Echo Spacing	0.64 ms
Asymmetric Echo	Off
Turbo Factor	52
Segmentation	1
EPI Factor	50

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260602_THS_ses03\ldzne_ep3d_Frankfurt_6echoes_minimumMaxBW

TA: 6:29 min Coil Selection: Auto Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	49.9 ms
Vol. TR	383.232 s
TE 1	4.70 ms
TE 2	11.90 ms
TE 3	19.10 ms
TE 4	26.30 ms
TE 5	33.50 ms
TE 6	40.70 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	49.9 ms
Vol. TR	383.232 s
TE 1	4.70 ms
TE 2	11.90 ms
TE 3	19.10 ms
TE 4	26.30 ms
TE 5	33.50 ms
TE 6	40.70 ms
Multi-echo spacing	7.20 ms

Contrast - Common

MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fast Fat Saturation
Contrasts	6
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
Base Resolution	500
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	36
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %

Geometry - Common

Slice Thickness	0.35 mm
TR	49.9 ms
Vol. TR	383.232 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	R1.7 P9.3 H1.6 mm
Orientation	T > C-12.1 > S1.7
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
-----------------	-------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	2 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R2.2 P13.1 H20.7 mm
! Orientation	T > C-12.1 > S1.7
! Rotation	90.54 deg
! R >> L	175 mm
! A >> P	210 mm
! F >> H	96 mm

System - Adjust Volume

Reset	Off
-------	-----

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.20 ms
Asymmetric Echo	Off
Turbo Factor	7680
Segmentation	40
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\gradient_comparison\ep2d_bold_grad_1p8mm

TA: 38 sec Coil Selection: Auto Voxel Size: 1.8x1.8x1.8 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	80
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	98.2 %
Slice Thickness	1.8 mm
TR	1000.0 ms
TE	21.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	21.00 ms
MTC	Off
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
FOV Phase	98.2 %

Resolution - Common

Slice Thickness	1.8 mm
Base Resolution	112
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	1
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	80
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	98.2 %
Slice Thickness	1.8 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	197 mm
R >> L	200 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	On
Ignore Meas. at Start	0
Ignore After Transition	0

BOLD

Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	On
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	4464 Hz/Px
Echo Spacing	0.29 ms
Free Echo Spacing	Off
EPI Factor	110

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Special

Custom Grad Perf.	On
System type	TerraX Impulse IX

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\gradient_comparison\ep2d_bold_grad_1p8mm_SC72

TA: 53 sec Coil Selection: Auto Voxel Size: 1.8x1.8x1.8 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	80
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	98.2 %
Slice Thickness	1.8 mm
TR	1520.0 ms
TE	35.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1520.0 ms
TE	35.00 ms
MTC	Off
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
FOV Phase	98.2 %

Resolution - Common

Slice Thickness	1.8 mm
Base Resolution	112
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	1
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	80
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	98.2 %
Slice Thickness	1.8 mm
TR	1520.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	197 mm
R >> L	200 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1520.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0

BOLD

Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	On
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	2480 Hz/Px
Echo Spacing	0.51 ms
Free Echo Spacing	Off
EPI Factor	110

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Special

Custom Grad Perf.	On
System type	TerraX SC72 W60

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\gradient_comparison\ep2d_bold_1p4mm_TE19_TR1000

TA: 16 sec Coil Selection: Auto Voxel Size: 1.4×1.4×1.4 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	84
Distance Factor	0 %
Position	R0.4 P7.6 H2.8 mm
Orientation	T > C-8.7 > S0.1
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
TE	19.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	19.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
----------	--------

Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	48
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	84
Distance Factor	0 %
Position	R0.4 P7.6 H2.8 mm
Orientation	T > C-8.7 > S0.1
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R0.4 P7.6 H2.8 mm
Orientation	T > C-8.7 > S0.1
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.4 P7.6 H2.8
R	0.4 mm
P	7.6 mm
H	2.8 mm
Initial Orientation	T > C
T > C	-8.70
> S	0.10

Geometry - AutoAlign

Initial Rotation	-0.14 deg
------------------	-----------

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	3 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.4 P7.6 H2.8 mm
Orientation	T > C-8.7 > S0.1
Rotation	-0.14 deg
A >> P	200 mm
R >> L	200 mm
F >> H	118 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off

Physio - Signal

Concatenations	1
----------------	---

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	1
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	3156 Hz/Px
Echo Spacing	0.38 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\gradient_comparison\ep2d_bold_grad_1p4mm_SC72

TA: 52 sec Coil Selection: Auto Voxel Size: 1.4×1.4×1.4 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	84
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1430.0 ms
TE	31.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1430.0 ms
TE	31.00 ms
MTC	Off
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	48
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	84
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1430.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	200 mm
R >> L	200 mm
F >> H	118 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1430.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	On
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	2042 Hz/Px
Echo Spacing	0.66 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Special

Custom Grad Perf.	On
System type	TerraX SC72 W60

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\gradient_comparison\cmrr_mbep2d_bold_1p4mm_TE19_TR1000

TA: 40 sec Coil Selection: Auto Voxel Size: 1.4×1.4×1.4 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	84
Distance Factor	0 %
Position	R0.4 P7.6 H2.8 mm
Orientation	T > C-8.7 > S0.1
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
TE	19.00 ms
Averages	1
Multi-band accel. factor	4

Contrast - Common

TR	1000.0 ms
TE	19.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference scan mode	Single-shot
Acceleration Factor PE	2
Reference Lines PE	70
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	84
Distance Factor	0 %
Position	R0.4 P7.6 H2.8 mm
Orientation	T > C-8.7 > S0.1
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	4

Geometry - AutoAlign

Slice Group	1
Position	R0.4 P7.6 H2.8 mm
Orientation	T > C-8.7 > S0.1
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.4 P7.6 H2.8
R	0.4 mm
P	7.6 mm
H	2.8 mm
Initial Orientation	T > C
T > C	-8.70

Geometry - AutoAlign

> S	0.10
Initial Rotation	-0.14 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	3 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R0.4 P7.6 H2.8 mm
! Orientation	T > C-8.7 > S0.1
! Rotation	-0.14 deg
! A >> P	200 mm
! R >> L	200 mm
! F >> H	118 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms

Physio - Signal

Multi-band accel. factor	4
--------------------------	---

BOLD

GLM Statistics	On
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	On
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	3156 Hz/Px
Echo Spacing	0.38 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Ghost Reduction	Off

Sequence - Special

Online multi-band recon.	Online
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\gradient_comparison\ep2d_bold_grad_1p0mm_SC72

TA: 1:05 min Coil Selection: Auto Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 12 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	102
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
TR	1600.0 ms
TE	44.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1600.0 ms
TE	44.00 ms
MTC	Off
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
FOV Phase	96.0 %

Resolution - Common

Slice Thickness	1.0 mm
Base Resolution	200
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	48
SMS Factor	6
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	102
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
TR	1600.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	192 mm
R >> L	200 mm
F >> H	102 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1600.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	On
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	1562 Hz/Px
Echo Spacing	0.76 ms
Free Echo Spacing	Off
EPI Factor	192

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Special

Custom Grad Perf.	On
System type	TerraX SC72 W60

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\gradient_comparison\ep2d_bold_1p0mm_TR1000_TE25_SMS2x6_BW3126_GC

TA: 33 sec Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc.: 12 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	102
Distance Factor	0 %
Position	L0.0 P22.7 H24.5 mm
Orientation	T > C-14.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
TR	1000.0 ms
TE	25.00 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	1000.0 ms
TE	25.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
Base Resolution	200
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	48
SMS Factor	6
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	102
Distance Factor	0 %
Position	L0.0 P22.7 H24.5 mm
Orientation	T > C-14.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P22.7 H24.5 mm
Orientation	T > C-14.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P22.7 H24.5
L	0.0 mm
P	22.7 mm
H	24.5 mm
Initial Orientation	T > C
T > C	-14.50

Geometry - AutoAlign

> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	25 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P22.7 H24.5 mm
! Orientation	T > C-14.5
! Rotation	0.00 deg
! A >> P	200 mm
! R >> L	200 mm
! F >> H	101 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
-----------------	------

Physio - Signal

TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	10
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	Off
EPI Factor	192

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260805-S012-2\AAhead_scout

TA: 14 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.25 ms
TE	1.53 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.25 ms
TE	1.53 ms
Flip Angle	16 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.3 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.25 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Initial Position	Isocenter
L	0.0 mm

Geometry - AutoAlign

P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
AutoAlign	Head

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off
AutoAlign MPR	On

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	16 deg
Measurements	1
Contrasts	1
TE	1.53 ms
TR	3.25 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260805-S012-2\ptxps_AC_PA8_PW12_3mmISO_46s

TA: 26 sec Coil Selection: Auto Voxel Size: 3.0×3.0×3.0 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	40
Distance Factor	0 %
Position	L0.9 A5.6 H5.7 mm
Orientation	T > C-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	900.0 ms
TE	25.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	900.0 ms
TE	25.00 ms
MTC	Off
Flip Angle	45 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	6
Delay in TR	0.00 ms

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	3.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	2
Reference Lines PE	62
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Hamming	Off
Distortion Correction	2D
Normalize	Prescan

Geometry - Common

Slice Group	1
Slices	40
Distance Factor	0 %
Position	L0.9 A5.6 H5.7 mm
Orientation	T > C-0.6
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	900.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.9 A5.6 H5.7 mm
Orientation	T > C-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.0 P1.9 H8.7
R	0.0 mm
P	1.9 mm
H	8.7 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	6 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	Patient-specific
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.9 A5.6 H5.7 mm
Orientation	T > C-0.6
Rotation	-3.72 deg
A >> P	192 mm
R >> L	192 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	Patient-specific
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	900.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	6
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	4112 Hz/Px
Echo Spacing	0.31 ms
Free Echo Spacing	Off
EPI Factor	64

Sequence - Part 2

Introduction	Off
Ghost Reduction	Off

Sequence - Special

ABS Flip Angle	8.00 degree
ABS Pulse Width	12 ms
CP Mode	Off
Delta TE	1.00 ms
ABS Spoilers Width	2520 us
ABS Spoilers Mask Bit	3
ABS Spoilers M1 Null Mode	On
No of Pre Dummy TRs	2
No of Post Dummy TRs	2
Ramp Sampling Factor	5.43 %
Grad Stim Ratio	75.07 %
Ref Amplitude	0.00 V
ABS Tx Ref Amp	0.00 V

Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\260805-S012-2\ptxps_CP_PA4_PW10_3mmISO_34s

TA: 43 sec Coil Selection: Auto Voxel Size: 3.0×3.0×3.0 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	40
Distance Factor	0 %
Position	L0.9 A5.6 H5.7 mm
Orientation	T > C-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	2280.0 ms
TE	25.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	2280.0 ms
TE	25.00 ms
MTC	Off
Flip Angle	30 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	11
Delay in TR	0.00 ms

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	3.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	2
Reference Lines PE	62
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Hamming	Off
Distortion Correction	2D
Normalize	Prescan

Geometry - Common

Slice Group	1
Slices	40
Distance Factor	0 %
Position	L0.9 A5.6 H5.7 mm
Orientation	T > C-0.6
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	2280.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.9 A5.6 H5.7 mm
Orientation	T > C-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.0 P1.9 H8.7
R	0.0 mm
P	1.9 mm
H	8.7 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	6 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	Patient-specific
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.9 A5.6 H5.7 mm
Orientation	T > C-0.6
Rotation	-3.72 deg
A >> P	192 mm
R >> L	192 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	Patient-specific
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2280.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	11
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	4112 Hz/Px
Echo Spacing	0.31 ms
Free Echo Spacing	Off
EPI Factor	64

Sequence - Part 2

Introduction	Off
Ghost Reduction	Off

Sequence - Special

ABS Flip Angle	4.00 degree
ABS Pulse Width	10 ms
CP Mode	On
Delta TE	1.00 ms

Sequence - Special

ABS Spoilers Width	2520 us
ABS Spoilers Mask Bit	3
ABS Spoilers M1 Null Mode	On
No of Pre Dummy TRs	2
No of Post Dummy TRs	2
Ramp Sampling Factor	5.43 %
Grad Stim Ratio	75.07 %
Ref Amplitude	0.00 V
ABS Tx Ref Amp	0.00 V

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260805-S012-2\ep2d_bold_1p8mm_TE21_TR1000

TA: 38 sec Coil Selection: Auto Voxel Size: 1.8×1.8×1.8 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	80
Distance Factor	0 %
Position	L1.1 A7.5 H6.1 mm
Orientation	T > C-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	98.2 %
Slice Thickness	1.8 mm
TR	1000.0 ms
TE	21.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	21.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
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Resolution - Common

FOV Phase	98.2 %
Slice Thickness	1.8 mm
Base Resolution	112
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	1
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	80
Distance Factor	0 %
Position	L1.1 A7.5 H6.1 mm
Orientation	T > C-0.6
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	98.2 %
Slice Thickness	1.8 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L1.1 A7.5 H6.1 mm
Orientation	T > C-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H9.0
L	0.0 mm
P	0.0 mm
H	9.0 mm
Initial Orientation	Transversal
Initial Rotation	0.18 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	6 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

I Position	L0.9 A5.6 H5.7 mm
I Orientation	T > C-0.6
I Rotation	-3.72 deg
I A >> P	192 mm
I R >> L	192 mm
I F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	4464 Hz/Px
Echo Spacing	0.29 ms
Free Echo Spacing	Off
EPI Factor	110

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\260805-S012-2\ep2d_bold_1p8mm_TE21_TR1000_MB1

TA: 1:45 min Coil Selection: Auto Voxel Size: 1.8×1.8×1.8 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Resolution - Common

FOV Phase	98.2 %
Slice Thickness	1.8 mm
Base Resolution	112
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	80
Distance Factor	0 %
Position	L1.1 A7.5 H6.1 mm
Orientation	T > C-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	98.2 %
Slice Thickness	1.8 mm
TR	4300.0 ms
TE	21.00 ms
Averages	1
Concatenations	1

Geometry - Common

Slice Group	1
Slices	80
Distance Factor	0 %
Position	L1.1 A7.5 H6.1 mm
Orientation	T > C-0.6
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	98.2 %
Slice Thickness	1.8 mm
TR	4300.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	4300.0 ms
TE	21.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	84 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
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Geometry - AutoAlign

Slice Group	1
Position	L1.1 A7.5 H6.1 mm
Orientation	T > C-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H9.0
L	0.0 mm
P	0.0 mm
H	9.0 mm
Initial Orientation	Transversal
Initial Rotation	0.18 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	6 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L1.1 A7.5 H6.1 mm
Orientation	T > C-0.6
Rotation	-3.54 deg
A >> P	197 mm
R >> L	200 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	4300.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0

BOLD

Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	4464 Hz/Px
Echo Spacing	0.29 ms
Free Echo Spacing	Off
EPI Factor	110

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\260805-S012-2\ep2d_bold_1p0mm_TE19_TR2000

TA: 1:07 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 9 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	126
Distance Factor	0 %
Position	L0.2 P1.8 H2.9 mm
Orientation	T > C-20.9 > S0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
TR	2000.0 ms
TE	19.00 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	2000.0 ms
TE	19.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	69 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
Base Resolution	200
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	3
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	126
Distance Factor	0 %
Position	L0.2 P1.8 H2.9 mm
Orientation	T > C-20.9 > S0.6
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
TR	2000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.2 P1.8 H2.9 mm
Orientation	T > C-20.9 > S0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.2 P9.4 H6.0
R	0.2 mm
P	9.4 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-20.40

Geometry - AutoAlign

> S	-0.60
Initial Rotation	-0.32 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	3 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.2 P1.8 H2.9 mm
Orientation	T > C-20.9 > S0.6
Rotation	-3.83 deg
A >> P	192 mm
R >> L	200 mm
F >> H	126 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
-----------------	------

Physio - Signal

TR	2000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	10
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	3126 Hz/Px
Echo Spacing	0.42 ms
Free Echo Spacing	Off
EPI Factor	192

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260805-S012-2\ep2d_bold_1p0mm_TE19_TR2000_MB1

TA: 2:10 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	126
Distance Factor	0 %
Position	L0.2 P1.8 H2.9 mm
Orientation	T > C-20.9 > S0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
TR	7140.0 ms
TE	19.00 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	7140.0 ms
TE	19.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	89 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
Base Resolution	200
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	1
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	126
Distance Factor	0 %
Position	L0.2 P1.8 H2.9 mm
Orientation	T > C-20.9 > S0.6
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	96.0 %
Slice Thickness	1.0 mm
TR	7140.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.2 P1.8 H2.9 mm
Orientation	T > C-20.9 > S0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.2 P9.4 H6.0
R	0.2 mm
P	9.4 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-20.40

Geometry - AutoAlign

> S	-0.60
Initial Rotation	-0.32 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	3 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.2 P1.8 H2.9 mm
Orientation	T > C-20.9 > S0.6
Rotation	-3.83 deg
A >> P	192 mm
R >> L	200 mm
F >> H	126 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
-----------------	------

Physio - Signal

TR	7140.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	10
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	3126 Hz/Px
Echo Spacing	0.42 ms
Free Echo Spacing	Off
EPI Factor	192

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260805-S012-2\ep2d_bold_1p4mm_TE19_TR1000

TA: 37 sec Coil Selection: Auto Voxel Size: 1.4×1.4×1.4 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	84
Distance Factor	0 %
Position	L0.2 A0.1 F0.2 mm
Orientation	T > C-9.3 > S0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
TE	19.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	19.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
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Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	48
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	84
Distance Factor	0 %
Position	L0.2 A0.1 F0.2 mm
Orientation	T > C-9.3 > S0.6
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.2 A0.1 F0.2 mm
Orientation	T > C-9.3 > S0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.4 P7.6 H2.8
R	0.4 mm
P	7.6 mm
H	2.8 mm
Initial Orientation	T > C
T > C	-8.70
> S	0.10

Geometry - AutoAlign

Initial Rotation	-0.14 deg
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Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.9 A5.6 H5.7 mm
! Orientation	T > C-0.6
! Rotation	-3.72 deg
! A >> P	192 mm
! R >> L	192 mm
! F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off

Physio - Signal

Concatenations	1
----------------	---

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	3156 Hz/Px
Echo Spacing	0.38 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260805-S012-2\ep2d_bold_1p4mm_TE19_TR1000_MB1

TA: 1:51 min Coil Selection: Auto Voxel Size: 1.4x1.4x1.4 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	84
Distance Factor	0 %
Position	L0.2 A0.1 F0.2 mm
Orientation	T > C-9.3 > S0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	4560.0 ms
TE	19.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	4560.0 ms
TE	19.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	85 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
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Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	48
SMS Factor	1
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	84
Distance Factor	0 %
Position	L0.2 A0.1 F0.2 mm
Orientation	T > C-9.3 > S0.6
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	4560.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.2 A0.1 F0.2 mm
Orientation	T > C-9.3 > S0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.4 P7.6 H2.8
R	0.4 mm
P	7.6 mm
H	2.8 mm
Initial Orientation	T > C
T > C	-8.70
> S	0.10

Geometry - AutoAlign

Initial Rotation	-0.14 deg
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Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.2 A0.1 F0.2 mm
Orientation	T > C-9.3 > S0.6
Rotation	-3.83 deg
A >> P	200 mm
R >> L	200 mm
F >> H	118 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	4560.0 ms
Log Signals	Off

Physio - Signal

Concatenations	1
----------------	---

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	3156 Hz/Px
Echo Spacing	0.37 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260805-S012-2\tof_fl3d_tra_0p25iso_TE1p67_TR5p09_GRAPPA_3x1_noFC

TA: 7:42 min Coil Selection: Manual Voxel Size: 0.3x0.3x0.3 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	288
Phase Oversampling	0 %
Slice Oversampling	11.1 %
FOV Read	200 mm
FOV Phase	90.0 %
Slice Thickness	0.25 mm
TR	5.58 ms
TE	1.71 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	5.58 ms
TE	1.71 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	200 mm
FOV Phase	90.0 %
Slice Thickness	0.25 mm
Base Resolution	800
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Phase Encoding Dir.	R >> L
Slices per Slab	288
Phase Oversampling	0 %
Slice Oversampling	11.1 %
FOV Read	200 mm

Geometry - Common

FOV Phase	90.0 %
Slice Thickness	0.25 mm
TR	5.58 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.8 P9.6 H14.0
R	0.8 mm
P	9.6 mm
H	14.0 mm
Initial Orientation	T > C
T > C	-18.30
> S	0.00
Initial Rotation	89.91 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	11 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Rotation	86.36 deg
R >> L	180 mm
A >> P	200 mm
F >> H	72 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.58 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	90.0 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.71 ms
TR	5.58 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	250 Hz/Px
Echo Spacing	5.58 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\260805-S012-2\tof_fl3d_tra_0p25iso_TE1p67_TR5p09_GRAPPA_3x1_noFC

TA: 3:54 min Coil Selection: Manual Voxel Size: 0.3x0.3x0.3 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	11.1 %
FOV Read	200 mm
FOV Phase	90.0 %
Slice Thickness	0.25 mm
TR	5.59 ms
TE	1.72 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	5.59 ms
TE	1.72 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	200 mm
FOV Phase	90.0 %
Slice Thickness	0.25 mm
Base Resolution	800
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Phase Encoding Dir.	R >> L
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	11.1 %
FOV Read	200 mm

Geometry - Common

FOV Phase	90.0 %
Slice Thickness	0.25 mm
TR	5.59 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.8 P9.6 H14.0
R	0.8 mm
P	9.6 mm
H	14.0 mm
Initial Orientation	T > C
T > C	-18.30
> S	0.00
Initial Rotation	89.91 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	11 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Rotation	86.36 deg
R >> L	180 mm
A >> P	200 mm
F >> H	36 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.59 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	90.0 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.72 ms
TR	5.59 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	250 Hz/Px
Echo Spacing	5.59 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\260805-S012-2\gre_3D_0p5mm_bipolar_4echoes_waterex

TA: 5:54 min Coil Selection: Manual Voxel Size: 0.5×0.5×0.5 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Common

Contrasts	4
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
Base Resolution	440
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L1.1 A7.6 H7.7 mm
Orientation	T > C-0.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
TR	16.0 ms
TE 1	2.00 ms
TE 2	4.08 ms
TE 3	6.16 ms
TE 4	8.24 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	48
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L1.1 A7.6 H7.7 mm
Orientation	T > C-0.6
Phase Encoding Dir.	R >> L
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm

Contrast - Common

TR	16.0 ms
TE 1	2.00 ms
TE 2	4.08 ms
TE 3	6.16 ms
TE 4	8.24 ms
MTC	Off
Magn. Preparation	None
Flip Angle	7 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off

Geometry - Common

TR	16.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L1.1 A7.6 H7.7 mm
Orientation	T > C-0.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.0 A0.1 H10.6
R	0.0 mm
A	0.1 mm
H	10.6 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	8 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L1.1 A7.6 H7.7 mm
Orientation	T > C-0.6
Rotation	86.28 deg
R >> L	169 mm
A >> P	220 mm

System - Adjust Volume

F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	16.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	76.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Sequence - Nuclei

Coil Elements	NC7
---------------	-----

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	7 deg
Measurements	1
Contrasts	4
TE 1	2.00 ms
TE 2	4.08 ms
TE 3	6.16 ms
TE 4	8.24 ms
TR	16.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Fast
Readout Mode	Bipolar
Gradient Mode	Fast
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Bandwidth 1	520 Hz/Px
Bandwidth 2	520 Hz/Px
Bandwidth 3	520 Hz/Px
Bandwidth 4	520 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Phase Stabilisation	On
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

\\USER\jon\Saskia\260805-S012-2\ep2d_bold_1p4mm_TE19_TR1000_80slices

TA: 37 sec Coil Selection: Auto Voxel Size: 1.4×1.4×1.4 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	80
Distance Factor	0 %
Position	L0.2 A0.1 F0.2 mm
Orientation	T > C-9.3 > S0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
TE	19.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	19.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
----------	--------

Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	48
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	80
Distance Factor	0 %
Position	L0.2 A0.1 F0.2 mm
Orientation	T > C-9.3 > S0.6
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.2 A0.1 F0.2 mm
Orientation	T > C-9.3 > S0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.4 P7.6 H2.8
R	0.4 mm
P	7.6 mm
H	2.8 mm
Initial Orientation	T > C
T > C	-8.70
> S	0.10

Geometry - AutoAlign

Initial Rotation	-0.14 deg
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Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.9 A5.6 H5.7 mm
! Orientation	T > C-0.6
! Rotation	-3.72 deg
! A >> P	192 mm
! R >> L	192 mm
! F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off

Physio - Signal

Concatenations	1
----------------	---

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	3156 Hz/Px
Echo Spacing	0.38 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260805-S012-2\tof_fl3d_tra_0p25iso_TE1p67_TR5p09_GRAPPA_3x1_noFC

TA: 5:38 min Coil Selection: Manual Voxel Size: 0.3x0.3x0.3 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	22.2 %
FOV Read	200 mm
FOV Phase	90.0 %
Slice Thickness	0.25 mm
TR	6.97 ms
TE	2.36 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	6.97 ms
TE	2.36 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	200 mm
FOV Phase	90.0 %
Slice Thickness	0.25 mm
Base Resolution	800
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	48
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Phase Encoding Dir.	R >> L
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	22.2 %
FOV Read	200 mm

Geometry - Common

FOV Phase	90.0 %
Slice Thickness	0.25 mm
TR	6.97 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.8 P9.6 H14.0
R	0.8 mm
P	9.6 mm
H	14.0 mm
Initial Orientation	T > C
T > C	-18.30
> S	0.00
Initial Rotation	89.91 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	11 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Rotation	86.36 deg
R >> L	180 mm
A >> P	200 mm
F >> H	36 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	6.97 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	90.0 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	2.36 ms
TR	6.97 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	250 Hz/Px
Echo Spacing	6.97 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	None

\\USER\jon\Saskia\260805-S012-2\gre_3D_0p5mm_bipolar_4echoes_waterex

TA: 5:54 min Coil Selection: Manual Voxel Size: 0.5×0.5×0.5 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Common

Contrasts	4
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
Base Resolution	440
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L1.1 A7.6 H7.7 mm
Orientation	T > C-0.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
TR	16.0 ms
TE 1	2.00 ms
TE 2	4.08 ms
TE 3	6.16 ms
TE 4	8.24 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	48
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L1.1 A7.6 H7.7 mm
Orientation	T > C-0.6
Phase Encoding Dir.	R >> L
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm

Contrast - Common

TR	16.0 ms
TE 1	2.00 ms
TE 2	4.08 ms
TE 3	6.16 ms
TE 4	8.24 ms
MTC	Off
Magn. Preparation	None
Flip Angle	7 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off

Geometry - Common

TR	16.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L1.1 A7.6 H7.7 mm
Orientation	T > C-0.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.0 A0.1 H10.6
R	0.0 mm
A	0.1 mm
H	10.6 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	8 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L1.1 A7.6 H7.7 mm
Orientation	T > C-0.6
Rotation	86.28 deg
R >> L	169 mm
A >> P	220 mm

System - Adjust Volume

F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	16.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	76.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Sequence - Nuclei

Coil Elements	NC7
---------------	-----

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	7 deg
Measurements	1
Contrasts	4
TE 1	2.00 ms
TE 2	4.08 ms
TE 3	6.16 ms
TE 4	8.24 ms
TR	16.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Fast
Readout Mode	Bipolar
Gradient Mode	Fast
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Bandwidth 1	520 Hz/Px
Bandwidth 2	520 Hz/Px
Bandwidth 3	520 Hz/Px
Bandwidth 4	520 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Phase Stabilisation	On
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

\\USER\jon\Saskia\260805-S012-2\gre_3D_0p5mm_bipolar_4echoes_waterex

TA: 5:54 min Coil Selection: Manual Voxel Size: 0.5×0.5×0.5 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Common

Contrasts	4
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
Base Resolution	440
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L1.1 A7.6 H7.7 mm
Orientation	T > C-0.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm
TR	16.0 ms
TE 1	2.00 ms
TE 2	4.08 ms
TE 3	6.16 ms
TE 4	8.24 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	48
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L1.1 A7.6 H7.7 mm
Orientation	T > C-0.6
Phase Encoding Dir.	R >> L
Slices per Slab	240
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	76.8 %
Slice Thickness	0.50 mm

Contrast - Common

TR	16.0 ms
TE 1	2.00 ms
TE 2	4.08 ms
TE 3	6.16 ms
TE 4	8.24 ms
MTC	Off
Magn. Preparation	None
Flip Angle	7 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off

Geometry - Common

TR	16.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L1.1 A7.6 H7.7 mm
Orientation	T > C-0.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.0 A0.1 H10.6
R	0.0 mm
A	0.1 mm
H	10.6 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	8 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L1.1 A7.6 H7.7 mm
Orientation	T > C-0.6
Rotation	86.28 deg
R >> L	169 mm
A >> P	220 mm

System - Adjust Volume

F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	16.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	76.8 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Sequence - Nuclei

Coil Elements	NC7
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Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	7 deg
Measurements	1
Contrasts	4
TE 1	2.00 ms
TE 2	4.08 ms
TE 3	6.16 ms
TE 4	8.24 ms
TR	16.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Fast
Readout Mode	Bipolar
Gradient Mode	Fast
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Bandwidth 1	520 Hz/Px
Bandwidth 2	520 Hz/Px
Bandwidth 3	520 Hz/Px
Bandwidth 4	520 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Phase Stabilisation	On
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

\\USER\jon\Saskia\260805-S012-2\tof_fl3d_tra_0p25iso_TE1p67_TR5p09_GRAPPA_3x1_noFC

TA: 3:35 min Coil Selection: Manual Voxel Size: 0.3x0.3x0.3 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	200 mm
FOV Phase	90.0 %
Slice Thickness	0.25 mm
Base Resolution	800
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	22.2 %
FOV Read	200 mm
FOV Phase	90.0 %
Slice Thickness	0.25 mm
TR	5.47 ms
TE	1.66 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	24
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Contrast - Common

TR	5.47 ms
TE	1.66 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Phase Encoding Dir.	R >> L
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	22.2 %

Geometry - Common

FOV Read	200 mm
FOV Phase	90.0 %
Slice Thickness	0.25 mm
TR	5.47 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.8 P9.6 H14.0
R	0.8 mm
P	9.6 mm
H	14.0 mm
Initial Orientation	T > C
T > C	-18.30
> S	0.00
Initial Rotation	89.91 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	11 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Rotation	86.36 deg
R >> L	180 mm
A >> P	200 mm
F >> H	36 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.47 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	90.0 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.66 ms
TR	5.47 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	250 Hz/Px
Echo Spacing	5.47 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\260805-S012-2\tof_fl3d_tra_0p25iso_TE1p67_TR5p09_GRAPPA_3x1_noFC

TA: 8:19 min Coil Selection: Manual Voxel Size: 0.2x0.2x0.2 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	200 mm
FOV Phase	90.1 %
Slice Thickness	0.20 mm
Base Resolution	1008
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	22.2 %
FOV Read	200 mm
FOV Phase	90.1 %
Slice Thickness	0.20 mm
TR	5.88 ms
TE	1.84 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Contrast - Common

TR	5.88 ms
TE	1.84 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-20 %
Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Phase Encoding Dir.	R >> L
Slices per Slab	144
Phase Oversampling	0 %
Slice Oversampling	22.2 %
FOV Read	200 mm

Geometry - Common

FOV Phase	90.1 %
Slice Thickness	0.20 mm
TR	5.88 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.8 P9.6 H14.0
R	0.8 mm
P	9.6 mm
H	14.0 mm
Initial Orientation	T > C
T > C	-18.30
> S	0.00
Initial Rotation	89.91 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	11 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.3 P2.0 H11.0 mm
Orientation	T > C-18.8 > S1.1
Rotation	86.36 deg
R >> L	181 mm
A >> P	200 mm
F >> H	29 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5.88 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	90.1 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.84 ms
TR	5.88 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Linear
Bandwidth	248 Hz/Px
Echo Spacing	5.88 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\epirecon\ldzne_ep3d

TA: 8 sec Coil Selection: Auto Voxel Size: 2.5×2.5×2.5 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P9.0 H3.6 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	48
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
TR	130.0 ms
Vol. TR	6240 ms
TE 1	48.00 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	130.0 ms
Vol. TR	6240 ms
TE 1	48.00 ms
Multi-echo spacing	85.10 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
Base Resolution	84
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P9.0 H3.6 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
TR	130.0 ms
Vol. TR	6240 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P9.0 H3.6 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	4 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P9.0 H3.6 mm
Orientation	Transversal
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.

Sequence - Part 1

RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1044 Hz/Px
Echo Spacing	1.01 ms
Asymmetric Echo	Off
Turbo Factor	48
Segmentation	1
EPI Factor	84

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\epirecon\ldzne_ep3d

TA: 8 sec Coil Selection: Auto Voxel Size: 2.5×2.5×2.5 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	48
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
TR	130.0 ms
Vol. TR	6240 ms
TE 1	48.00 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	130.0 ms
Vol. TR	6240 ms
TE 1	48.00 ms
Multi-echo spacing	85.10 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
Base Resolution	84
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
TR	130.0 ms
Vol. TR	6240 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.

Sequence - Part 1

RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1044 Hz/Px
Echo Spacing	1.01 ms
Asymmetric Echo	Off
Turbo Factor	48
Segmentation	1
EPI Factor	84

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\epirecon\ldzne_ep3d

TA: 28 sec Coil Selection: Auto Voxel Size: 1.0×1.0×1.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P7.8 H20.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	48
Phase Oversampling	2.4 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	13.0 ms
Vol. TR	26.832 s
TE 1	2.70 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	13.0 ms
Vol. TR	26.832 s
TE 1	2.70 ms
Multi-echo spacing	3.10 ms
MTC	Off
Flip Angle	13 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	210
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P7.8 H20.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	2.4 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	13.0 ms
Vol. TR	26.832 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P7.8 H20.3 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	20 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P7.8 H20.3 mm
Orientation	Transversal
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	48 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.

Sequence - Part 1

RF Pulse Type	Optimized
Gradient Mode	Fast
Reordering	Linear
Bandwidth	2976 Hz/Px
Echo Spacing	0.52 ms
Asymmetric Echo	Off
Turbo Factor	2064
Segmentation	43
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260921-VascProt\AAhead_scout

TA: 14 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.25 ms
TE	1.53 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.25 ms
TE	1.53 ms
Flip Angle	16 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.3 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.25 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Initial Position	Isocenter
L	0.0 mm

Geometry - AutoAlign

P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
AutoAlign	Head

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off
AutoAlign MPR	On

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	16 deg
Measurements	1
Contrasts	1
TE	1.53 ms
TR	3.25 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260921-VascProt\ldzne_ep3d_6echoes_R1

TA: 6:52 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L1.3 P18.7 H19.3 mm
Orientation	T > C-28.5 > S-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	4.2 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	50.6 ms
Vol. TR	404.8 s
TE 1	5.00 ms
TE 2	12.20 ms
TE 3	19.40 ms
TE 4	26.60 ms
TE 5	33.80 ms
TE 6	41.00 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	50.6 ms
Vol. TR	404.8 s
TE 1	5.00 ms
TE 2	12.20 ms
TE 3	19.40 ms
TE 4	26.60 ms
TE 5	33.80 ms
TE 6	41.00 ms
Multi-echo spacing	7.20 ms

Contrast - Common

MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fat Saturation
Contrasts	6
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
Base Resolution	500
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	36
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L1.3 P18.7 H19.3 mm
Orientation	T > C-28.5 > S-0.6
Phase Encoding Dir.	A >> P
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	4.2 %
FOV Read	175 mm
FOV Phase	120.0 %

Geometry - Common

Slice Thickness	0.35 mm
TR	50.6 ms
Vol. TR	404.8 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L1.3 P18.7 H19.3 mm
Orientation	T > C-28.5 > S-0.6
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.2 P8.6 H9.6
R	0.2 mm
P	8.6 mm
H	9.6 mm
Initial Orientation	T > C
T > C	-15.70
> S	0.20
Initial Rotation	0.19 deg

Geometry - Saturation

Saturation Mode	Standard
-----------------	----------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	19 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L1.3 P24.3 H29.3 mm
! Orientation	T > C-28.5 > S-0.6
! Rotation	91.00 deg
! R >> L	175 mm

System - Adjust Volume

! A >> P	210 mm
! F >> H	108 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Optimized
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.20 ms
Asymmetric Echo	Off
Turbo Factor	8000
Segmentation	40
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\260921-VascProt\11_mp2rage_sag_Op7_R5

TA: 8:50 min Coil Selection: Manual Voxel Size: 0.7×0.7×0.7 mm³ Acc:: 5 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.2 A0.3 F27.5 mm
Orientation	S > T1.6 > C-1.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Slices per Slab	224
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	6000.0 ms
TE	1.99 ms
Averages	1
Concatenations	1
Coil Elements	NC7

Contrast - Common

TR	6000.0 ms
TE	1.99 ms
Magn. Preparation	Non-sel. IR
TI 1	870 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	5
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Allowed
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.2 A0.3 F27.5 mm
Orientation	S > T1.6 > C-1.5
Phase Encoding Dir.	A >> P
Slices per Slab	224
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	6000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.2 A0.3 F27.5 mm
Orientation	S > T1.6 > C-1.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis
Initial Position	L0.6 P5.1 F0.5
L	0.6 mm
P	5.1 mm
F	0.5 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	27 mm
Table Position	F

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.2 A0.3 F27.5 mm
Orientation	S > T1.6 > C-1.5
Rotation	-13.83 deg
A >> P	224 mm
F >> H	224 mm
R >> L	157 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	6000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI 1	870 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	224 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	1.99 ms
TR	6000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	240 Hz/Px
Echo Spacing	5.86 ms
Asymmetric Echo	Allowed
Turbo Factor	224

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Custom pTx Pulse Duration	Off
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Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\260921-VascProt\tof_HR_fl3d_tra_0p2_TE1p83_TR10_GRAPPA_3x1_FA15

TA: 6:32 min Coil Selection: Auto Voxel Size: 0.2x0.2x0.2 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Centric

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	100 %

Resolution - Common

FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm
Base Resolution	1040
Phase Resolution	100 %
Slice Resolution	100 %
Trajectory	Cartesian
Interpolation	1.00

Routine

Slab Group	1
Slabs	1
Distance Factor	-10 %
Position	R2.6 P25.9 H50.5 mm
Orientation	T > C-12.6 > S1.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Slices per Slab	104
Phase Oversampling	0 %
Slice Oversampling	15.4 %
FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm
TR	10.00 ms
TE	1.83 ms
Averages	1
Concatenations	1

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	36
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Contrast - Common

TR	10.00 ms
TE	1.83 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	15 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-10 %
Position	R2.6 P25.9 H50.5 mm
Orientation	T > C-12.6 > S1.6
Phase Encoding Dir.	R >> L
Slices per Slab	104
Phase Oversampling	0 %
Slice Oversampling	15.4 %
FOV Read	208 mm
FOV Phase	86.2 %
Slice Thickness	0.20 mm

Contrast - Dynamic

Dynamic Mode	Standard
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Geometry - Common

TR	10.00 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R2.6 P25.9 H50.5 mm
Orientation	T > C-12.6 > S1.6
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	R0.2 P12.7 H54.6
R	0.2 mm
P	12.7 mm
H	54.6 mm
Initial Orientation	T > C
T > C	-4.50
> S	0.00
Initial Rotation	89.88 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	50 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.6 P25.9 H50.5 mm
Orientation	T > C-12.6 > S1.6

System - Adjust Volume

Rotation	89.51 deg
R >> L	180 mm
A >> P	208 mm
F >> H	21 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	10.00 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	208 mm
FOV Phase	86.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	1.83 ms
TR	10.00 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - Open Recon**

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Boost*
Flow Compensation	None
Reordering	Centric
Bandwidth	253 Hz/Px
Echo Spacing	5.83 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Boost Assistant	Off
Optimization	None

\\USER\jon\Saskia\THS_S02_ses01\localizer

TA: 15 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	All Segments
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
TE	3.69 ms
Averages	2
Concatenations	3

Contrast - Common

TR	8.6 ms
TE	3.69 ms

Contrast - Common

TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slice Group	3

Geometry - Common

Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg
Slice Group	2
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
Slice Group	3
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.6 ms
Segments	1
Concatenations	3

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None

Physio - Cardiac

Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	20 deg
Measurements	1
Contrasts	1
TE	3.69 ms
TR	8.6 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	qfl
---------------	-----

Sequence - Part 1

Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\THS_S02_ses01\AAhead_scout_md

TA: 15 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.53 ms
TE	1.53 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.53 ms
TE	1.53 ms
Flip Angle	16 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.7 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.53 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Initial Position	Isocenter
L	0.0 mm

Geometry - AutoAlign

P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
AutoAlign	Head

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off
AutoAlign MPR	On

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	16 deg
Measurements	1
Contrasts	1
TE	1.53 ms
TR	3.53 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS_S02_ses01\gre_b0map_4iso_sag

TA: 31 sec Coil Selection: Auto Voxel Size: 4.0×4.0×4.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	44
Distance Factor	0 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
Averages	1
Concatenations	1

Contrast - Common

TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
MTC	Off
Flip Angle	25 deg
Fat-Water Contrast	Standard
Contrasts	2
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	44
Distance Factor	0 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
---------	----------

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fm_r
Dimension	2D
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	On
Bandwidth	908 Hz/Px
Asymmetric Echo	Off

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS_S02_ses01\ldzne_b1map_5iso_sag

TA: 7 sec Coil Selection: Auto Voxel Size: 5.0×5.0×5.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Averages	1
Concatenations	1

Contrast - Common

TR	3000.0 ms
TE 1	0.86 ms
TE 2	1.99 ms
Flip Angle 1	60 deg
Flip Angle 2	6 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
Base Resolution	44
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal

Geometry - AutoAlign

Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	b1map 9e96902
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	1000 Hz/Px
Asymmetric Echo	Off
Turbo Factor	440
Echo Train Duration	1298 ms

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Mapping Technique	DREAM
Timing Scheme	STE*
Mixing Time	1130 us
Scale risetime	1.2

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS_S02_ses01\gre_FA10_acpc

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	1 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Rotation	0.93 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\THS_S02_ses01\gre_FA0_acpc

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5x0.5x5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	1 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Rotation	0.93 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
! Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\THS_S02_ses01\gre_FA10_plus20mm

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.7 P3.9 H20.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.7 P3.9 H20.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R2.7 P3.9 H20.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L1.4 A1.2 H19.9
L	1.4 mm
A	1.2 mm
H	19.9 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	21 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.7 P3.9 H20.7 mm
Orientation	T > S4.1 > C-3.4
Rotation	0.93 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\THS_S02_ses01\gre_FA0_plus20mm

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.7 P3.9 H20.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.7 P3.9 H20.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R2.7 P3.9 H20.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L1.4 A1.2 H19.9
L	1.4 mm
A	1.2 mm
H	19.9 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	21 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.7 P3.9 H20.7 mm
Orientation	T > S4.1 > C-3.4
Rotation	0.93 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.168790 MHz
! Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Nuclei

TX/RX Nucleus	¹ H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\jon\Saskia\THS_S02_ses01\mp2rage_0p7iso_TrueForm

TA: 8:27 min Coil Selection: Auto Voxel Size: 0.7×0.7×0.7 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
TE	3.37 ms
Averages	1
Concatenations	1

Contrast - Common

TR	5000.0 ms
TE	3.37 ms
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
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Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	6/8
Slice Partial Fourier	7/8
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	224 mm
F >> H	224 mm
R >> L	180 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	224 mm
FOV Phase	100.0 %
Phase Resolution	100 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	3.37 ms
TR	5000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	200 Hz/Px
Echo Spacing	7.68 ms
Asymmetric Echo	Off
Turbo Factor	224

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS_S02_ses01\mp2rage_0p7iso_TrueForm

TA: 8:27 min Coil Selection: Auto Voxel Size: 0.7×0.7×0.7 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
TE	3.37 ms
Averages	1
Concatenations	1

Contrast - Common

TR	5000.0 ms
TE	3.37 ms
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
--------------	----------

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	6/8
Slice Partial Fourier	7/8
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	224 mm
F >> H	224 mm
R >> L	180 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	224 mm
FOV Phase	100.0 %
Phase Resolution	100 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	3.37 ms
TR	5000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	200 Hz/Px
Echo Spacing	7.68 ms
Asymmetric Echo	Off
Turbo Factor	224

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS_S02_ses01\ep2d_diff_1p3iso_b2k5_PA

TA: 9:18 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	1 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Rotation	-179.07 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
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Physio - PACE

Concatenations	1
----------------	---

Diff

Diffusion Mode	MDDW
Diff. Directions	64
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	2500 s/mm ²
Averages 1	8
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\jon\Saskia\THS_S02_ses01\ep2d_diff_1p3iso_b1k0_PA

TA: 5:20 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	1 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

I Position	R2.7 P3.9 H0.7 mm
I Orientation	T > S4.1 > C-3.4
I Rotation	-179.07 deg
I A >> P	208 mm
I R >> L	208 mm
I F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
---------------	-----

Physio - PACE

Concatenations	1
----------------	---

Diff

Diffusion Mode	MDDW
Diff. Directions	30
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	1000 s/mm ²
Averages 1	8
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\jon\Saskia\THS_S02_ses01\ep2d_diff_1p3iso_AP

TA: 1:50 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R2.7 P3.9 H0.7 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	1 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

I Position	R2.7 P3.9 H0.7 mm
I Orientation	T > S4.1 > C-3.4
I Rotation	-179.07 deg
I A >> P	208 mm
I R >> L	208 mm
I F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
---------------	-----

Physio - PACE

Concatenations	1
----------------	---

Diff

Diffusion Mode	MDDW
Diff. Directions	30
Diffusion Scheme	Monopolar
Diff. Weightings	1
b-value	0 s/mm ²
Averages	8
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\jon\Saskia\THS_S02_ses01lep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC

TA: 6:37 min Coil Selection: Auto Voxel Size: 1.4x1.4x1.4 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R4.5 P5.4 H25.6 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
TE	24.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	24.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	380
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
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Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	66
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R4.5 P5.4 H25.6 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R4.5 P5.4 H25.6 mm
Orientation	T > S4.1 > C-3.4
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H25.0
L	0.0 mm
P	0.0 mm
H	25.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	26 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R4.5 P5.4 H25.6 mm
Orientation	T > S4.1 > C-3.4
Rotation	0.93 deg
A >> P	200 mm
R >> L	200 mm
F >> H	101 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	380
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	2480 Hz/Px
Echo Spacing	0.50 ms
Free Echo Spacing	Off
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\jon\Saskia\THS_S02_ses01\dzne_ep3d_vasoNIH

TA: 3:14 min Coil Selection: Auto Voxel Size: 0.8x0.8x0.9 mm³ Acc:: 16 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Multi-echo spacing	32.50 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	87
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
Base Resolution	236
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	80
Acceleration Factor 3D	4
Reference Lines 3D	36
Reordering Shift 3D	2
Phase Partial Fourier	7/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	13 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Rotation	0.00 deg
A >> P	194 mm
R >> L	200 mm
F >> H	177 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1766 Hz/Px
Echo Spacing	0.64 ms
Asymmetric Echo	Off
Turbo Factor	52
Segmentation	1
EPI Factor	50

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\THS_S02_ses01\dzne_ep3d_vasoNIH

TA: 3:14 min Coil Selection: Auto Voxel Size: 0.8x0.8x0.9 mm³ Acc:: 16 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Multi-echo spacing	32.50 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	87
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
Base Resolution	236
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	80
Acceleration Factor 3D	4
Reference Lines 3D	36
Reordering Shift 3D	2
Phase Partial Fourier	7/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P11.4 F12.6 mm
Orientation	T > C-14.7
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	13 mm
Table Position	F

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P11.4 F12.6 mm
! Orientation	T > C-14.7
! Rotation	0.00 deg
! A >> P	194 mm
! R >> L	200 mm
! F >> H	177 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1766 Hz/Px
Echo Spacing	0.64 ms
Asymmetric Echo	Off
Turbo Factor	52
Segmentation	1
EPI Factor	50

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
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\\USER\jon\Saskia\THS_S02_ses01\dzne_ep3d_Frankfurt_6echoes_minimumMaxBW_optR

TA: 6:52 min Coil Selection: Auto Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	R4.0 P18.9 H21.9 mm
Orientation	T > C-16.3 > S3.8
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	4.2 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	50.6 ms
Vol. TR	404.8 s
TE 1	5.00 ms
TE 2	12.20 ms
TE 3	19.40 ms
TE 4	26.60 ms
TE 5	33.80 ms
TE 6	41.00 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	50.6 ms
Vol. TR	404.8 s
TE 1	5.00 ms
TE 2	12.20 ms
TE 3	19.40 ms
TE 4	26.60 ms
TE 5	33.80 ms
TE 6	41.00 ms
Multi-echo spacing	7.20 ms

Contrast - Common

MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fat Saturation
Contrasts	6
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
Base Resolution	500
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	36
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R4.0 P18.9 H21.9 mm
Orientation	T > C-16.3 > S3.8
Phase Encoding Dir.	A >> P
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	4.2 %
FOV Read	175 mm
FOV Phase	120.0 %

Geometry - Common

Slice Thickness	0.35 mm
TR	50.6 ms
Vol. TR	404.8 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	R4.0 P18.9 H21.9 mm
Orientation	T > C-16.3 > S3.8
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P13.8 H22.1
L	0.0 mm
P	13.8 mm
H	22.1 mm
Initial Orientation	T > C
T > C	-12.90
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	22 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R4.0 P18.9 H21.9 mm
! Orientation	T > C-16.3 > S3.8
! Rotation	91.81 deg
! R >> L	175 mm

System - Adjust Volume

! A >> P	210 mm
! F >> H	91 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.168790 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Optimized
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.20 ms
Asymmetric Echo	Off
Turbo Factor	8000
Segmentation	40
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
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